TH series/TS2000/TS2100

Industrial Robot

INSTRUCTION MANUAL

INDUSTRIAL ROBOT

TRANSPORTATION AND INSTALLATION MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

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NUMAZU, JAPAN

Shibaura Machine

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Preface

This manual describes how to unpack and install the TH series industrial robot and controller. Specifically, it describes how to unpack a shipment containing the equipment, how to connect wiring and air piping, and how to attach tools. Be sure to look through this manual before unpacking the shipment.

Before beginning the work according to this manual, read through the Setup Manual and Safety Manual so that you can learn the names and functions of various components and the safety measures.

This manual is divided into the following four (4) sections:

Section 1 Transportation
This section describes how to remove the robot and controller from their boxes and how to transport them to the installation site. This section also discusses how to temporarily store the equipment after unpacking the shipment.

Section 2 Installation
This section discusses the equipment installation environment, space requirements, and how to install the equipment.

Section 3 System Connections
This section describes how to connect the robot, controller and peripheral equipment.

Section 4 Tool Interlace
This section describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool. This section also discusses maximum permissible loads of the tool.

Precautions on Safety

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
! **DANGER**	This means that "incorrect handling will lead to fatalities or major injuries".
! **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to major fires due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited(must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory(must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means caution and danger. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Always read through the Safety Manual provided separately before starting actual work to ensure safety work covering from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

DANGER	
 Prohibited	• DO NOT install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. • DO NOT install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. • Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
 Mandatory	• Always secure the robot with the attached clamps before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended. • Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. • Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. • Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
 Always ground	• Completely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause mis-operation by noise.

! CAUTION	
 Prohibited	• NEVER lift the robot by the arm 2 cover or arm 2. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. • For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
 Mandatory	• When lifting the robot, lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. • When storing the robot, securely fix it on the base. The robot will be unstable if just set down, and it could tilt over.

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Section 1

Transportation

1.1 Unpacking

The robot and controller are shipped separately in wooden crates and corrugated cardboards. The packaging states are shown in Figs. 1.1 and 1.2. The teach pendant and other accessories such as the instruction manual are packaged separately.

Open the packages in a location easily accessible to where the equipment is to be installed. Take care not to damage the robot or controller. After opening the packages, check to make sure that all the accessories are present and that nothing has been damaged in transit.

! DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden crate, corrugated cardboard, polyethylene shipping bag and cushion material according to your standards.

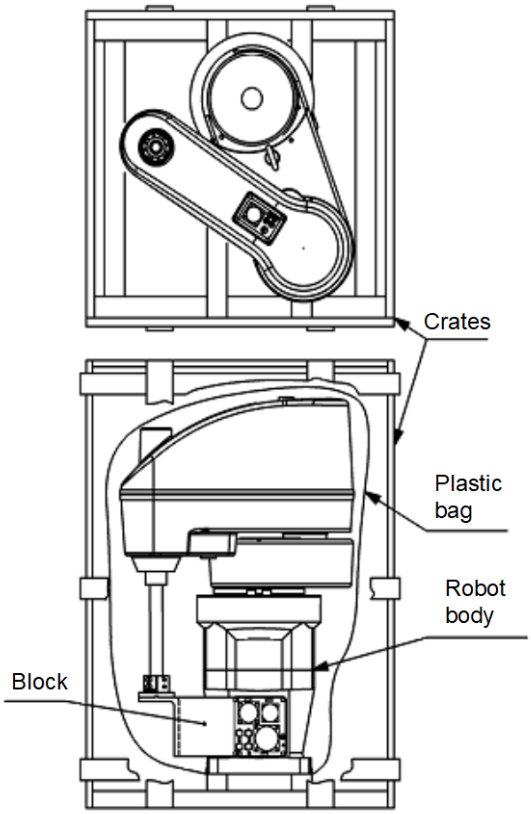


Fig. 1.1 Robot package

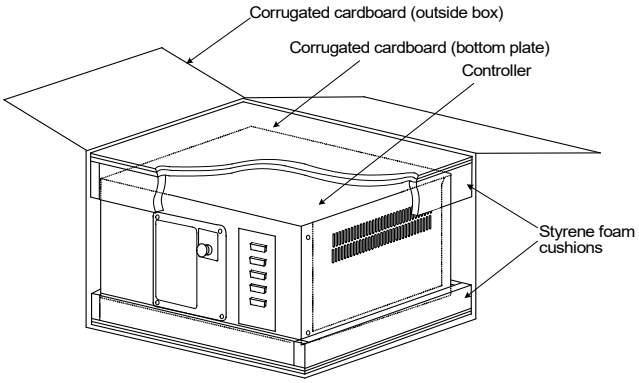


Fig. 1.2 Controller package

1.2 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact (9.8m/s^2 or more) or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the nuts and screws.
If the equipment is to be moved to a location some distance from where it was unpacked, put the equipment back into the wooden crates and corrugated cardboards and reposition the cushions as they were.

1.2.1 Mass and Dimensions

The mass and dimensions of the robot and controller are shown in Table 1.1.

Table 1.1

Type	Mass	Outer dimensions (during transport)
Robot TH450	27 kg	See Fig. 1.3.
Robot TH550	29 kg	See Fig. 1.4.
Robot TH650	60 kg	See Fig. 1.5.
Robot TH850	72 kg	See Fig. 1.6.
Robot TH1050	75 kg	See Fig. 1.7.
Controller TS2000	12 kg	See Fig. 2.19.
Controller TS2100	16 kg	See Fig. 2.20.

1.2.2 Transporting the Robot

In principle, transport the robot in the posture as shown in Figs. 1.3 to Fig. 1.7. The robot is shipped with the arm folded back and secured with a clamp. (After you have unpacked the shipment, you can move it as it is.)

! DANGER

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is suspended.

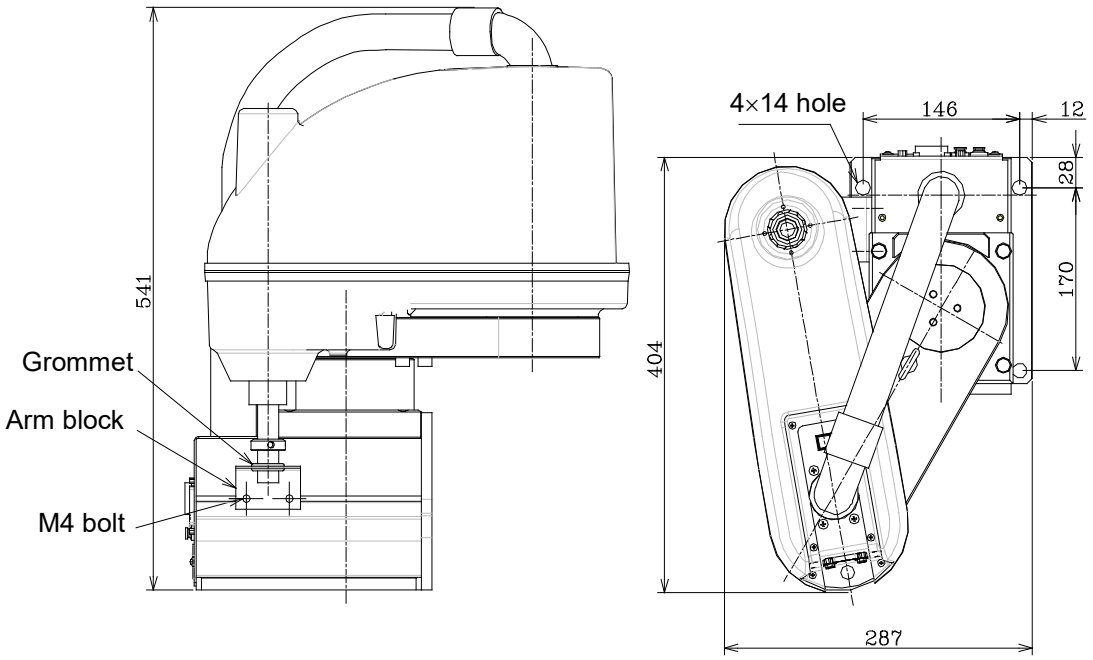


Fig. 1.3 Securing robot TH450 for transportation

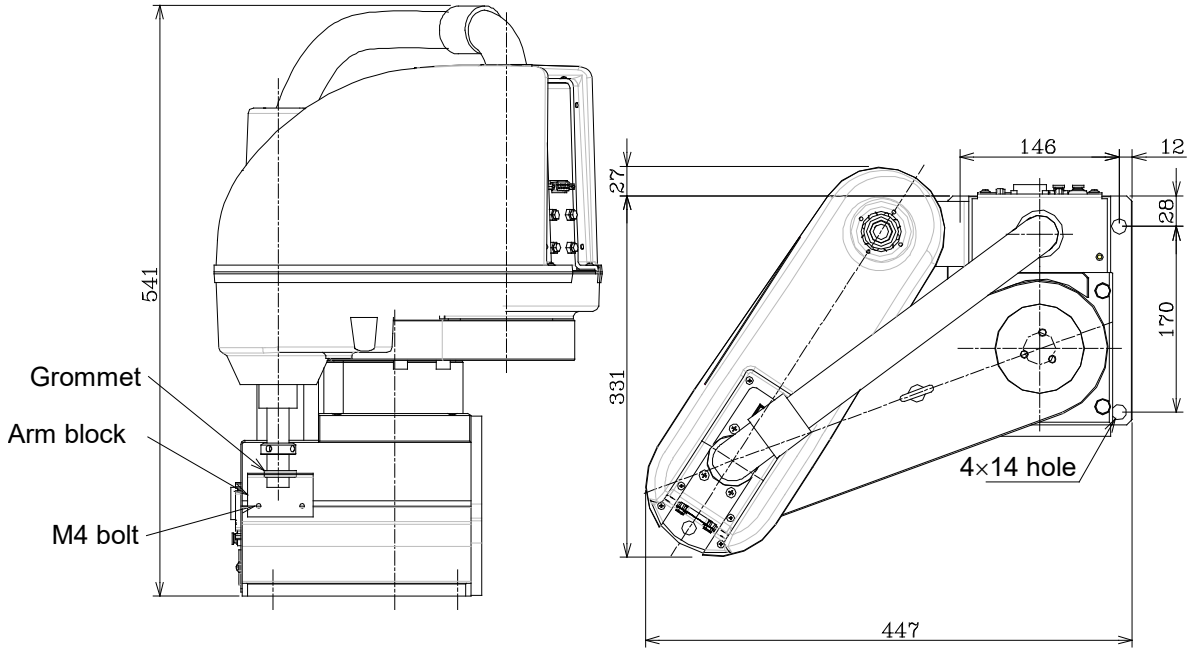


Fig. 1.4 Securing robot TH550 for transportation

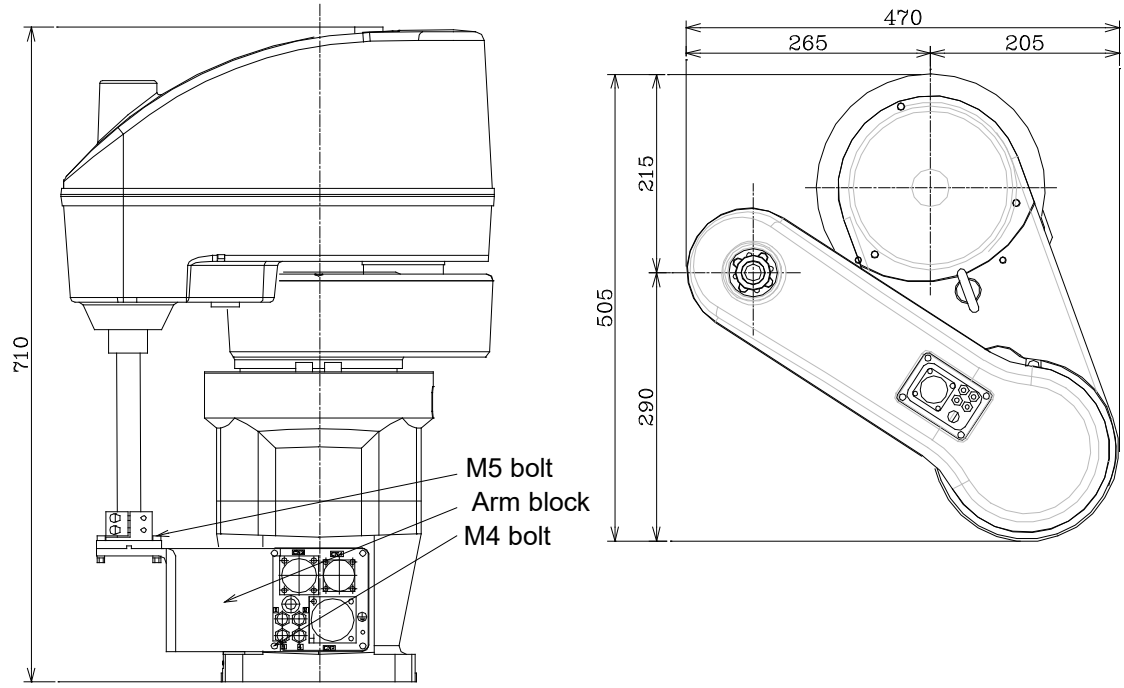


Fig. 1.5 Securing robot TH650 for transportation

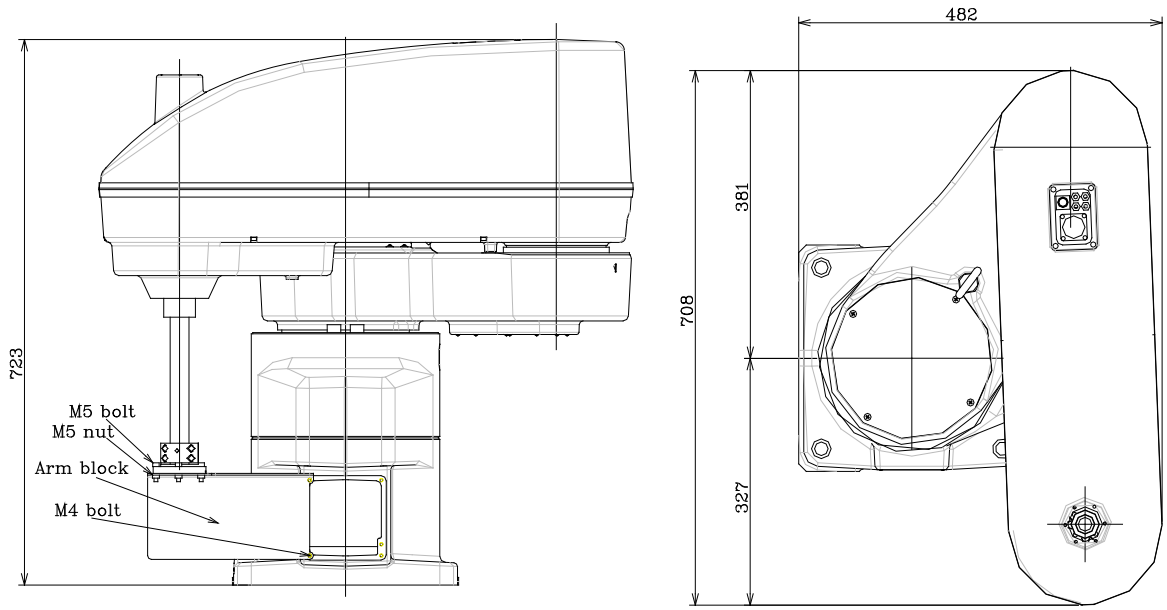


Fig. 1.6 Securing robot TH850 for transportation

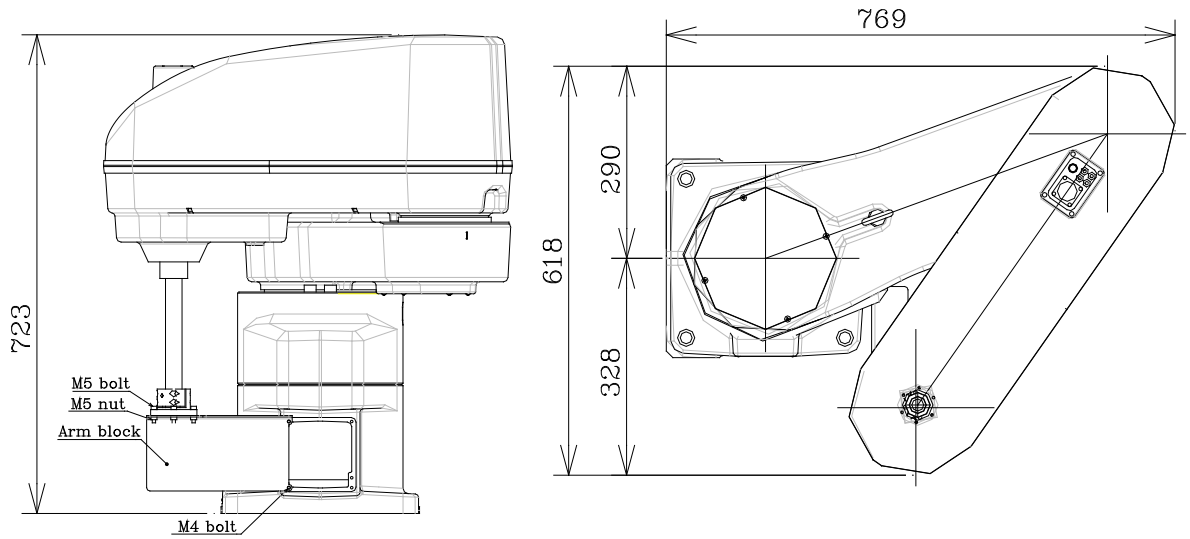
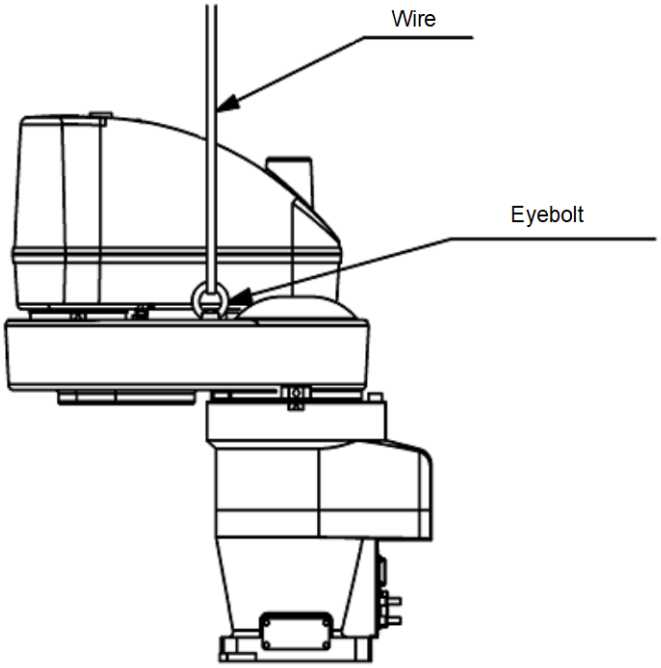


Fig. 1.7 Securing robot TH1050 for transportation

It is also possible to move the robot by crane. As shown in Fig. 1.8, pass a wire to the attached eyebolt, then lift up the robot carefully.



Mass of robot

TH450:27kg
TH550:29kg
TH650:60kg
TH850:72kg
TH1050:75kg

Fig. 1.8 Lifting up the robot by crane

! CAUTION

- Use a wire with a limit load that can sufficiently withstand the robot mass.
- The robot will tilt somewhat when lifted. Lift it up carefully and slowly.
- When raising and lowering the robot, DO NOT exert any excessive impact (9.8 m/s^2 or over) on it. Carefully treat the robot.
- When lifting the robot by workers, hold the positions shown in Fig. 1.9 by hands. If the arm 2 or cover is held by hands, an unusually large force is exerted, resulting in a trouble.
- If the robot has been carried manually by the workers, take utmost care to prevent their hand or leg from being caught in the robot when placing it on the floor.

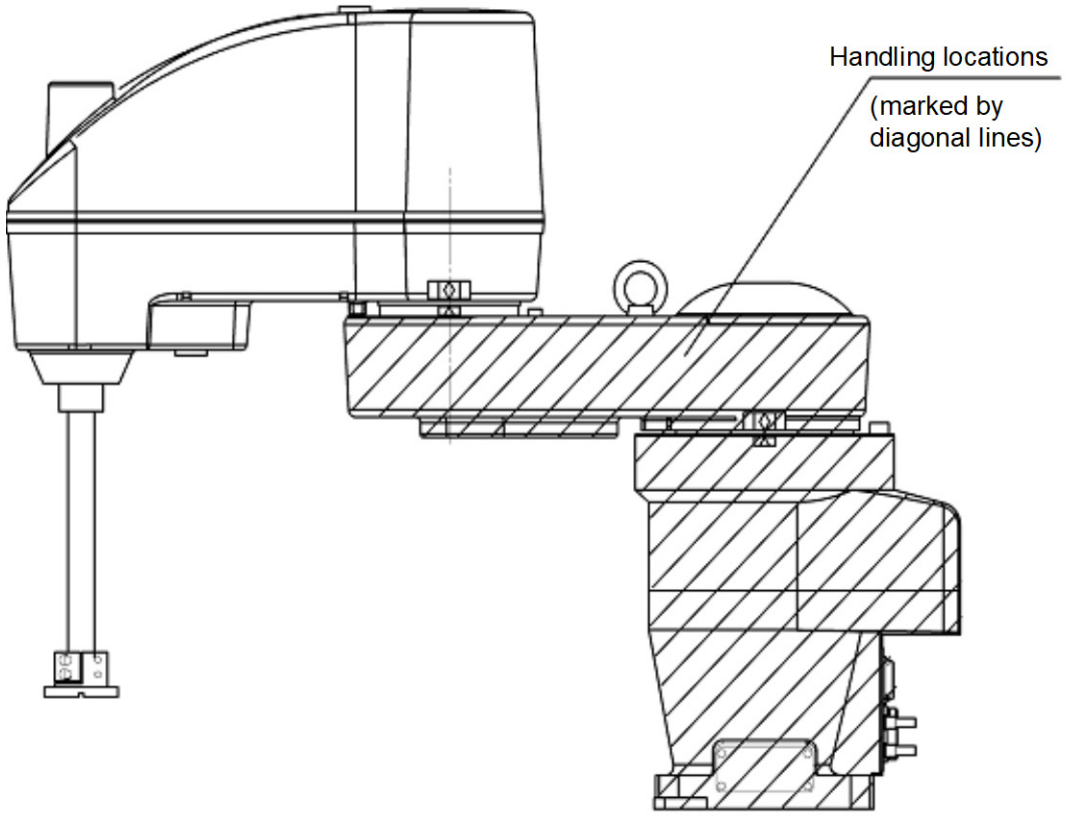


Fig. 1.9 Robot handling areas

After installing the robot, remove the clamps and eyebolt.

1.2.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

DANGER

- When placing the controller on the floor, etc., make sure not to catch your hands or feet.

1.3 Storage

We do not recommend storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

1.3.1 Storage Precautions for the Robot

CAUTION

- Secure the base completely to prevent the robot from falling over. When placed directly, the robot is unstable and will fall over.
- Keep the robot out of direct sunlight. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the base bottom and tool shaft are susceptible to rust development, coat the tool shaft with grease and the other surfaces with rust-preventive agent.
- Grease the robot before use.
- When the robot power cable was disconnected from the controller, connect the short-circuit connector to the same cable.

1.3.2 Storage Precautions for the Controller

CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.

Section 2

Installation

2.1 Installation Environment

Table 2.1 shows the environmental conditions for the location in which the TH series Robot and TS2000/TS2100 controller are to be installed.

Table 2.1 Environmental conditions for the robot and controller

Item	Specifications
Temperature	At operation : 0 ~ 40°C At storage : –10°C ~ 50°C
Humidity	20 to 90 % (Non-condensing) DO NOT install the robot where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation : 0.98 m/s^2 or less In transport : 9.8 m/ s^2 or less
Dust	No inductive dust should exists. Consult with SHIBAURA MACHINE first should you wish to use the robot and controller in an especially dusty environment.
Gas	No corrosive or combustible gas should exist.
Sunlight	The robot must not be subject to direct sunlight.
Electromagnetic noise	A heavy noise source should not exist nearby.
Magnetic field	A heavy magnetic field source should not exist nearby.

DANGER

- Do not place the robot or controller near combustible. Doing so could lead to fires if it ignites due to a fault, etc.

⚠ CAUTION

- In the case where batteries for detecting the motor position are of alkaline type, the batteries can overheat, leak battery fluid, or rupture when used under high temperatures. Also, high temperatures can reduce the performance and lifespan of the battery. If using the robot under high temperatures, please consult with the SHIBAURA MACHINE sales office.

2.2 Robot Installation

Before installing the robot, you should plan a layout, fully considering the working envelope, coordinate system and space for maintenance.

2.2.1 External Dimensions

The external views of the robot are shown in Figs. 2.1 to 2.5.

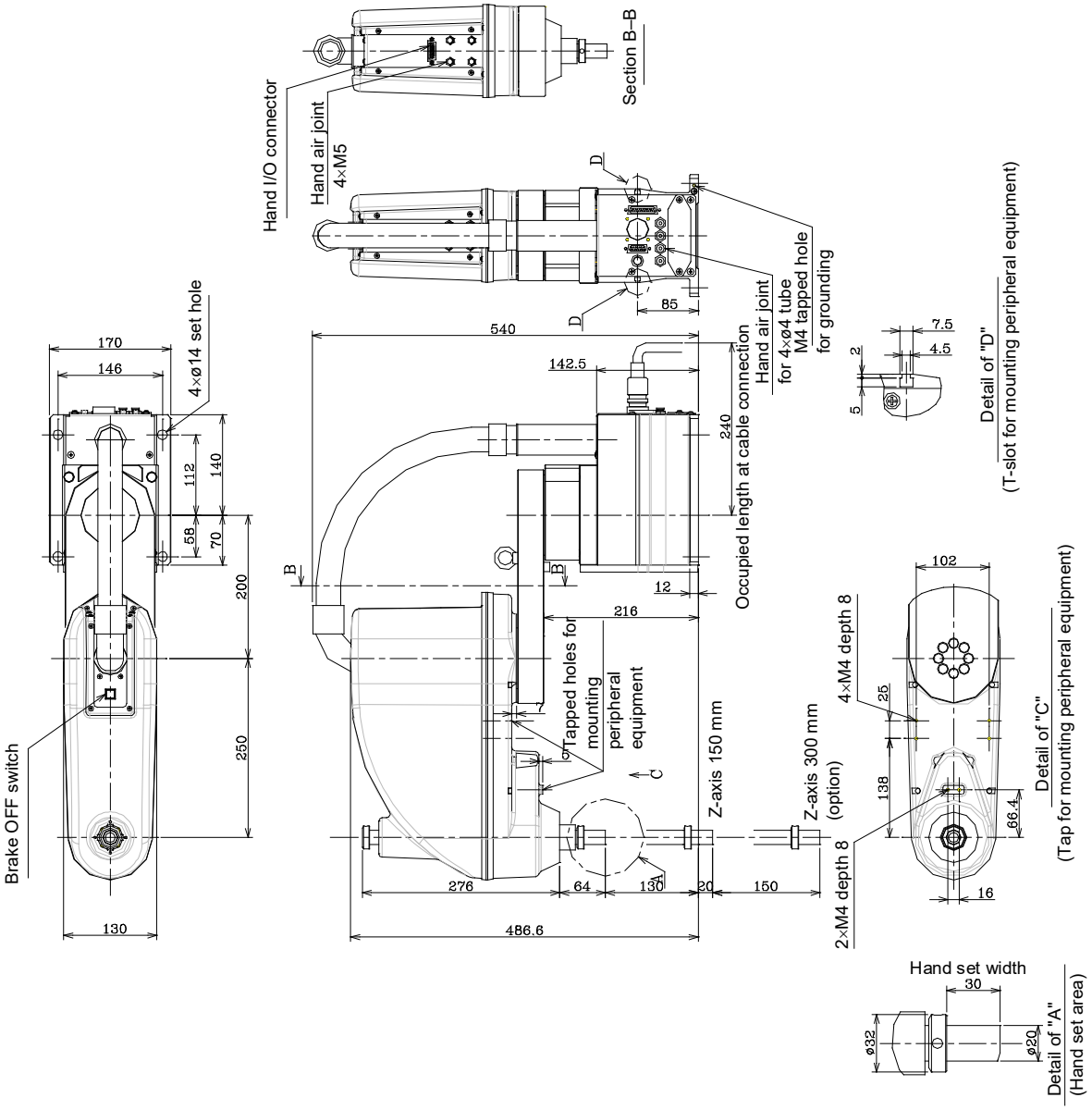


Fig. 2.1 External view of TH450 robot

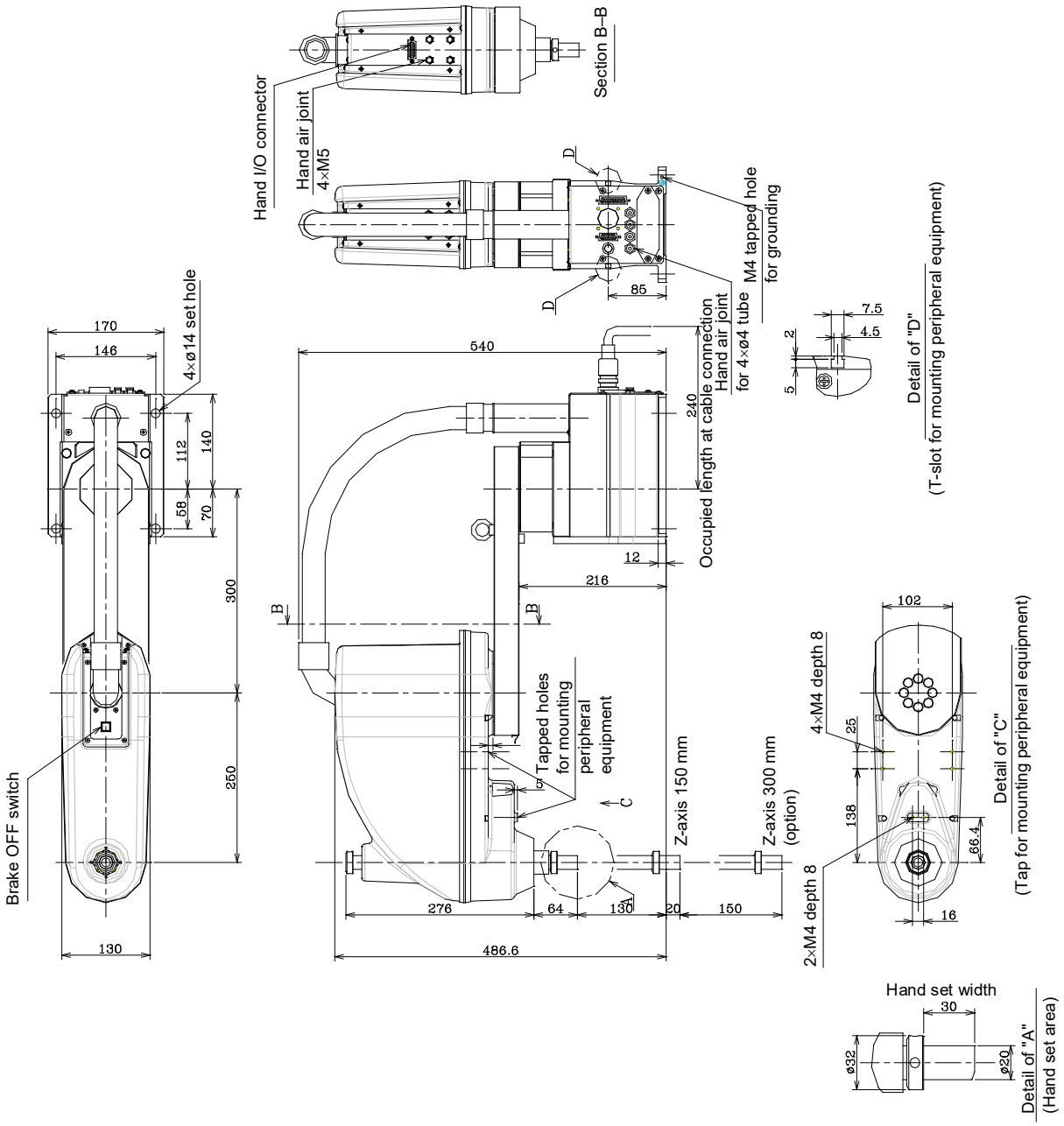


Fig. 2.2 External view of TH550 robot

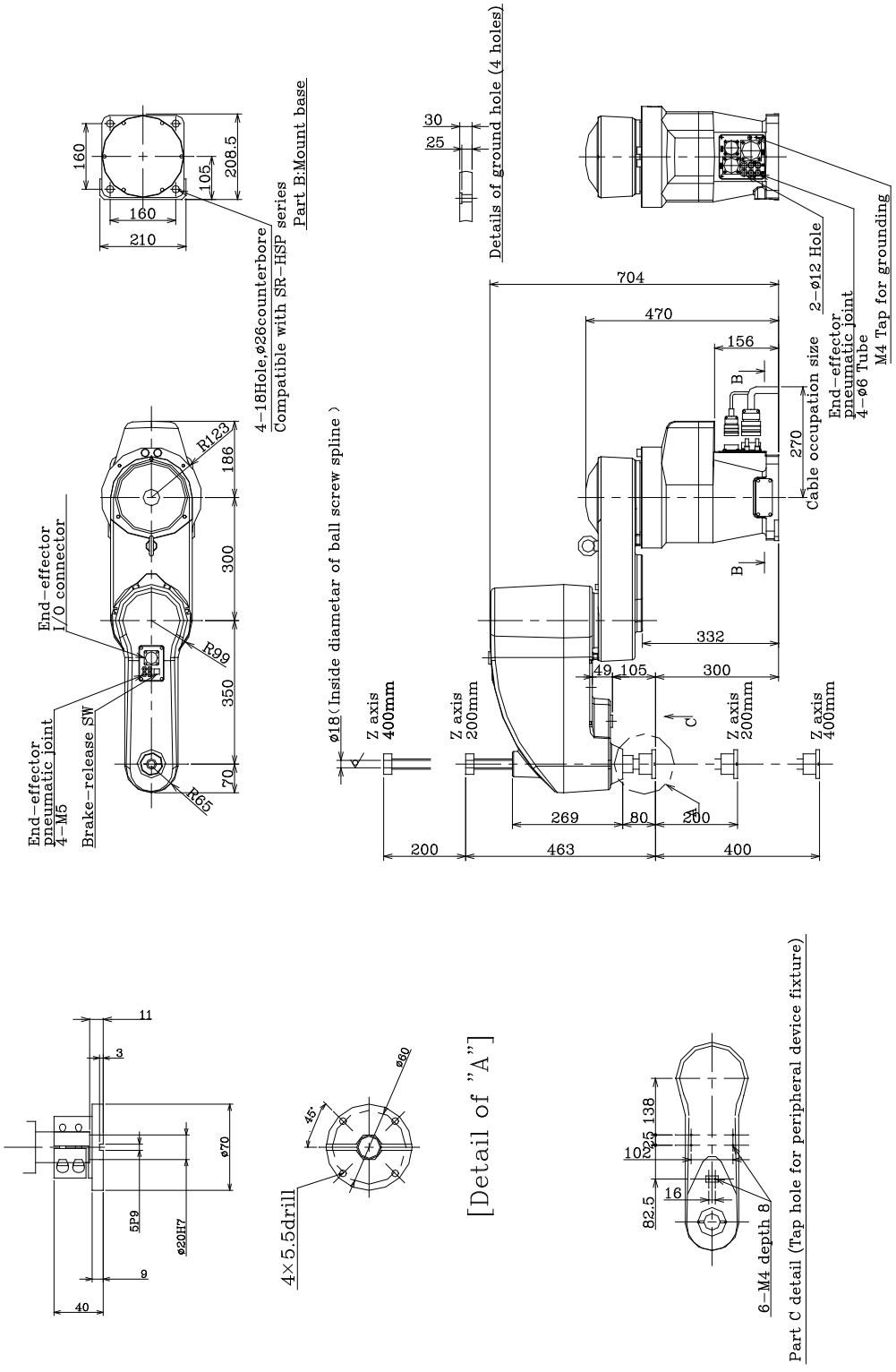


Fig. 2.3 External view of TH650 robot

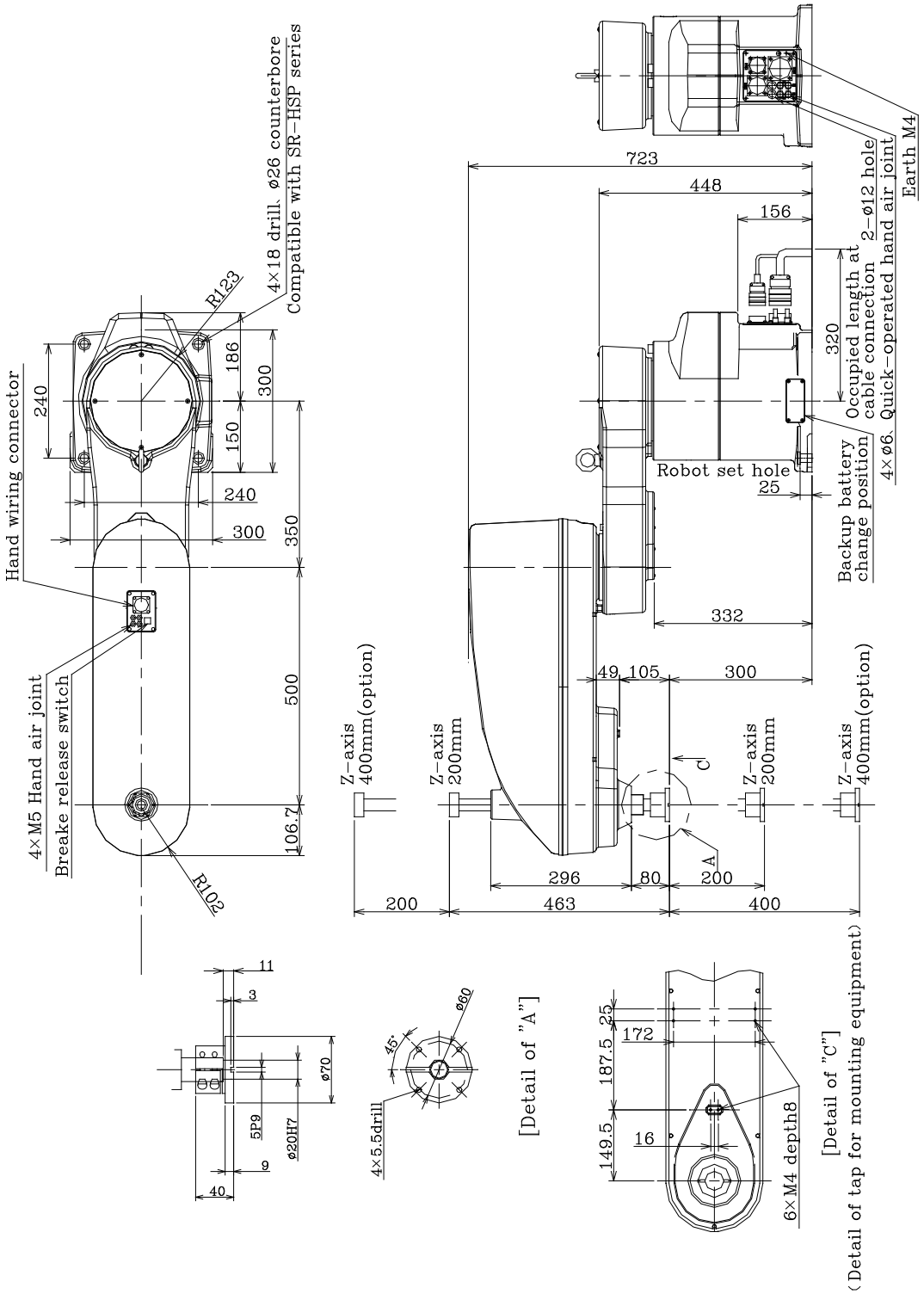


Fig. 2.4 External view of TH850 robot

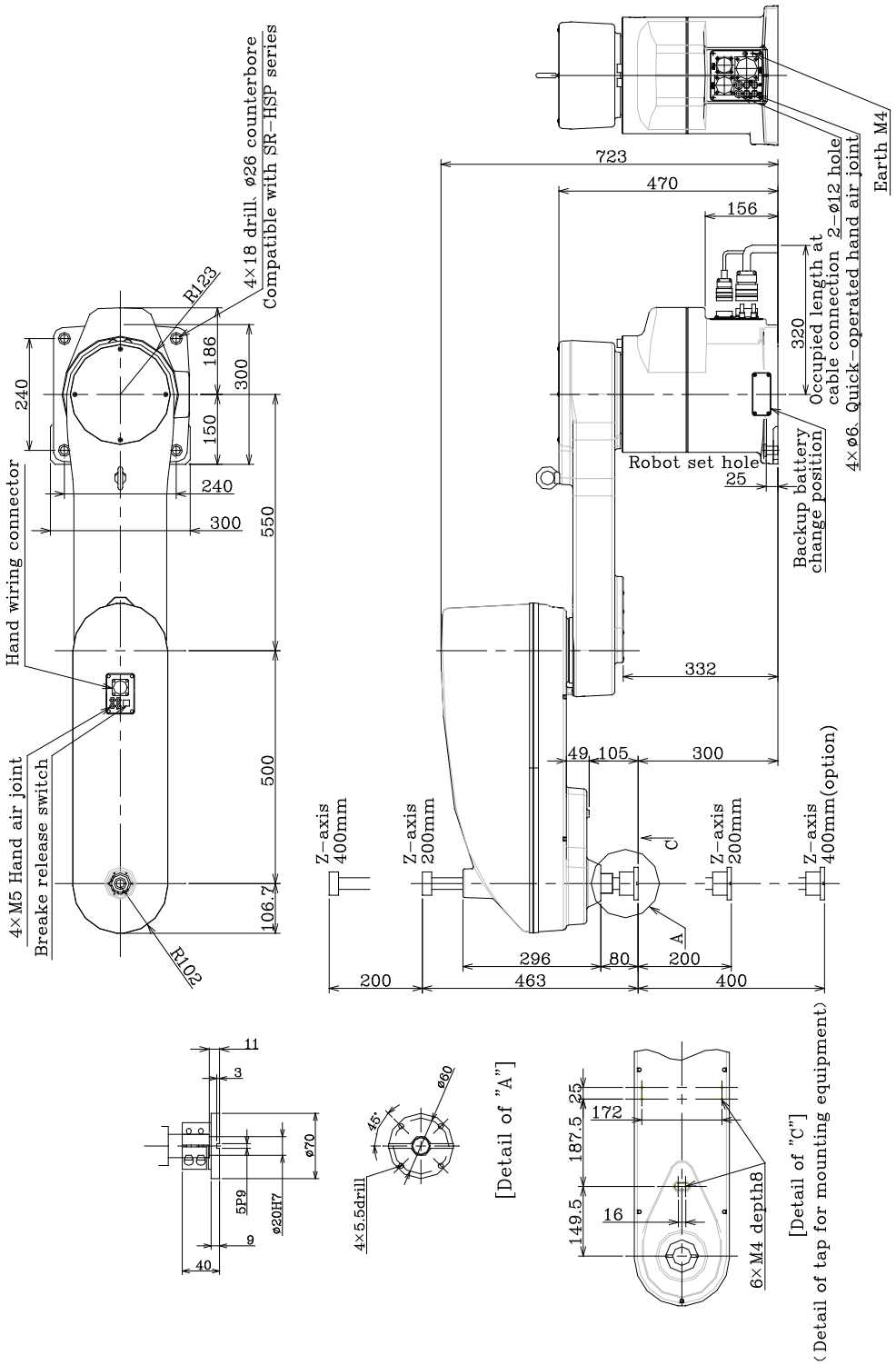
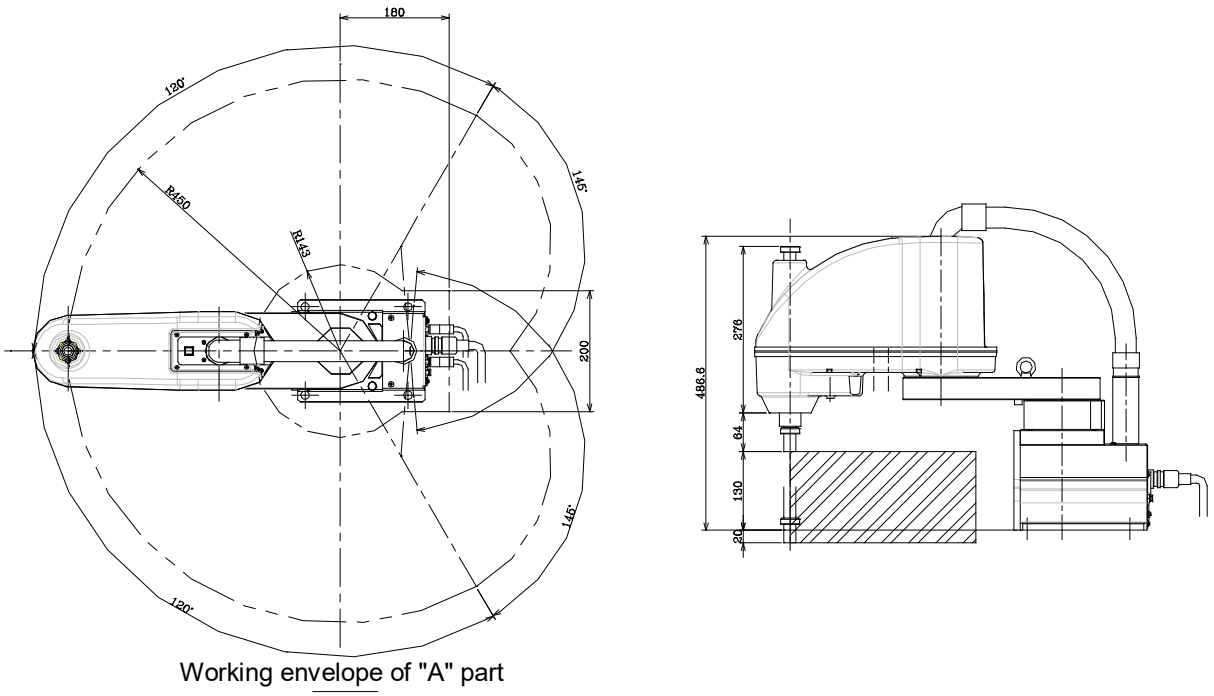


Fig. 2.5 External view of TH1050 robot

2.2.2 Working Envelope

Figs. 2.6 through 2.10 show the working envelope of the robot. Each axis can operate within the working envelope. To keep the robot from moving out of the working envelope by mis-operation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits are provided which may be set as desired by the user.



Working envelope of "A" part

Fig. 2.6 TH450 robot working envelope

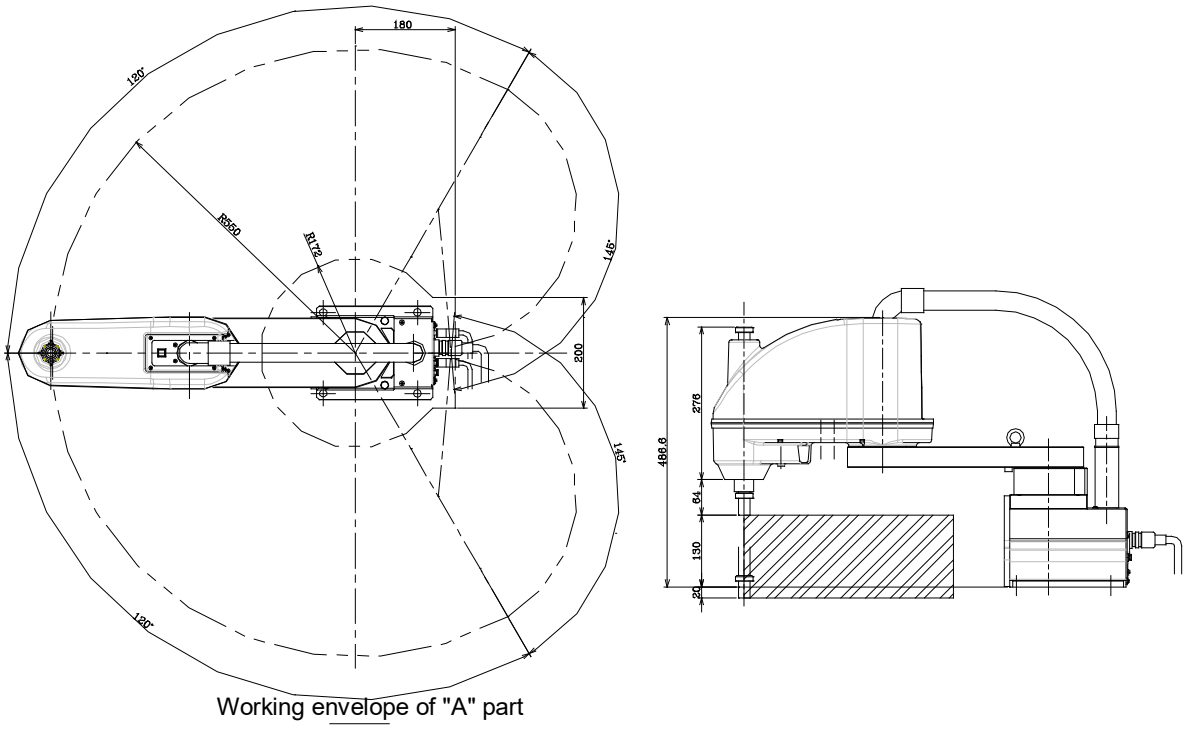


Fig. 2.7 TH550 robot working envelope

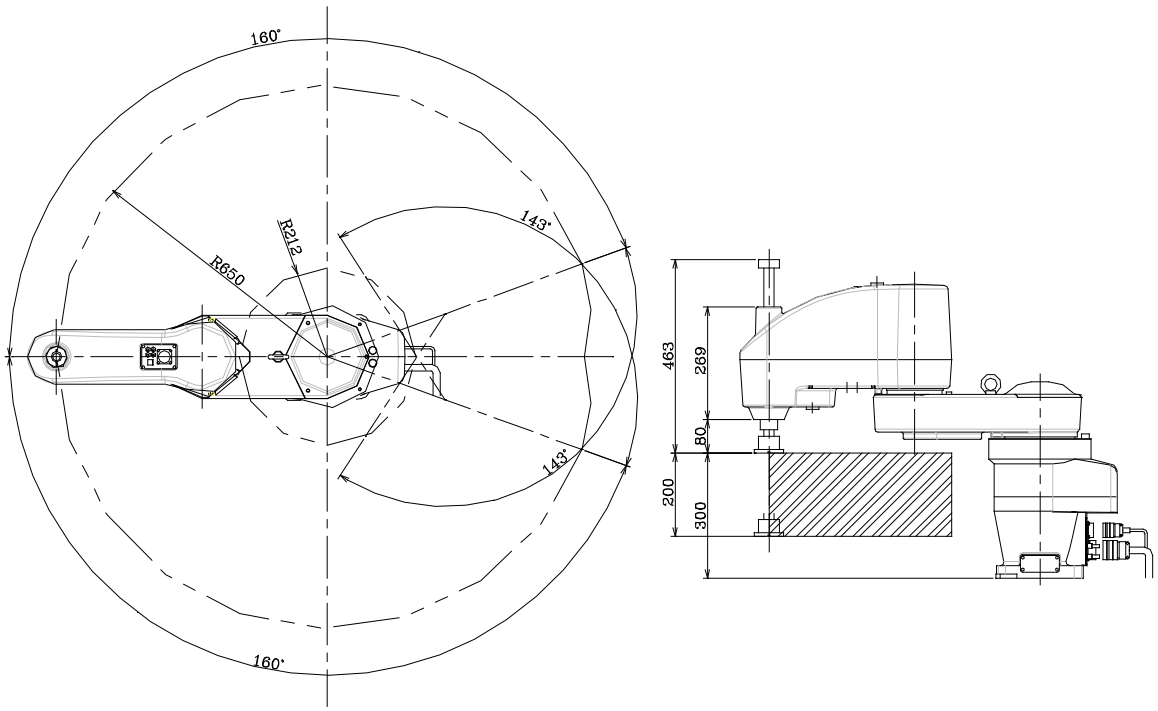


Fig. 2.8 TH650 robot working envelope

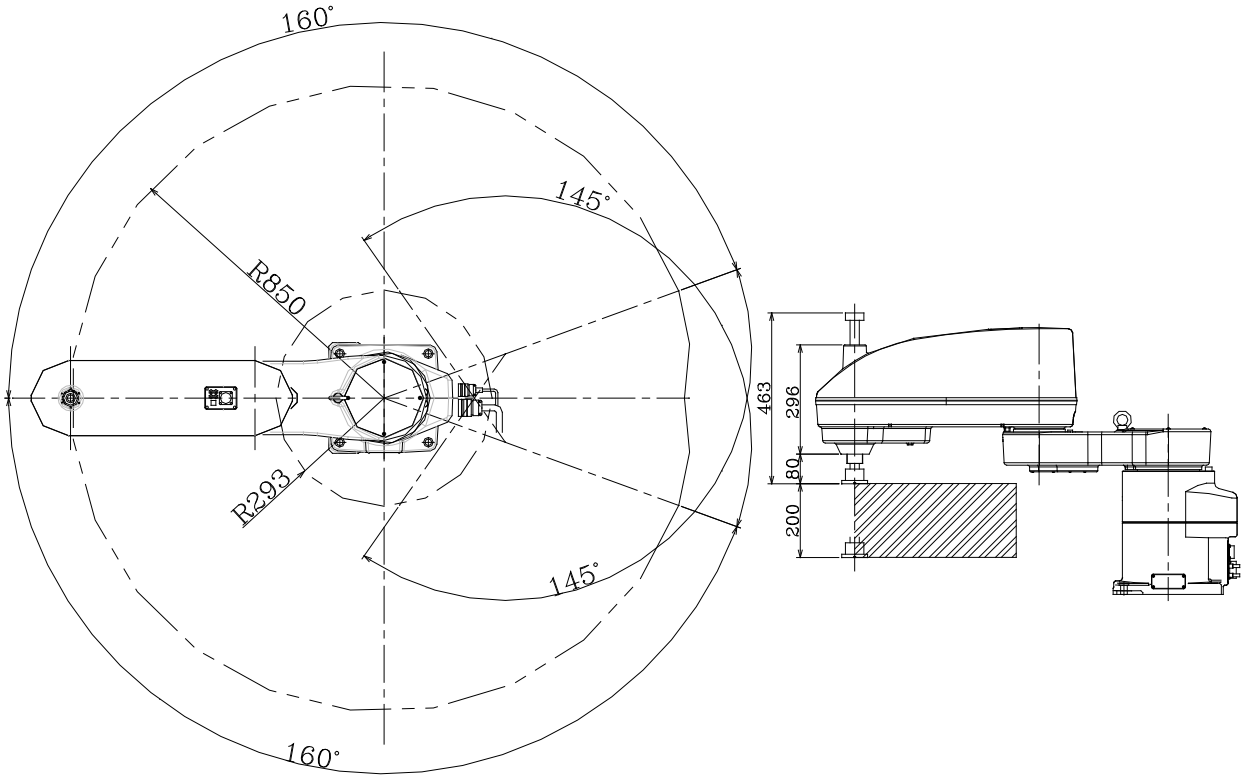


Fig. 2.9 TH850 robot working envelope

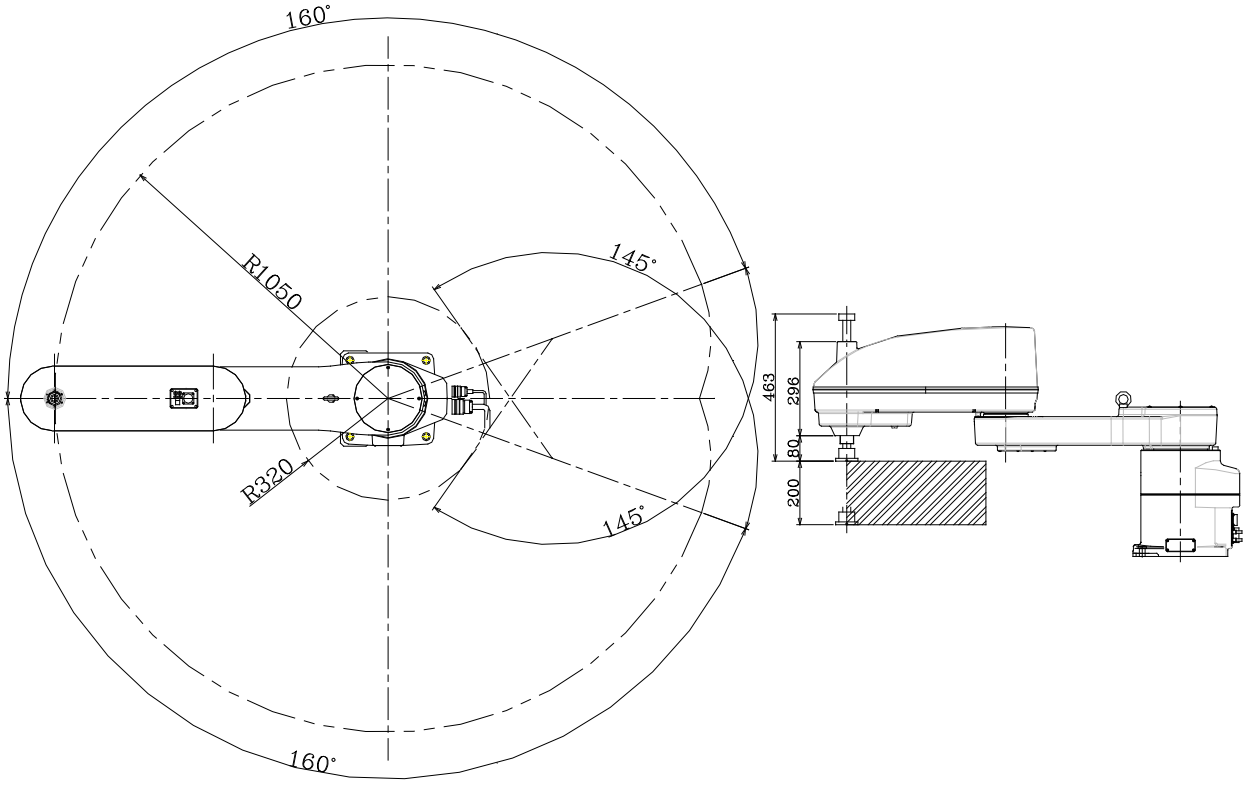


Fig. 2.10 TH1050 robot working envelope

2.2.3 Coordinate System

The robot's joint angle zero point (0° or 0 mm position) is factory-calibrated according to the base reference planes. The base coordinate system is determined according to this calibration. Figs. 2.11 to 2.15 show the base coordinate system and zero position of each axis joint angle.

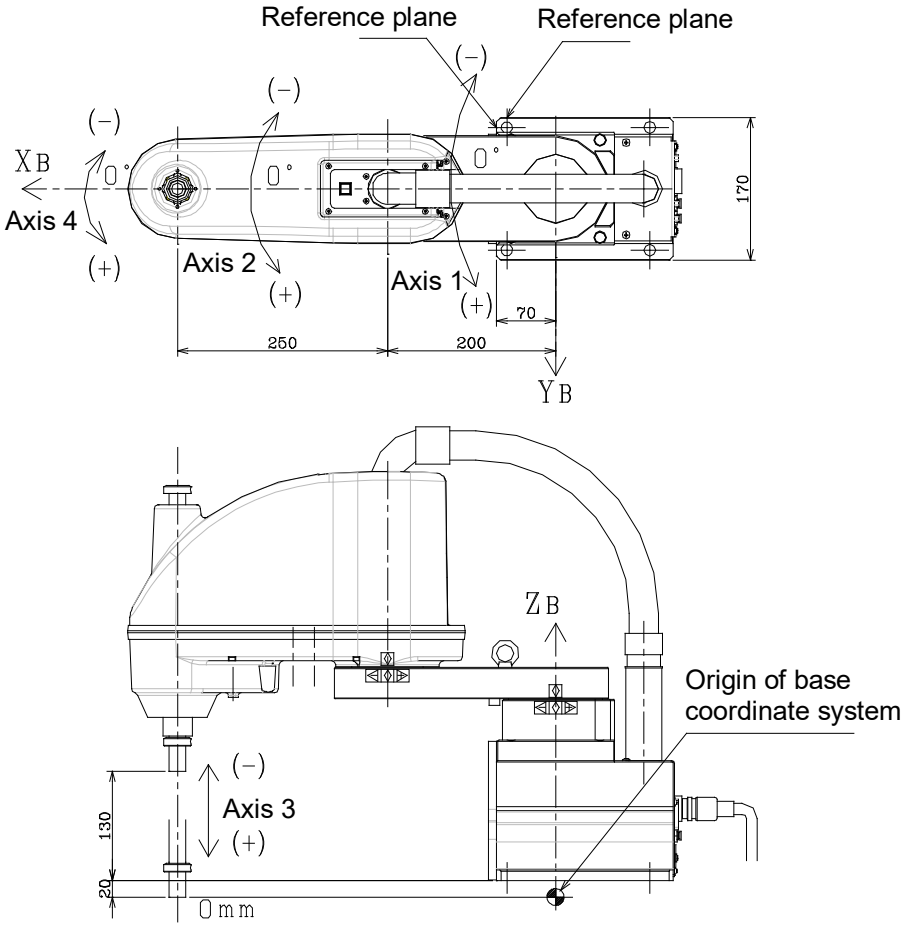


Fig. 2.11 TH450 base coordinate system and joint angle zero point

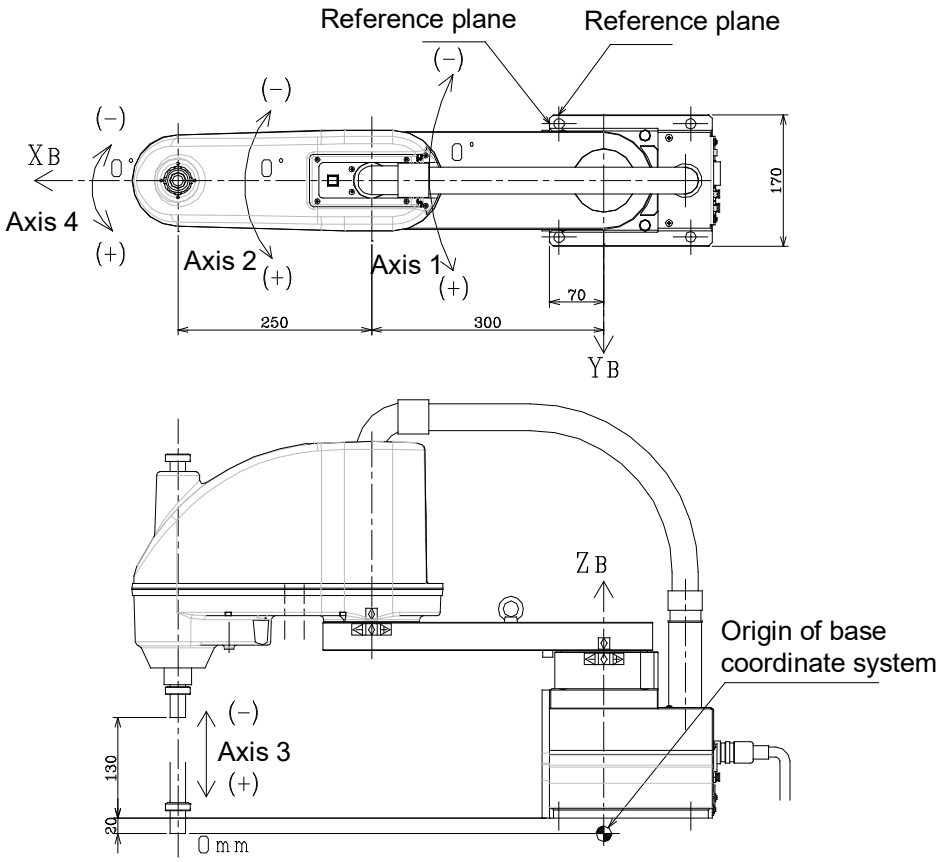


Fig. 2.12 TH550 base coordinate system and joint angle zero point

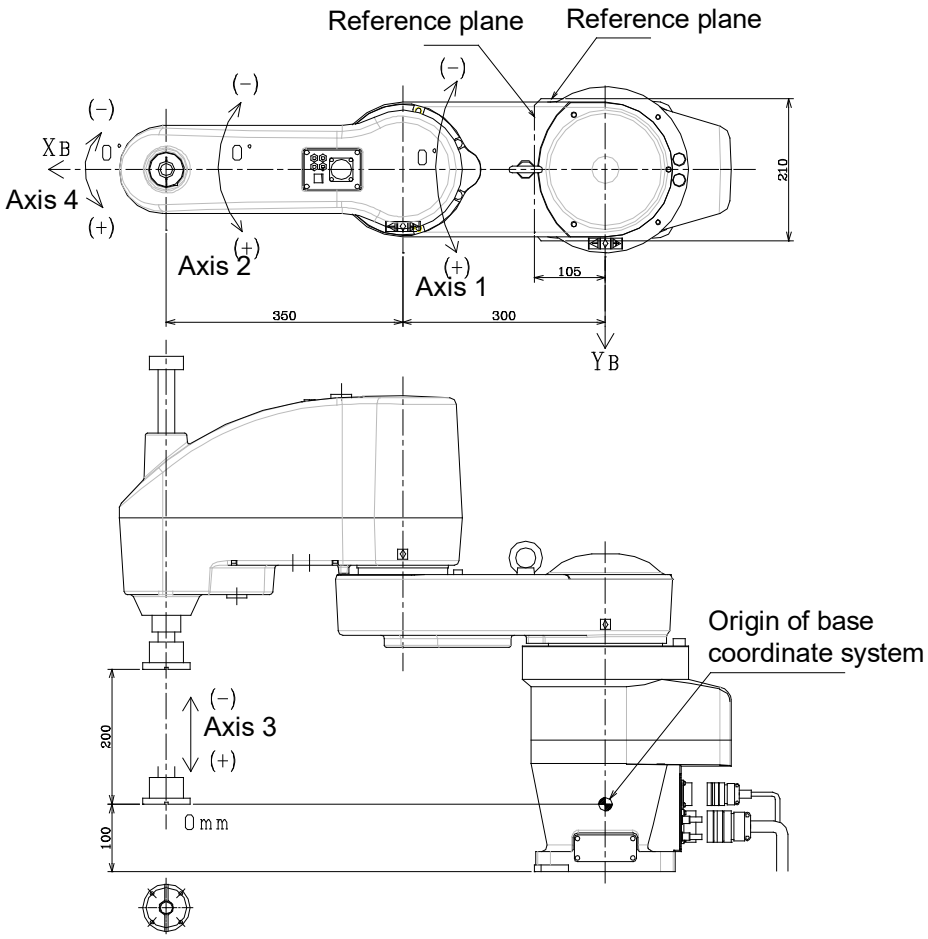


Fig. 2.13 TH650 base coordinate system and joint angle zero point

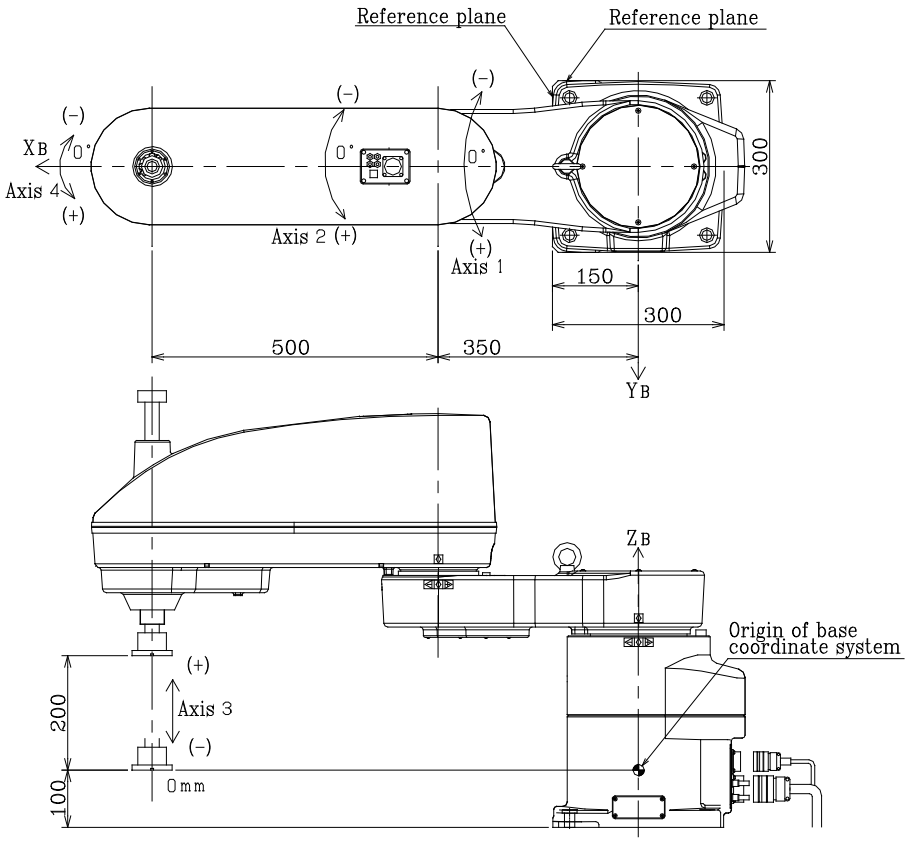


Fig. 2.14 TH850 base coordinate system and joint angle zero point

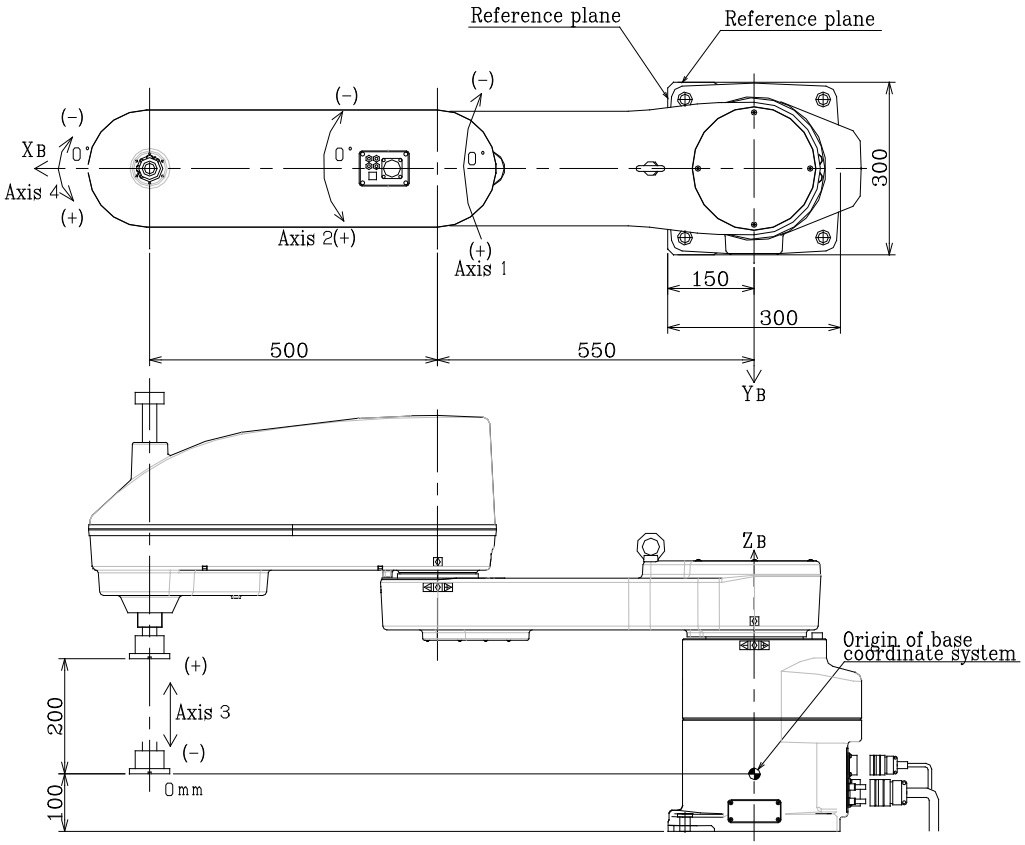


Fig. 2.15 TH1050 base coordinate system and joint angle zero point

2.2.4 Installing the Robot

The robot is secured, using the four (4) set holes on the base. Use the M12 hexagon socket head cap screws for the TH450 and TH550, and the M16 hexagon socket head cap screws for the TH650 and TH850 and TH1050.
The robot installation method is shown in Figs. 2.16 to 2.18. The reference planes are provided on the base section.
To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then contact them to the base reference planes and secure the robot.
The reference planes are factory-coated with anticorrosive agent. Before use, wipe out the agent with thinner. (Excluding the TH450 and TH550.)

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.

  If the robot is installed on a frame that does not have sufficient rigidity, the vibration will occur while the robot is operating, and could lead to faults.

  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

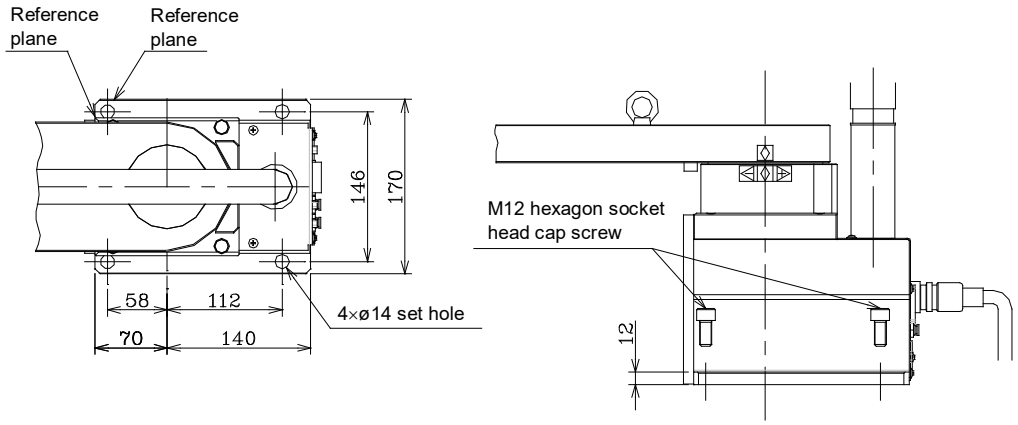


Fig. 2.16 Installing the TH450 and TH550 robots

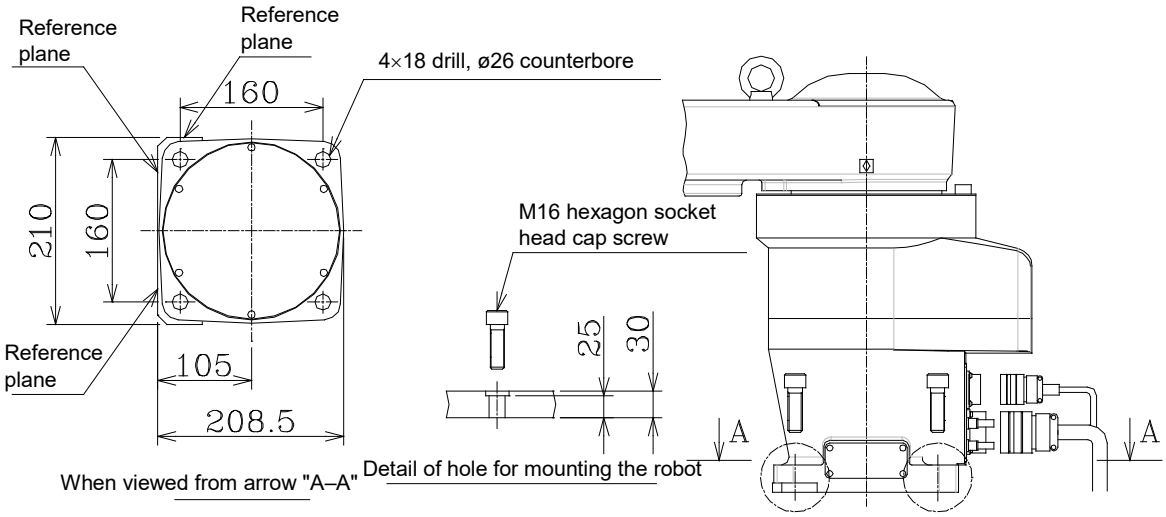


Fig. 2.17 Installing the TH650 robot

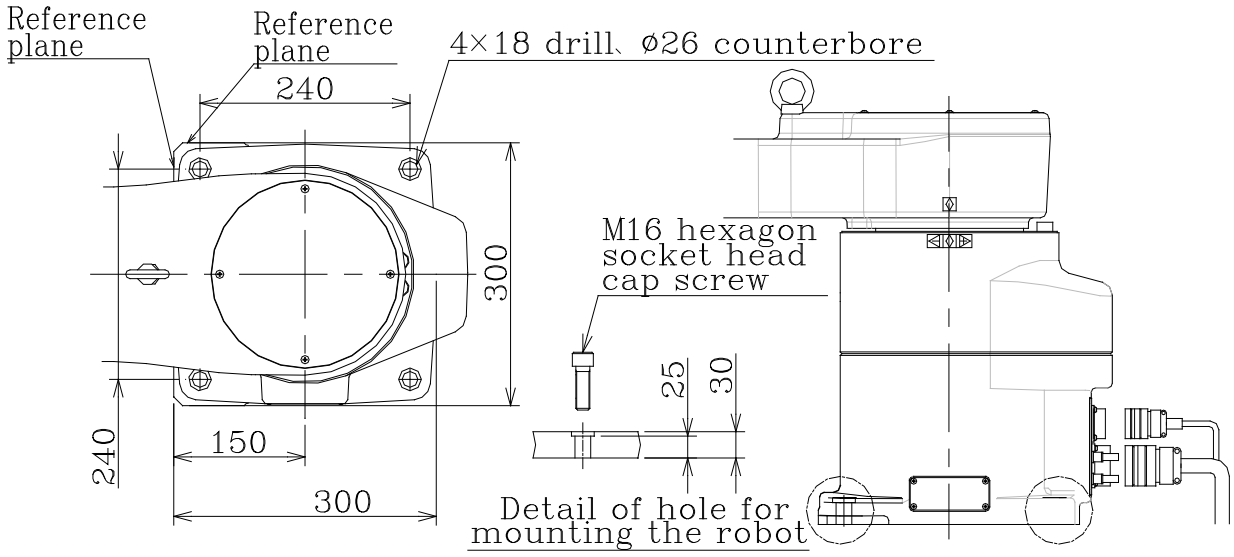
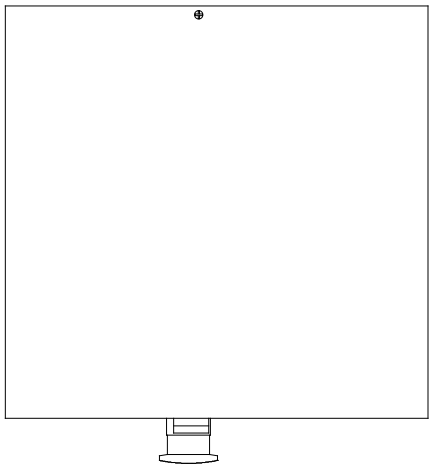


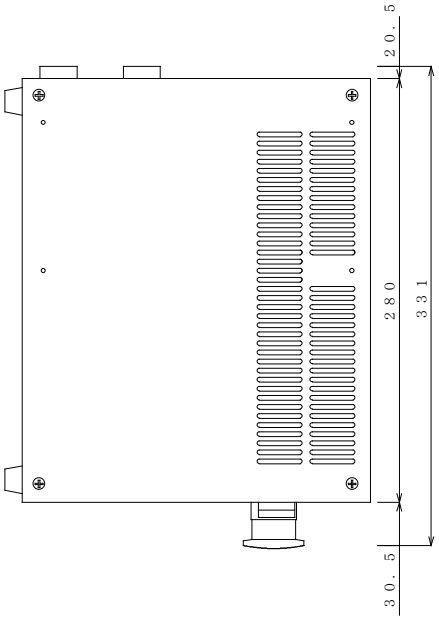
Fig. 2.18 Installing the TH850 and TH1050 robots

2.3 Installing the Controller

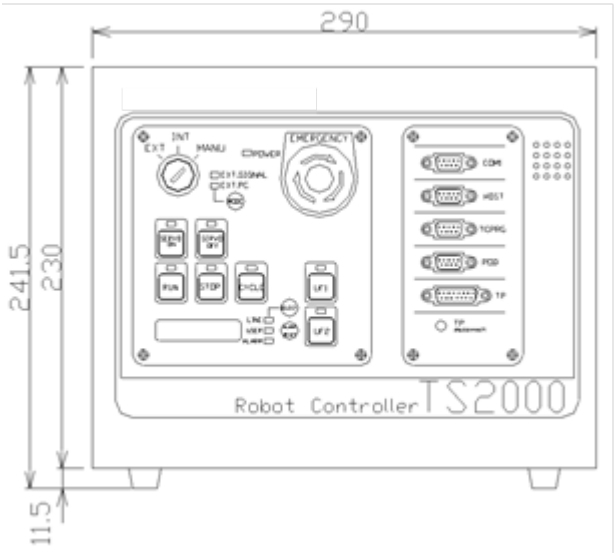
2.3.1 External Dimensions

External views of the TS2000 controller and TS2100 controller are shown in Fig. 2.19 and Fig. 2.20, respectively.

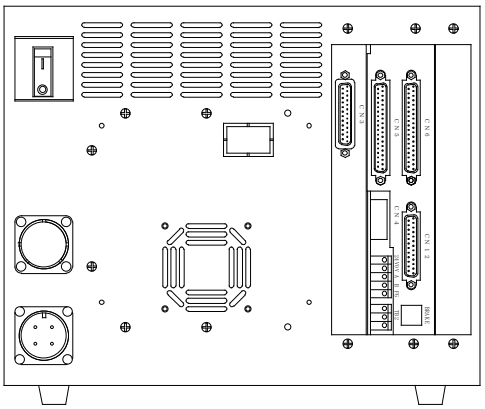




Left side view

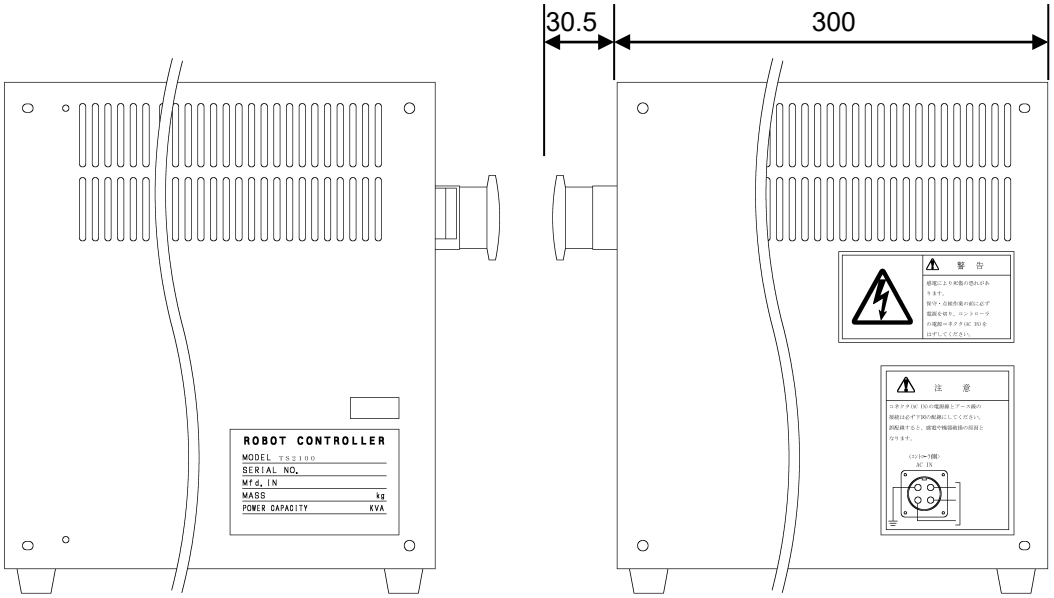


Front view

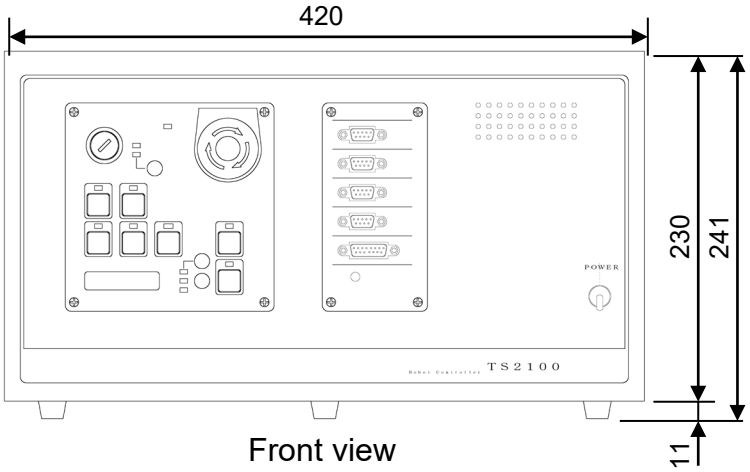


Rear view

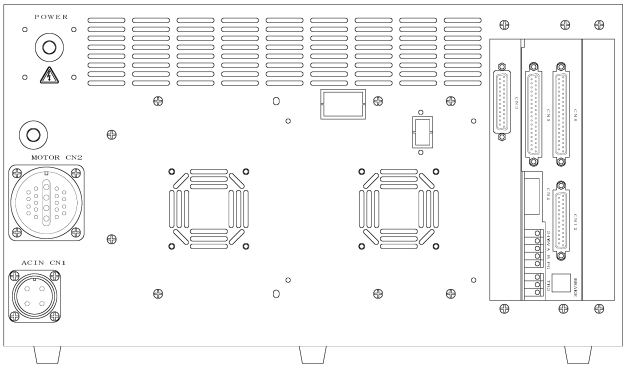
Fig. 2.19 External view of TS2000 controller



Left side view　　Right side view



Front view



Rear view

Fig. 2.20　External view of TS2100 controller

2.3.2 Rack Mounting Dimensions

When mounting the TS2000/TS2100 robot controller in the rack, set the side brackets using the screw holes provided on both ends of the front panel, and secure the controller to the rack. The side brackets [1] in Fig. 2.21 are optional.

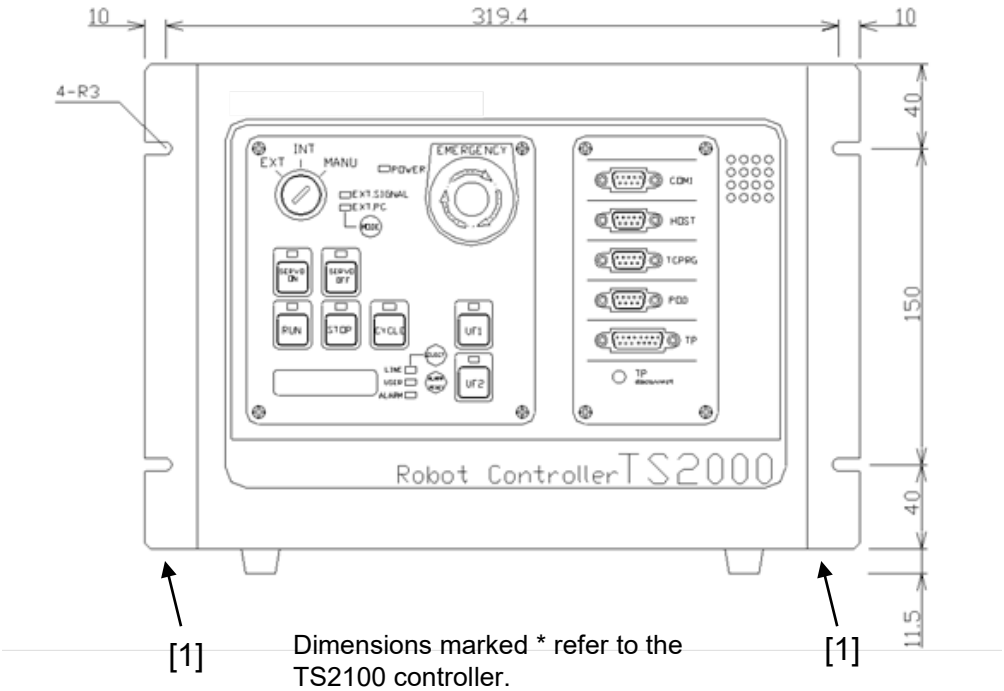


Fig. 2.21 Dimensions of controller rack mounting holes

2.3.3 Precautions for Rack Mounting

Pay attention to the following matters when rack-mounting the TS2000/TS2100 robot controller.

(a) When mounting the controller in a rack, use the screw holes provided at both ends of the front panel, and secure the controller. (Optional side brackets are required.)

! CAUTION

- If the rack is completely sealed, provide holes to allow the air to be let out, force-ventilate the rack with a fan, or cool it indirectly so that the heat will not be trapped in the rack.

  If the heat is trapped in the rack or controller, faults could occur.

(b) As the cable connectors are connected to the rear side of the controller, provide a space of 110 mm on the rear side.
For maintenance, the upper cover should be removed. (See Fig. 2.22.)

Keep this in mind when installing the controller. Specifically, the controller should be easily removable from the rack. Practically, be careful of the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).
2) Arrange the cables between the controller and control panel when the control panel is separated.
3) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

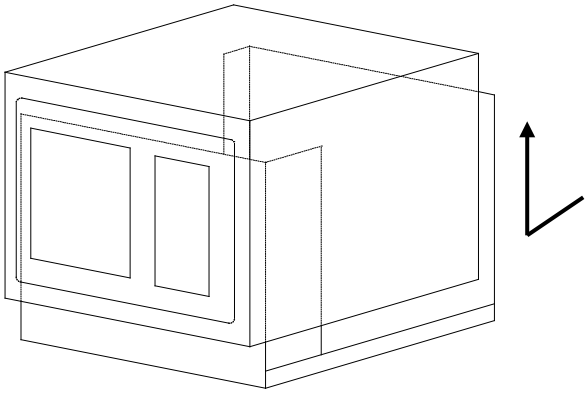


Fig. 2.22 Removing upper cover

(c) When the controller is mounted on the rack, the weight of the controller should be supported with the legs of the controller. Screw holes for rack-mounting the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with the screws.

(d) On the front side of the controller, a clearance of approx. 90 mm should be provided for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of approx. 50 mm is required for connecting a dummy plug.

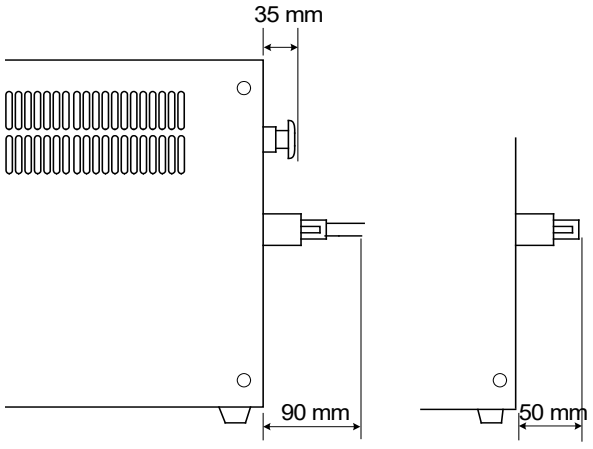


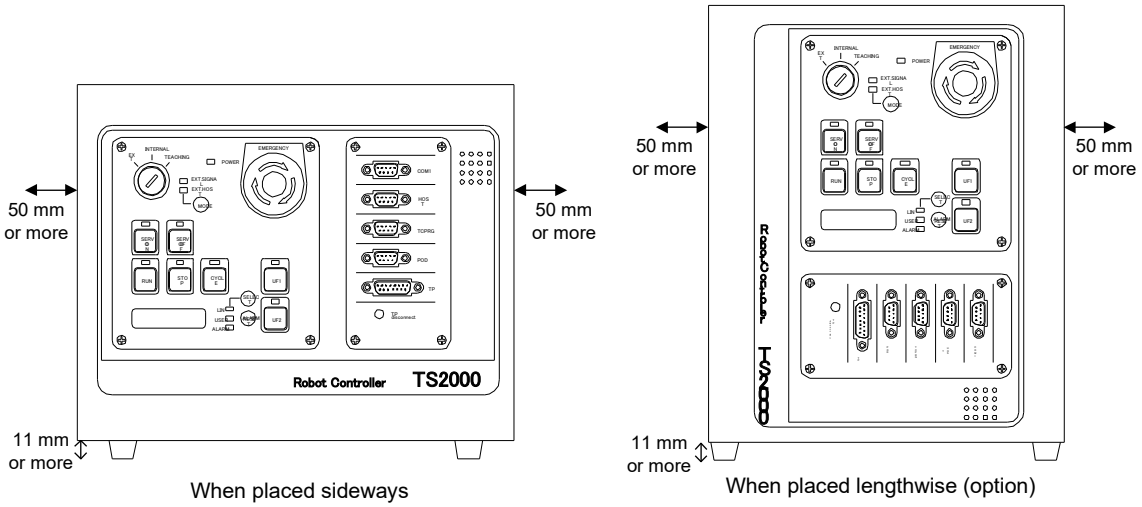
Fig. 2.23 Clearance in front of controller

2.3.4 Precautions for Direct Installation

When two (2) or more controllers are used, it is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the vertical direction between the controllers.
When the controller is set lengthwise, make sure that the right side of the controller when placed sideways faces down. (Lengthwise installation of the controller is an option.)

CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the legs should be kept below the lower surface.
  Failure to do so could cause the cooling performance to drop or to faults.
- DO NOT stack the controllers. DO NOT place any object on top of them.



* When using the TS2100 controller, keep the same spaces also.

Fig. 2.24 Controller ventilation space

2.4 Precautions for Handling the Teach Pendant

Be careful of the following matters when handling the teach pendant.

! CAUTION

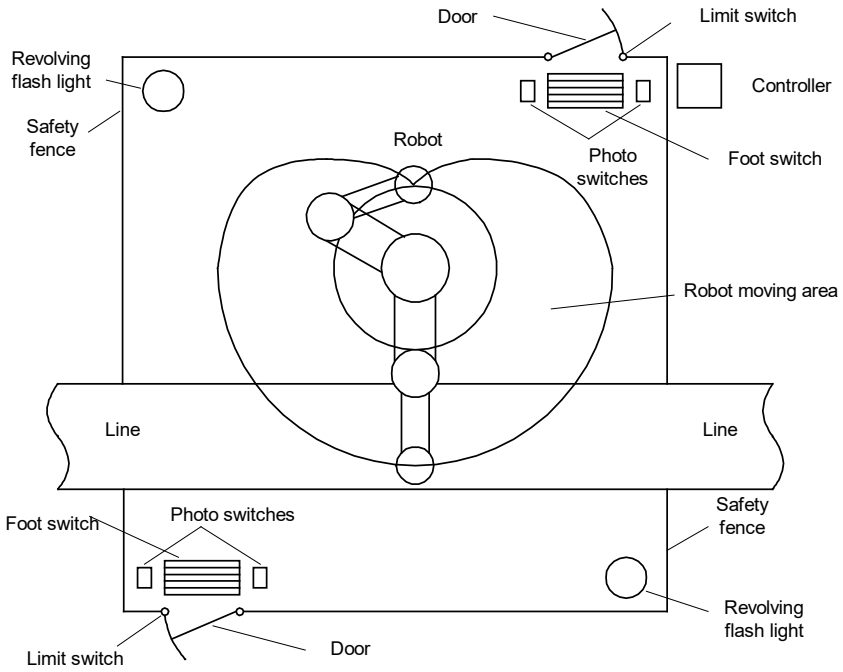
- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable running from the teach pendant.
- DO NOT press the switches on the teach pendant with anything sharp (like the tip of a knife, pencil, ball-point pen, etc.).
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

2.5 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence so that other persons cannot enter the zone easily. The hazard zone is the zone near the robot's working space where a hazardous state could occur if a person enters.

c) Provide a limit switch, photo switch or foot switch, etc., at the entrance of the safety fence to provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent close contact (closed in normal operation) with compulsive opening function and must not be automatically recovered.



d) The controller should be installed at a place outside the hazard zone where the operator can view the robot movement.

2.6 Position Detector Backup Batteries

The robot has batteries to back up positional information on position detectors in its base. Unless the robot is used for a long period of time, the battery voltage will drop and the home point data will be lost.
Also, if the battery voltage for backup of X8HC printed board SRAM of the controller has dropped, absolute position data set by the user will be lost. To avoid this, replace each battery on a regular basis.

If the master power voltage of battery has dropped, an alarm of level 1 as shown below occurs. When this happens, replace the battery immediately.

Error code	Descriptions
1–049	Axis1 Battery alarm
1–065	Axis2 Battery alarm
1–081	Axis3 Battery alarm
1–097	Axis4 Battery alarm
1–113	Axis5 Battery alarm
1–129	Axis6 Battery alarm
1–145	MAIN Battery alarm

If the master power voltage has dropped further, an alarm of level 8 as shown below occurs. When this happens, replace the battery with a new one and adjust the home point of the robot, if shifted, because there is a fear that the position encoder backup data has been lost. For the adjustment, see the Maintenance Manual.

Error code	Descriptions
8–065	Axis1 Encoder abnormal
8–097	Axis2 Encoder abnormal
8–129	Axis3 Encoder abnormal
8–161	Axis4 Encoder abnormal
8–193	Axis5 Encoder abnormal
8–225	Axis6 Encoder abnormal

Section 3

System Connections

3.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be connected.

3.1.1 Connector Arrangement on the Controller

The cables connected to the TS2000 and TS2100 robot controllers are shown in Figs. 3.1A and 3.1B.

[1] Power cable
[2] Motor cable
[3] Encoder cable
[4] Robot control signal cable (HAND I/O)
[5] External I/O signal cables
[6] Brake signal cable
[7] Distribution I/O signal cable

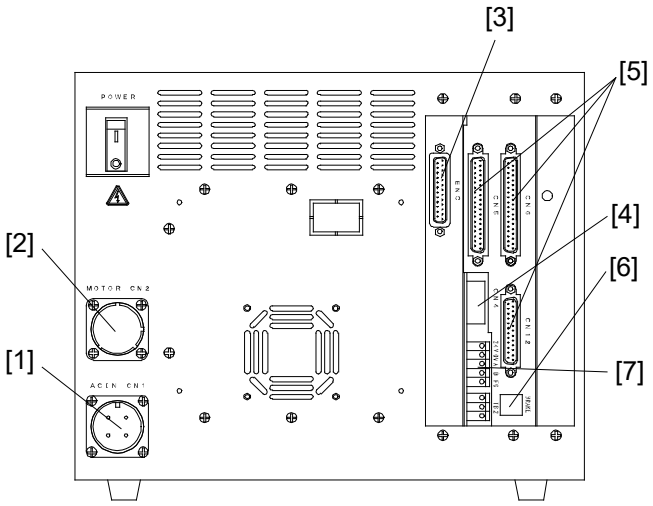


Fig. 3.1A TS2000 robot controller connector layout

[1] Power cable
[2] Motor cable
[3] Encoder cable
[4] Robot control signal cable (HAND I/O)
[5] External I/O signal cables
[6] Brake signal cable
[7] Distribution I/O signal cable

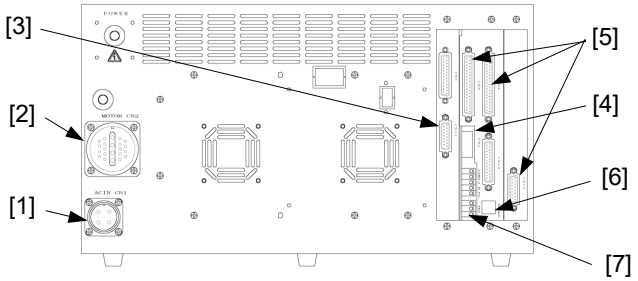


Fig. 3.1B TS2100 robot controller connector layout

In this section, we explain how to connect cables [1] to [4]. For information on how to connect cables [5] to [7], refer to the Interface Manual.

3.1.2 Connecting the Power Cable "CN1"

([1] of Fig. 3.1; plug connector attached)

The power cable is used to supply the main AC power to the controller.

Table 3.1 Power supply specifications

Item	TH450, TH550	TH650,TH850,TH1050
Controller	TS2000	TS2100
Power supply	Single phase, AC 190 ~ 250 V, 50/60 Hz±1 Hz	Three (3)-phase, AC 190 ~ 250 V, 50/60 Hz±1 Hz
Power capacity	2.3 kVA	3.5 kVA
Instantaneous power failure	40 msec or less	
Grounding	Grounding with grounding resistance of 100 Ω or less	

The connector is CN1 ([1] of Fig. 3.1)
The TS2000 and TS2100 share the same specifications.

CN1 plug connector	Type: JL04V–6A18–10SE–EB	Maker:	Japan Aviation Electronics Industry
CN1 cable clamp	Type: JL04–18CK	Maker:	Japan Aviation Electronics Industry
Wire	3.5 mm^2 ~ 5.5 mm^2		

The cable is not included in the accessories. Fabricate the cable, using the attached plug connector which is to be connected to CN1 on the controller side. Wires are to be soldered to the connector.

DANGER

- Be sure to use the designated wires. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 3.1.7.

CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of 8–027 “Slow Charge error” occurs. When this happens, make sure that the master power voltage at the controller power connector satisfies the specified input power of the controller, and that the same master power voltage is stabilized.
- For details of the 8–027 error, see the descriptions on error codes contained in the Operator’s Manual.

3.1.3 Connecting the Motor Cable "CN2"

([2] of Fig. 3.1) (Cable attached)

The motor cable connects the controller and robot and supplies the power required to rotate the motor from the controller servo driver to each axis feed motor of the robot. The connector for connecting the motor cable is CN2 ([2] of Fig. 3.1). Location of the motor cable on the robot side is [1] in Fig. 3.2 or [1] in Fig. 3.3.

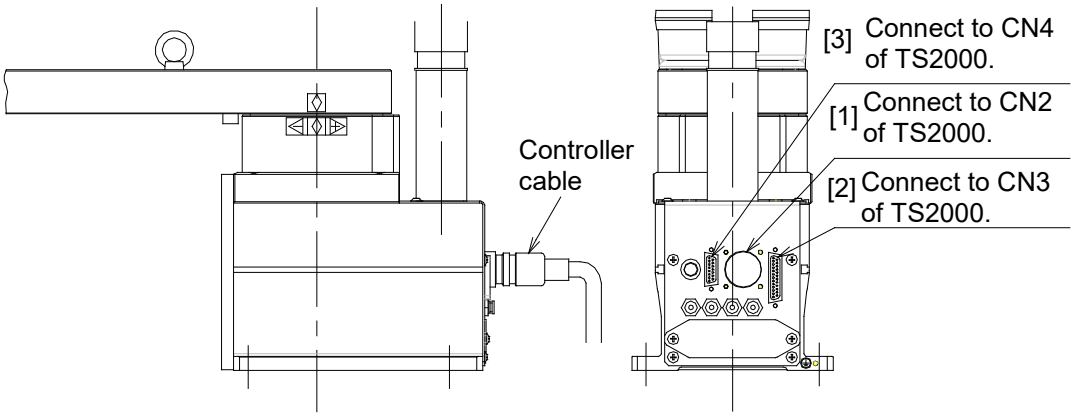


Fig. 3.2 Connector arrangement on the robot side (TH450, TH550)

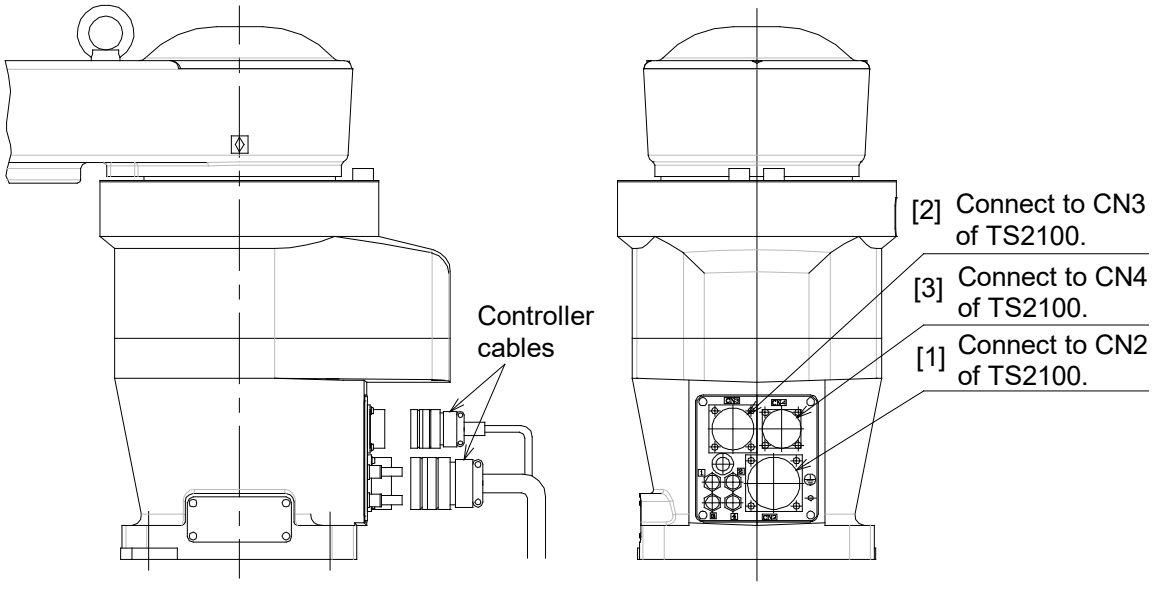


Fig. 3.3 Connector arrangement on the robot side (TH650,TH850,TH1050)

3.1.4 Connecting the Encoder Cable "CN3"

([3] of Fig. 3.1) (Cable attached)

The encoder cable is a signal cable used to transmit a signal from the rotation angle detecting encoder of each robot axis to the controller.
The connector for connecting the encoder cable is CN3 ([3] of Fig. 3.1). Location of the encoder cable on the robot side is [2] in Fig. 3.2 or [2] in Fig. 3.3.

3.1.5 Connecting the Robot Control Signal Cable "CN4"

([4] of Fig. 3.1) (Cable attached)

The robot control signal cable is used for the brake ON/OFF signals for locking the motor shaft, hand control signals and others.
The connector for connecting the robot control signal cable is CN4 ([4] of Fig. 3.1). Location of the robot control signal cable on the robot side is [3] in Fig. 3.2 or [3] in Fig. 3.3.

3.1.6 Inserting and Removing Connectors

! CAUTION

- Before inserting or removing any controller cable, be sure to turn off the "POWER" switch.
- When removing a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: CN1, CN2

First, completely insert the connector on the cable side (plug) into the controller connector (receptacle). Then, turn the cable side lock screw to the right to clamp the cable. A loose plug can cause an accident due to contact failure. To avoid this, make sure that they are tightened completely.

To remove the connectors, turn the lock screw to the left and pull out the cable side connector (plug).

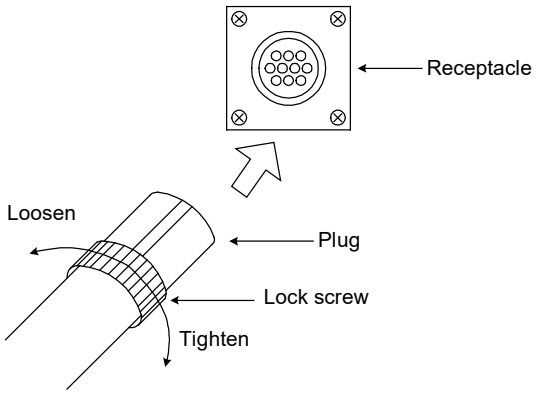


Fig. 3.4 Inserting and removing a circular connector

b) Square connectors: CN3, CN4, CN5, CN6, CN12

First, completely insert the cable side connector into the controller connector. Then, tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose plug can cause a contact failure or other accident. To avoid this, make sure that they are tightened completely.

To remove the connectors, first loosen the lock screws, then pull out the cable side connector. CN4 is the quick-operated lock type connector.

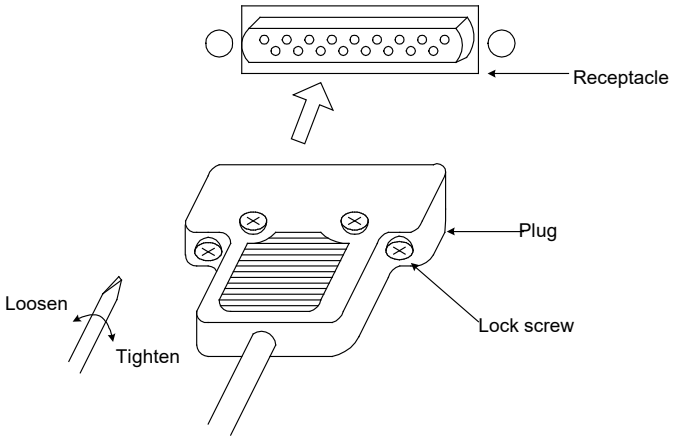
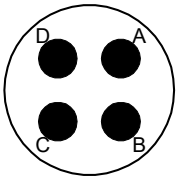


Fig. 3.5 Inserting and removing a square connector

3.1.7 Examples of Connector Terminal Arrangement

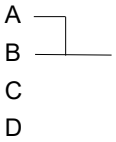
a) Power cable connector CN1 (for TS2000)



Connects to controller.

Type: JL04V–2E18–10PE–B

Manufacturer:
Japan Aviation
Electronics Industry



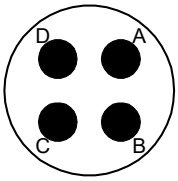
Single phase, AC 190~250 V, 50/60 Hz

Grounding with grounding resistance of 100 Ω or less.

! DANGER

- Completely connect the grounding cable. Otherwise, an electric shock or fire may be caused if a fault or electric leak has occurred. Or mis-operation may be caused by noise.

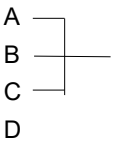
b) Power cable connector CN1 (for TS2100)



Connects to controller.

Type: JL04V–2E18–10PE–B

Manufacturer:
Japan Aviation
Electronics Industry



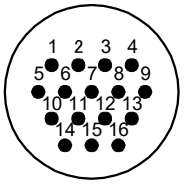
Three phases, AC 190~250 V, 50/60 Hz

Grounding with grounding resistance of 100 Ω or less.

! DANGER

- Completely connect the grounding cable. Otherwise, an electric shock or fire may be caused if a fault or electric leak has occurred. Or mis-operation may be caused by noise.

c) Motor cable connector CN2 (for TS2000)

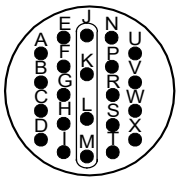


Connects to controller.

Type: JR25RK–16S

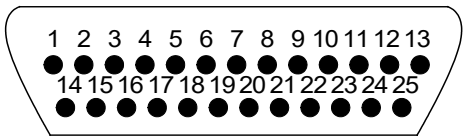
Manufacturer: Hirose Electric

d) Motor cable connector CN2 (for TS2100)



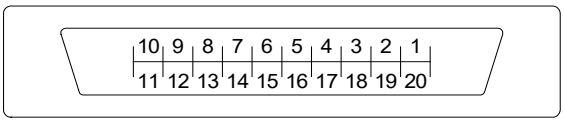
Connects to controller.
Type: JL04V–2A28–11SE
Manufacturer: Japan Aviation Electronics Industry

e) Encoder cable connector CN3



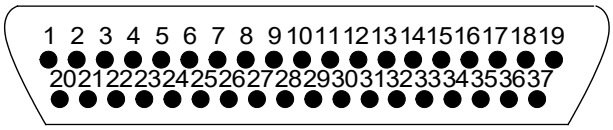
Connects to controller.
Type: XM2A–2501
Manufacturer: OMRON

f) Robot control signal cable connector CN4



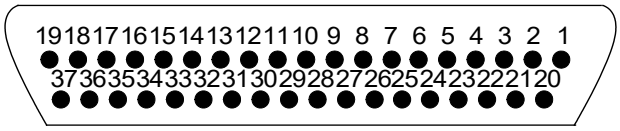
Connects to controller.
Type: PCR–E20LMD
Manufacturer: Honda Tsushin Kogyo

g) External input signal cable connector CN5



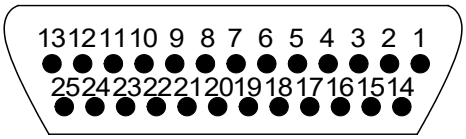
Connects to controller.
Type: XM2C–3712–112
Manufacturer: OMRON

h) External output signal cable connector CN6



Connects to controller.
Type: XM3B–3722–112
Manufacturer: OMRON

i) External I/O signal cable connector CN12



Connects to controller.
Type: XM3B–2522–112
Manufacturer: OMRON

3.2 Controller Connector Signals

3.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the Interface Manual.

3.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	CN5–16	(STOP)
	CN6–34	(SVOFF)
	CN5–17	(BREAK)
	CN6–16, 35	(EMS2B ~ EMS2C)
	CN6–17, 36	(EMS1B ~ EMS1C)

These signals are already jumpered for the connectors provided for the TS2000/TS2100 controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side CN5, CN6 connectors.
Also, if the following signals are not used as the system signals, be sure to jumper them.

CN5–36 (LOW_SPD)
CN5–35 (CYCLE)

Connector jumpers

CN5		CN6	
16–18	17–18	18–34	—
(35–37)	(36–19)	16–35	17–36

CAUTION

1. Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
2. Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
3. Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
4. Unless the STOP signal is jumpered, automatic operation of the robot is not possible.

3.3 Separating Control Panel from Controller

3.3.1 Removing Control Panel

Remove the control panel in the following manner.

a) Loosen the four (4) screws at the four (4) corners, which secure the control panel.

b) Remove these screws, then carefully draw out the control panel toward your side. (Be careful of the cable connected on the rear side.)

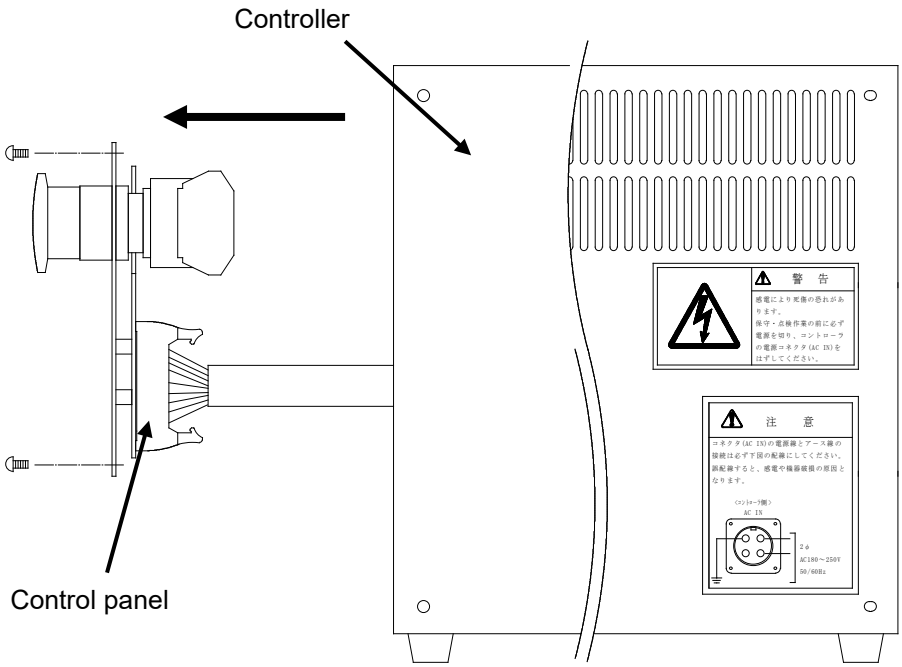


Fig. 3.6 Removing control panel

3.3.2 Cable between Controller and Control Panel

The cable required to connect the control panel and controller when they are installed separately can be provided optionally.

3.3.3 Control Panel Mounting Dimensions

The dimensions of mounting the control panel are shown in Fig. 3.6.
Cross truss head screws (ø3×6, ZN3–B) are used.

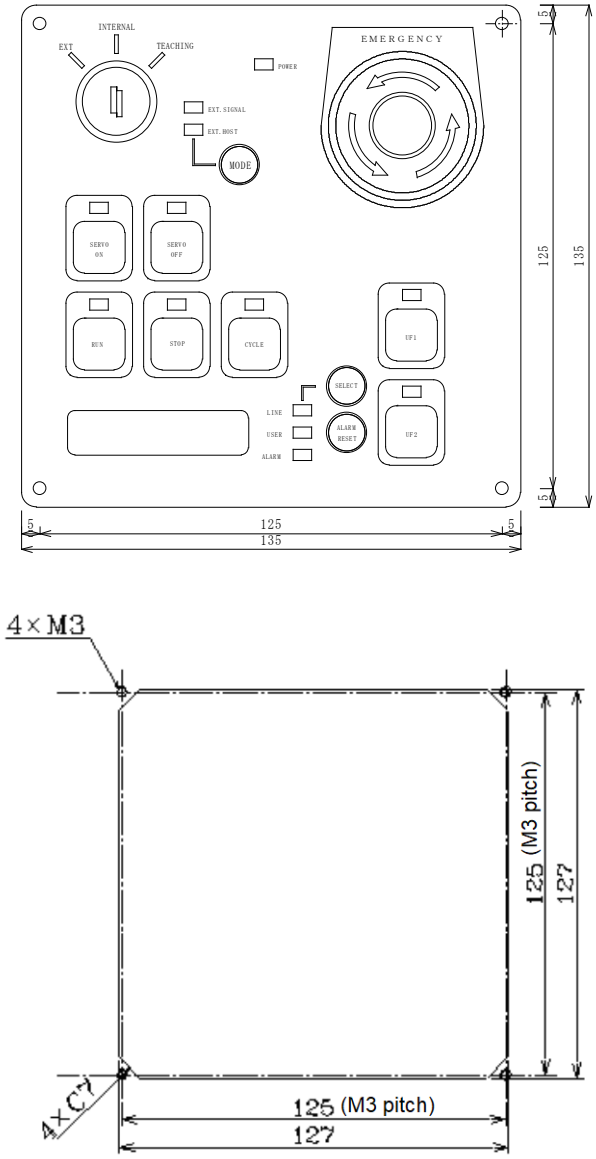


Fig. 3.7 Control panel mounting dimensions

3.3.4 Mounting Dummy Panel on Controller

When the control panel has been disengaged from the controller, mount a dummy panel on the place where the control panel was set before. The dummy panel, set parts, etc. are provided optionally.

a) Connect the cable connector which was disconnected when separating the controller from the control panel, to the rear side of the dummy panel, then screw both ends of the connector. When mounting the connector, use the cross truss head screws (ø3×6, ZN3–B).

b) Screw the dummy panel into the controller.

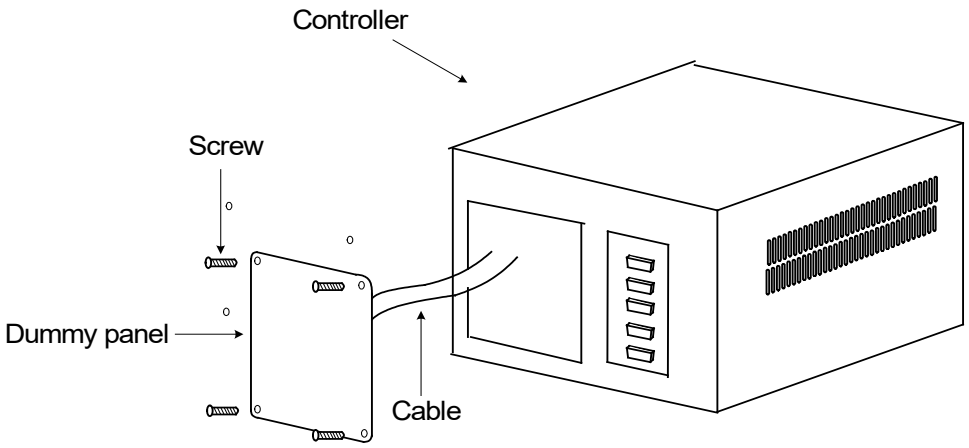


Fig. 3.8 Mounting dummy panel

3.3.5 Dimensions when Separating Control Panel

Fig. 3.9 shows the connections of the control panel and dummy panel. Provide a clearance of about 50 mm (or approximately 60 mm when the cover is attached) on the rear side of the separated control panel.
When the cable is connected to the dummy panel of the controller, provide a clearance of 80 mm or so in front of the controller as the cable connector sticks out of the panel surface.

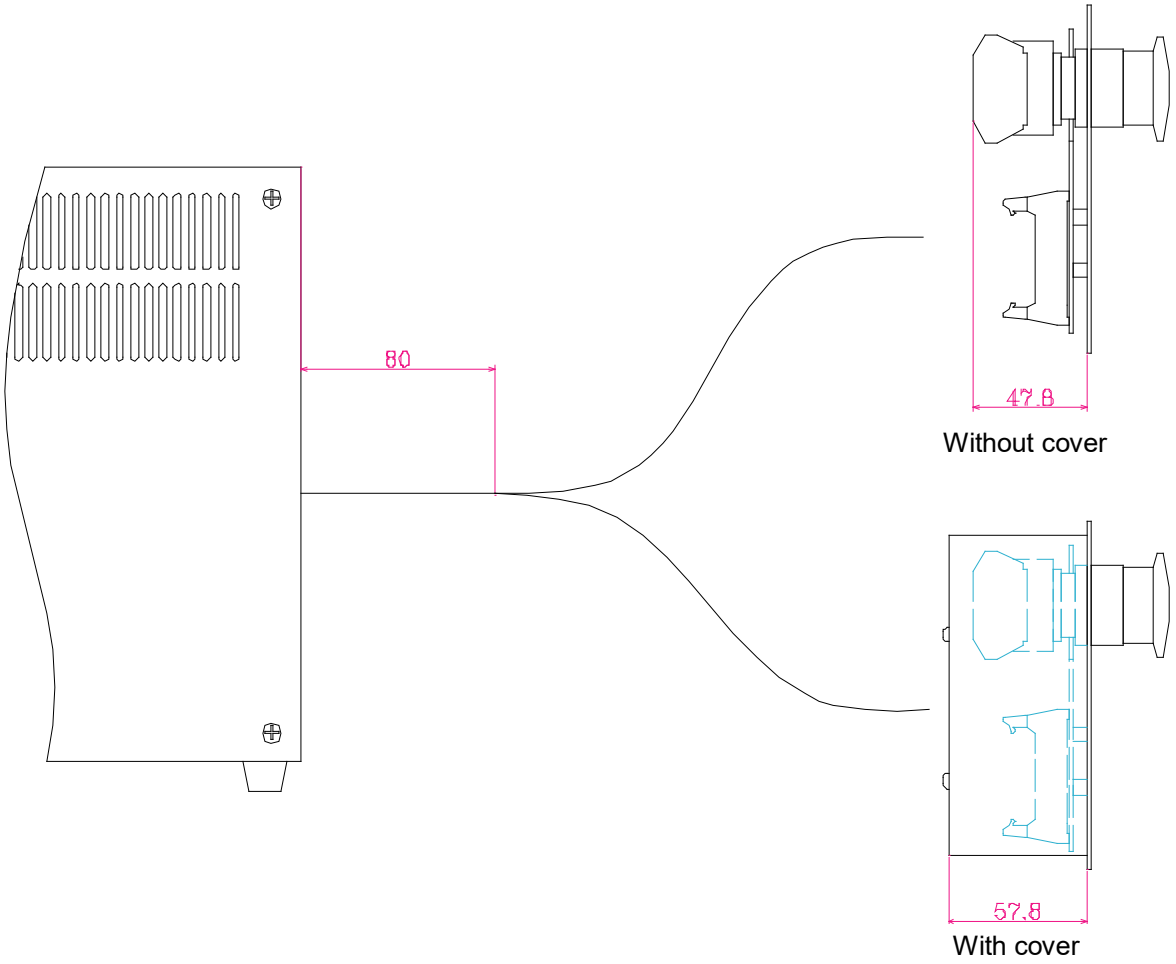
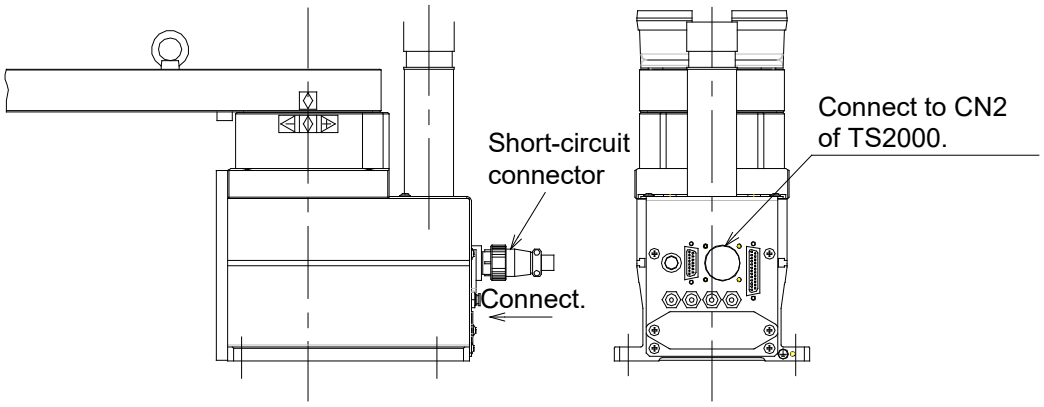


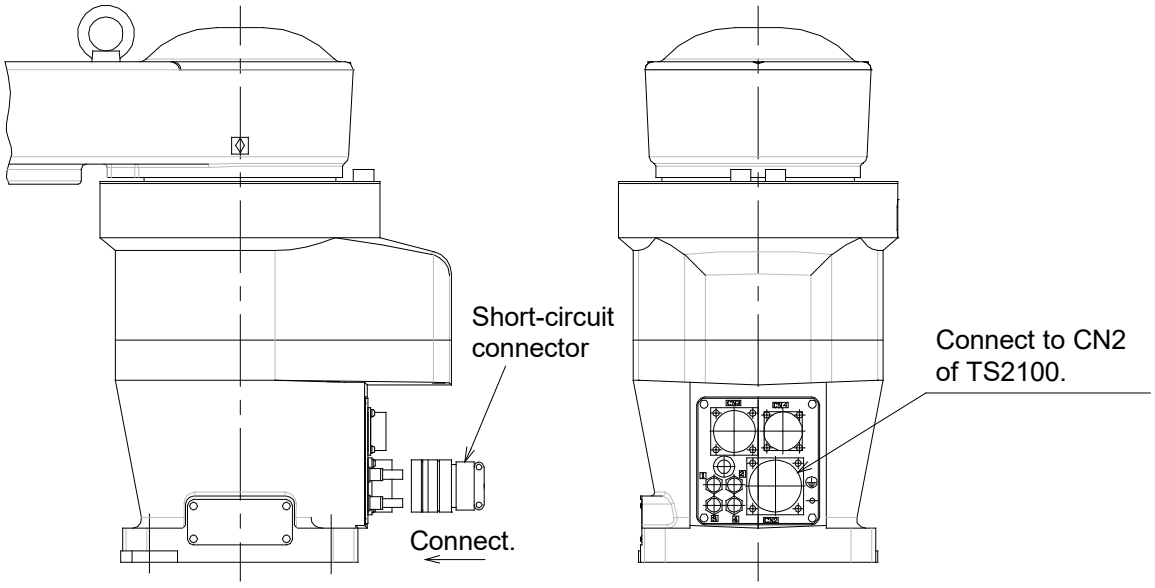
Fig. 3.9 Cable connections of dummy panel and control panel

3.3.6 Cautions on Transport

When the robot power cable has been disconnected from the controller, attach the following short-circuit connector (accessory) to the robot power cable (i.e., CN2 connector of the robot body).
If the robot has been moved suddenly or heavy vibration has been imposed on the robot at power OFF, without connecting the short-circuit connector, the absolute coordinate counter of the encoder cannot trace any further, resulting in generation of a position sensor error or loss of home position data.



TH450 and TH550



TH650 and TH850 and TH1050

Section 4

Tool Interface

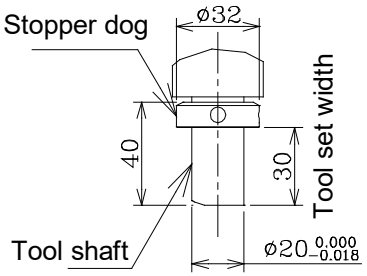
4.1 Mounting Tool

a) TH450 and TH550

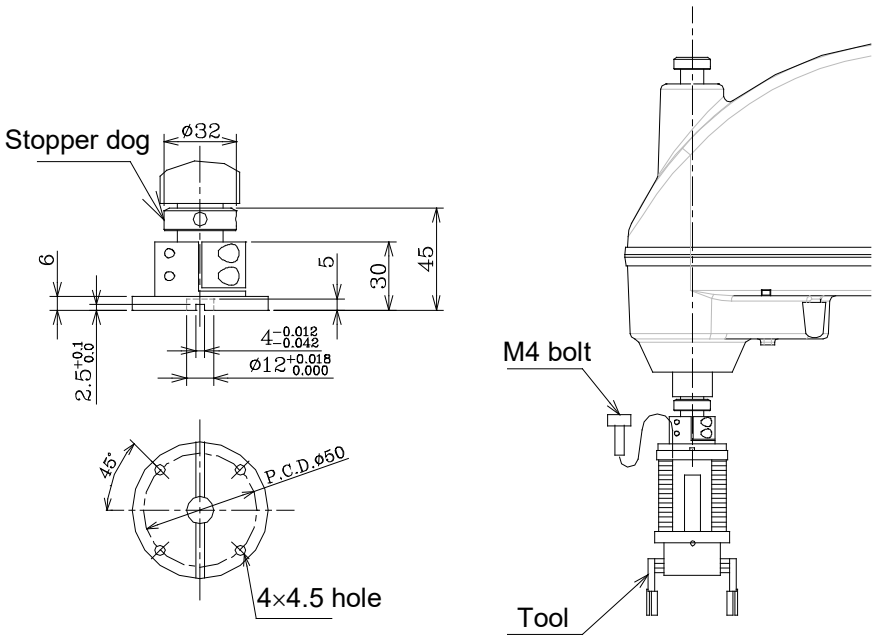
The tool is mounted on the end of the tool shaft (ø20). The tool shaft has a dog for the stopper, and the section in front of the dog is used for mounting the tool. At this time, the dog should not be removed. Dimensions of the tool shaft section are shown in Fig. 4.1 (a).

As shown in Fig. 4.1, the tool is secured to the tool shaft with split clamp. As there is not a reference such as keyway on the tool shaft, however, take care of the tool direction when mounting.

When using the tool flange (option), the tool center should be adjusted with the ø12H7 mating section and the tool direction with 4 × 4 keys, then the tool should be secured to the flange with four (4) M4 bolts.



(a) Dimensions of tool shaft section



(b) Dimensions of tool mounting flange (option)

Fig. 4.1 Mounting TH450 and TH550 tools

b) TH650 and TH850 and TH1050

The tool should be mounted on the tool set flange located at the end of the tool shaft. Dimensions of the tool set flange are shown in Fig. 4.2.

The tool is centered with the ø20H7 mating section. The tool direction should be adjusted with 5 × 5 keys, then the tool should be secured to the flange with four (4) M5 bolts.

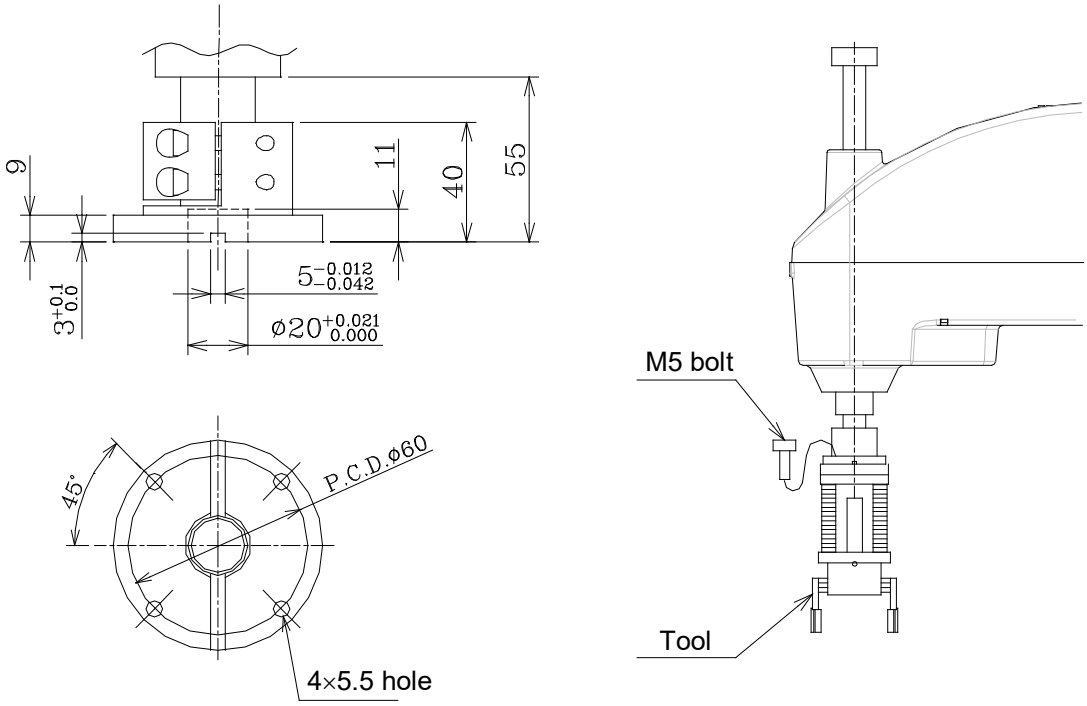


Fig. 4.2 Mounting TH650 and TH850 and TH1050 tool

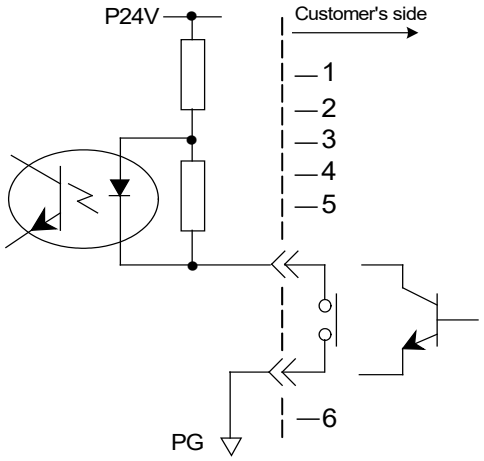
4.2 Tool Wiring and Piping

The robot is provided with wiring and air piping for the tool. These wiring and piping extend to the arm 2 and are used as follows:

4.2.1 Controller's Tool Signals (TS2000/TS2100)

The controller is provided with tool signals (i.e., five (5) input signals for sensor, etc., four (4) control signals for solenoid valve, etc., DC 24V and DC 24 VGND signals), which can be connected also from the controller side. (A cable between the controller and cable is an option.) The relevant signals are described below.

a) Input signal connector CN4

Pin	Signal name		Signal No.	Input circuit and example of connections
5	GRP1 OPN	Grip 1 open	201	
6	GRP1 CLS	Grip 1 close	202	
7	GRP2 OPN	Grip 2 open	203	
8	GRP2 CLS	Grip 2 close	204	
9	WORK	Workpiece	205	
20	PG	0 V		
Case	Shield			

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: DC24 V, 10 mA or over (circuit current: approx. 7 mA)
Minimum contact current: DC24 V, 1 mA
Contact impedance: 100 Ω or less

Transistor specifications:

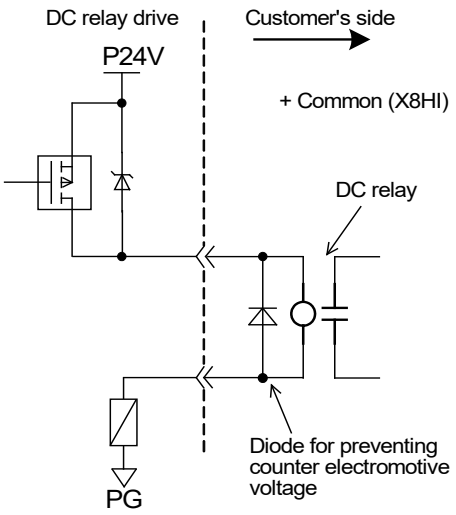
Withhold voltage between controller and emitter: 30 V or over

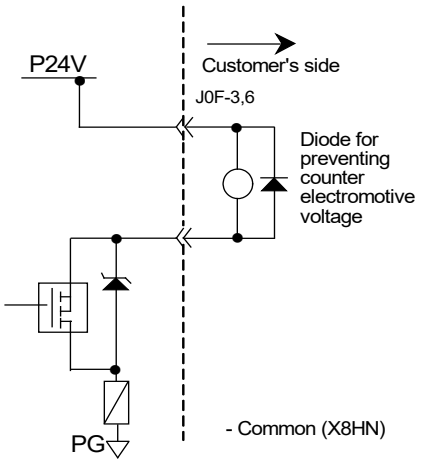
Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

b) Output signal connector CN4

Pin	Signal name		Signal No.	Output circuit and example of connections
11	GRP1	Grip 1	201	
12	$\overline{\text{GRP1}}$	– Grip 1	202	
13	GRP2	Grip 2	203	
14	$\overline{\text{GRP2}}$	– Grip 2	204	
17	P24V		DC24 V power	
18	P24V		DC24 V power	
19	P24V		DC24 V power	





By using the DC24 V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (PG) of the TS2000/TS2100 robot controller. NEVER connect +24 V with P24 V. Otherwise, the power supply may be damaged.

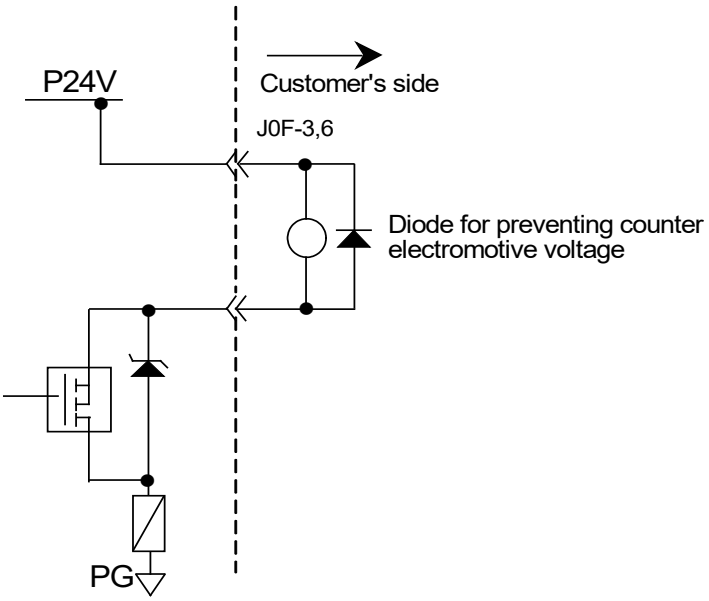
Output specifications:

Rated voltage	:	DC24 V (max. DC30 V)
Rated current	:	1 A (1.2 A or less including other I/Os when the internal power supply is used.)
Leak current	:	100 μA or less

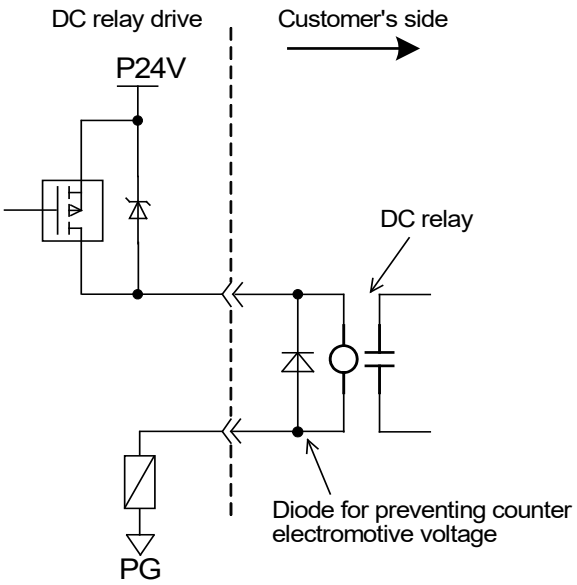
- If the DC24 V power supplied from the robot controller is used, the total current should be 1.2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.
- When a double solenoid is used, HO_1 and HO_2, and HO_3 and HO_4 are used as pairs.

The right figure shows the DC relay circuit when the external power is used.

– Common (X8HN):



+ Common (X8HI):



4.2.2 Tool Wiring (TH450, TH550)

Five (5) input signals are provided for sensors, etc., and four (4) signals for solenoid valves, etc. A DC24 V power supply signal is also provided. They are connected to the controller. The wiring arrangement for these cables is shown in Fig. 4.3.
The wires are connected to the connector on the rear side of the arm 2.
Provide the following connector and connect the cable by the user.

CN0 D-SUB connector	Type: XM2A–1501 (made by OMRON)
CN0 Connector hood	Type: XM2S–1501 (made by OMRON)
Adaptive cable	Conductor cross section area: 0.2 mm^2 ~ 0.5 mm^2

The connector and cable should be connected by soldering.

! DANGER

- Be sure to use the designated wires. Otherwise, fires or faults may be caused.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
  After the connection, use a tester, etc., to confirm the connection.

To connect the tool by means of the separately installed sequencer (PLC), disconnect the connector set plate from the base, disconnect the JOES and JOFS connectors on the back side, then connect the cable running from the PLC, etc., through the cable lead-in of the base connector section. (See Figs. 4.3 and 4.4.)
To connect the cables ahead of the JOES and JOFS connectors, provide the following plug connectors and connect the cables by the user.

Type of connector JOES	SMP–07V–BC (J.S.T. Mfg.)
Type of connector JOFS	SMP–06V–BC (J.S.T. Mfg.)
Type of contact	BHF–001T–0.8SS (J.S.T. Mfg.)
Adaptive cable	Conductor cross section area: 0.2 mm^2 ~ 0.3 mm^2

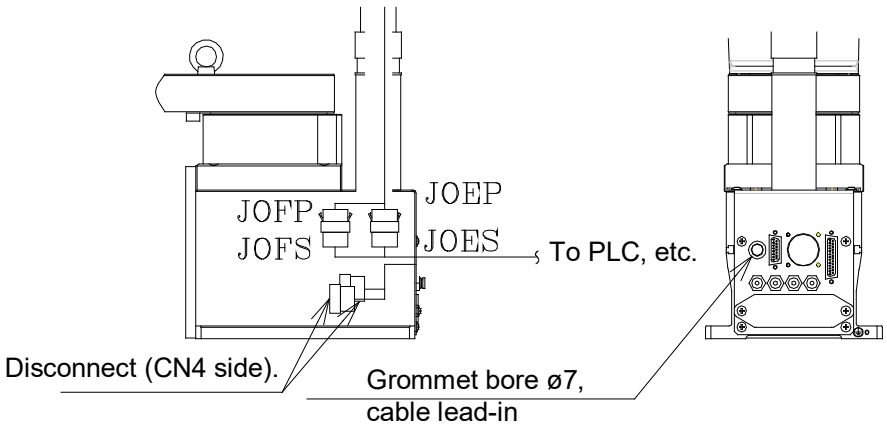
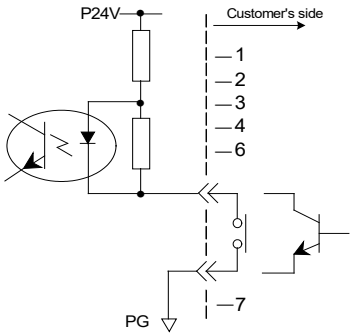
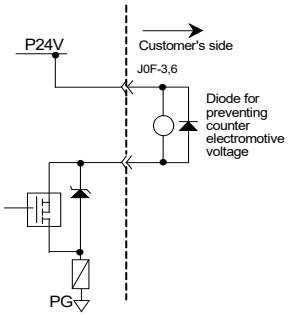
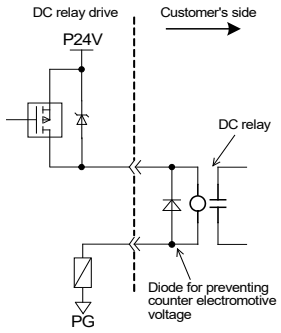


Fig. 4.4 Wiring to TH450/TH550 PLC, etc.

Input/output signal connector CN0

Pin	Signal name		Signal No.	Input/output circuit and example of connections
1	Unused			
2	Unused			
3	GRP1 OPN	Grip 1 open	201	
4	GRP1 CLS	Grip 1 close	202	
5	GRP2 OPN	Grip 2 open	203	
6	GRP2 CLS	Grip 2 close	204	
7	WORK	Workpiece	205	
8	PG	0 V		
9	SHIELD			– Common (X8HN) + Common (X8HI)
10	GRP1	Grip 1	201	
11	$\overline{\text{GRP1}}$	– Grip 1	202	
12	P24V	DC24 V power		
13	GRP2	Grip 2	203	
14	$\overline{\text{GRP2}}$	– Grip 2	204	
15	P24V	DC24 V power		





As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: DC24 V, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: DC24 V, 1 mA

Contact impedance: 100 Ω or less

Transistor specifications:

Withhold voltage between controller and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

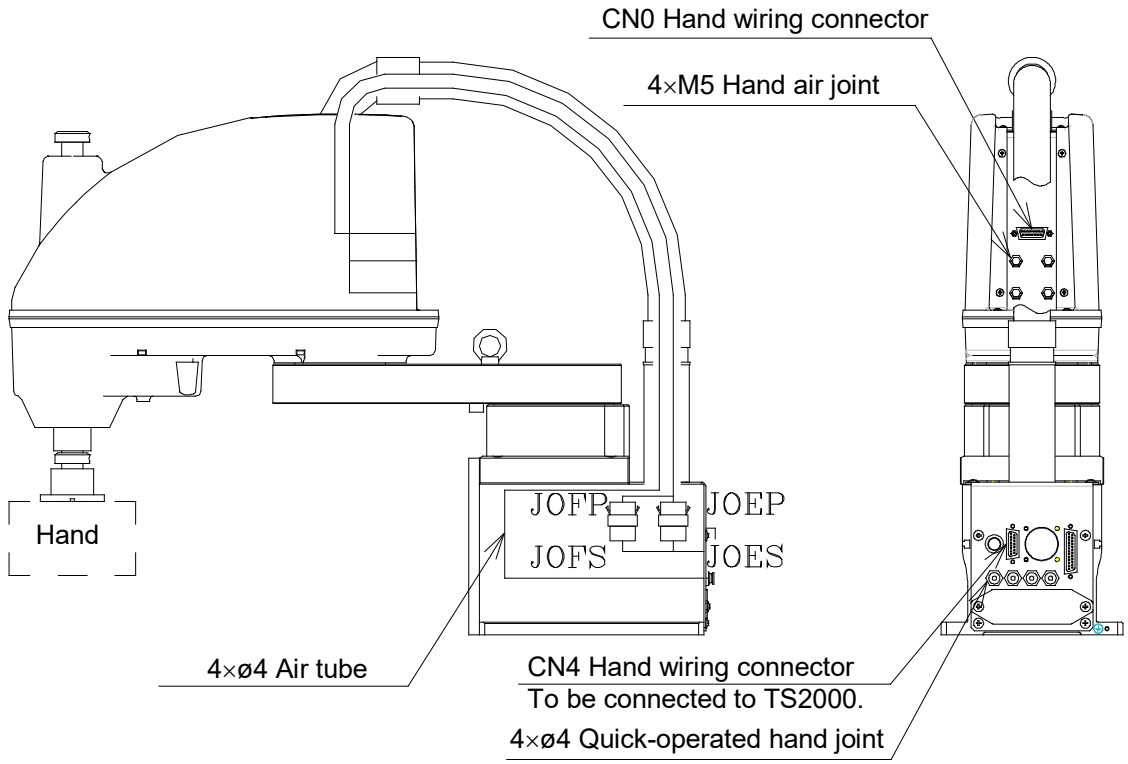
Leak current between collector and emitter: 100 μA or less

By using the DC24 V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (PG) of the TS2000/TS2100 robot controller.

Output specifications:

Rated voltage	:	DC24 V (max. DC30 V)
Rated current	:	1 A
Leak current	:	100 μA or less

- If the DC24 V power supplied from the robot controller is used, the total current should be 2 A or less.
- When the external power is used, the total current should also be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.



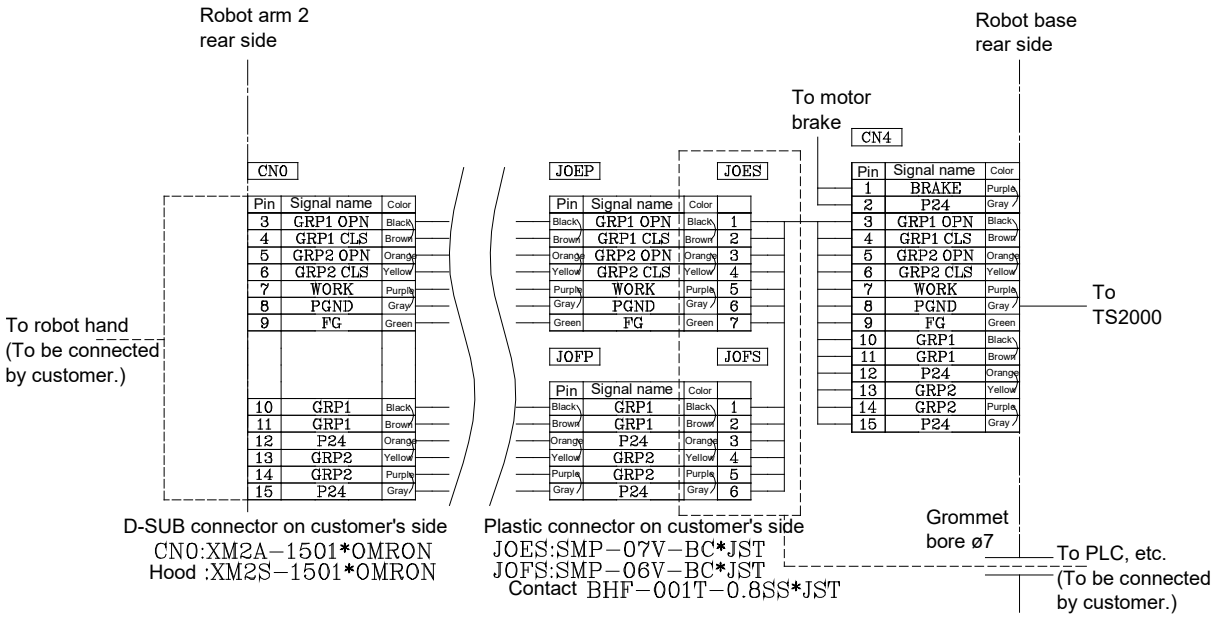


Fig. 4.3 TH450, TH550 tool wiring

The wires are connected to the connectors on the rear side of the arm 2.
The tool cable connection comes in the two (2) manners; connection through the hollow hole (ø14) on the Z-axis ball screw shaft and connection by using a clamp stay. The wiring method is shown in Fig. 4.5.

CAUTION

- Always use a highly flexible robot cable, which should be secured below the arm with a cable clamp, etc. Unless a robot cable is used, the wire may be broken.

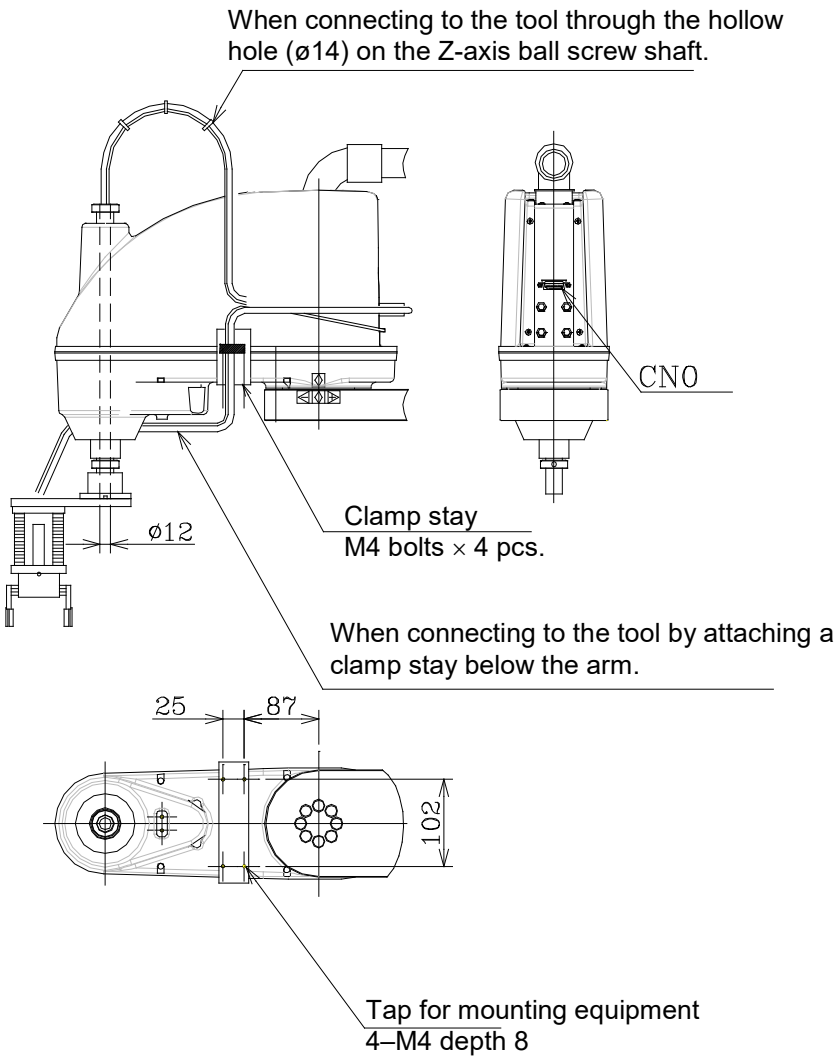


Fig. 4.5 TH450, TH550 wiring method

4.2.3 Tool Wiring (TH650/TH850/TH1050)

Five (5) input signals are provided for sensors, etc., and four (4) signals for solenoid valves, etc. A DC24 V power supply signal is also provided. They are connected to the controller. The wiring arrangement for these cables is shown in Fig. 4.6. The wires are connected to the connector on the upper side of the arm 2. Provide the following connector and connect the cable by the user.

CN0 cannon connector	Type: MS3106B–20–29–S (made by Daiichi Denshi Kogyo (DDK))
Adaptive cable	Conductor cross section area: 0.2 mm^2 ~ 0.5 mm^2

The connector and cable should be connected by soldering.

! DANGER

- Be sure to use the designated wires. Otherwise, fires or faults may be caused.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
  After the connection, use a tester, etc., to confirm the connection.

To connect the tool by means of the separately installed sequencer (PLC), disconnect the connector set plate from the base, disconnect the JOES and JOFS connectors on the back side, then connect the cable running from the PLC, etc., through the cable lead-in of the base connector section. (See Figs. 4.6 and 4.7.) To connect the cables ahead of the JOES and JOFS connectors, provide the following plug connectors and connect the cables by the user.

Type of connector JOES	SMP–07V–BC (J.S.T. Mfg.)
Type of connector JOFS	SMP–06V–BC (J.S.T. Mfg.)
Type of contact	BHF–001T–0.8SS (J.S.T. Mfg.)
Adaptive cable	Conductor cross section area: 0.2 mm^2 ~ 0.3 mm^2

Keep the current to 1 A or less per line.

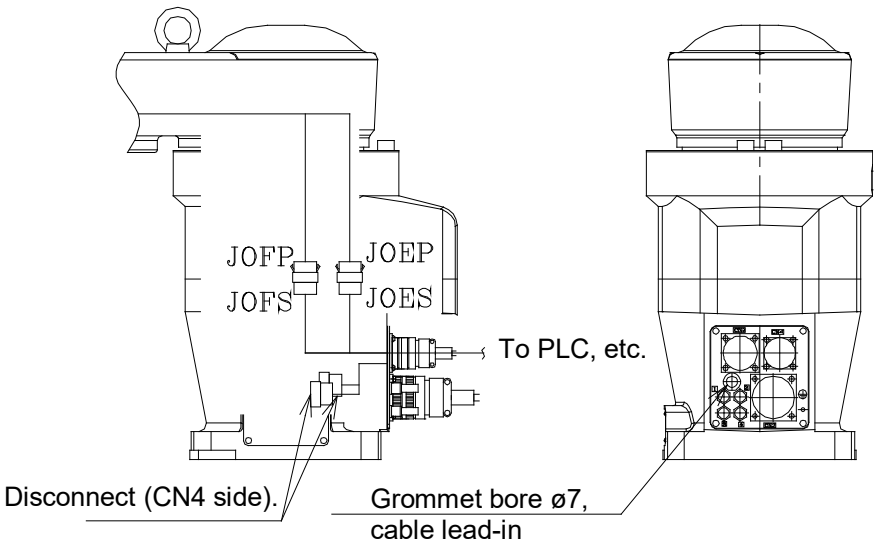
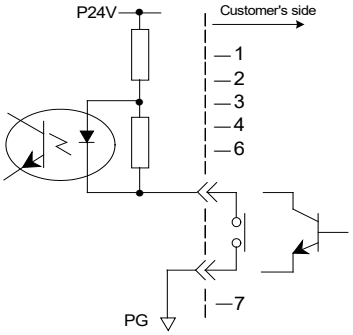
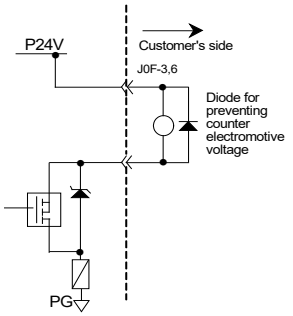
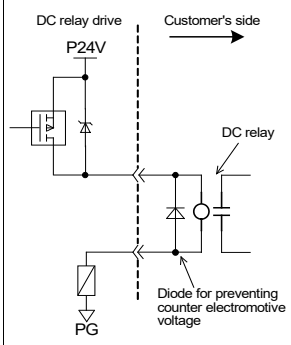


Fig. 4.7 Wiring to TH650/TH850/TH1050 PLC, etc.

Input/output signal connector CN0

Pin	Signal name		Signal No.	Input/output circuit and example of connections
A	Unused			
B	Unused			
C	Unused			
D	GRP1 OPN	Grip 1 open	201	
E	GRP1 CLS	Grip 1 close	202	
F	GRP2 OPN	Grip 2 open	203	
G	GRP2 CLS	Grip 2 close	204	
H	WORK	Workpiece	205	
J	PG	0 V		– Common (X8HN) 
K	Unused			
L	GRP1	Grip 1	201	
M	$\overline{\text{GRP1}}$	– Grip 1	202	
N	P24V	DC24 V power		
P	GRP2	Grip 2	203	
R	$\overline{\text{GRP2}}$	– Grip 2	204	
S	P24V	DC24 V power		
T	SHIELD			

+ Common (X8HI)



As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: DC24 V, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: DC24 V, 1 mA

Contact impedance: 100 Ω or less

Transistor specifications:

Withhold voltage between controller and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

By using the DC24 V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (PG) of the TS2000/TS2100 robot controller.

Output specifications:

Rated voltage	:	DC24 V (max. DC30 V)
Rated current	:	1 A
Leak current	:	100 μA or less

- If the DC24 V power supplied from the robot controller is used, the total current should be 2 A or less.
- When the external power is used, the total current should also be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.

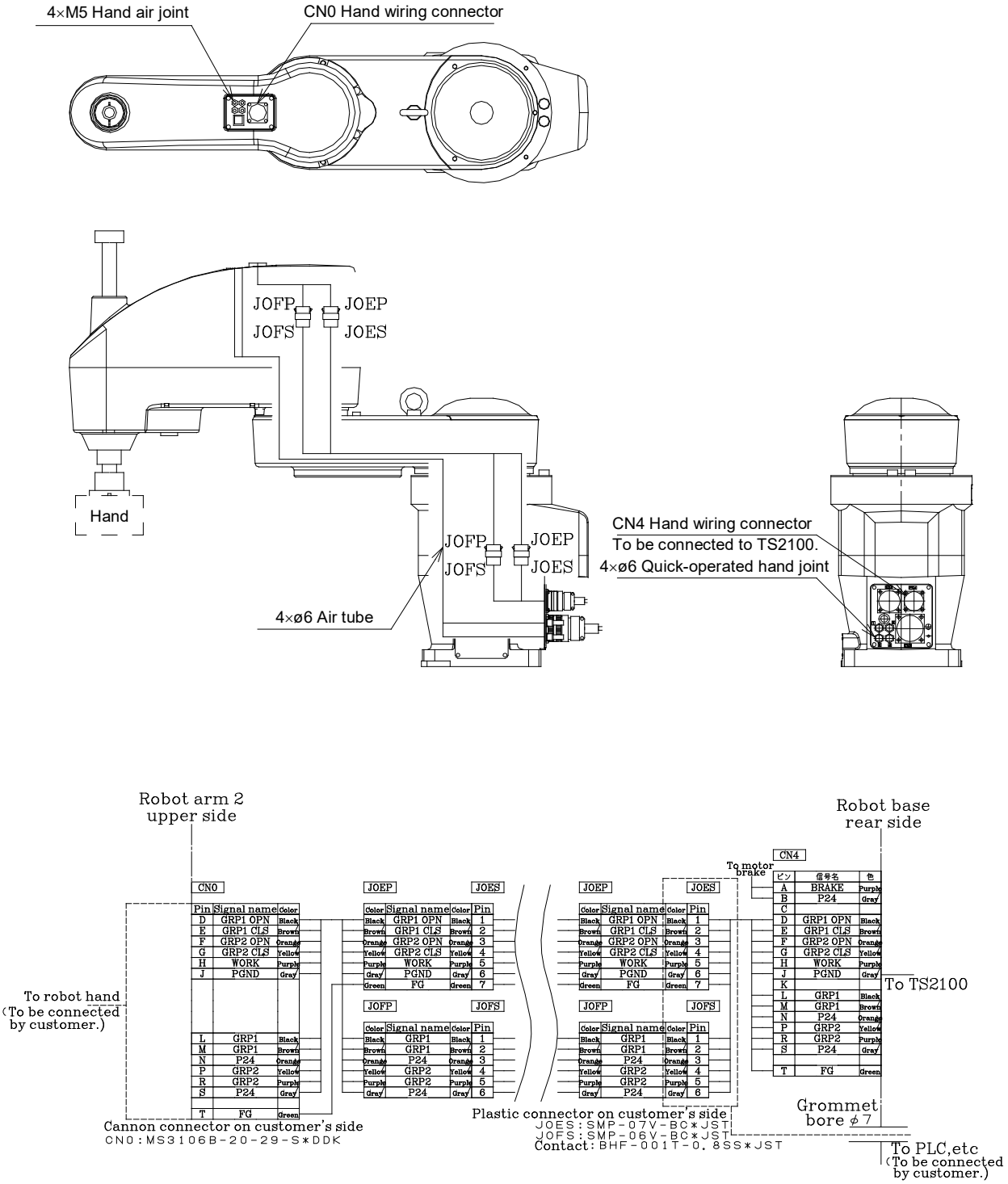


Fig. 4.6 TH650/TH850/TH1050 tool wiring

The wires are connected to the connectors on the upper side of the arm 2.
The tool cable connection comes in the two (2) manners; connection through the hollow hole (ø18) on the Z-axis ball screw shaft and connection by using a clamp stay. The wiring method is shown in Fig. 4.8.

CAUTION

- Always use a highly flexible robot cable, which should be secured below the arm with a cable clamp, etc. Unless a robot cable is used, the wire may be broken.

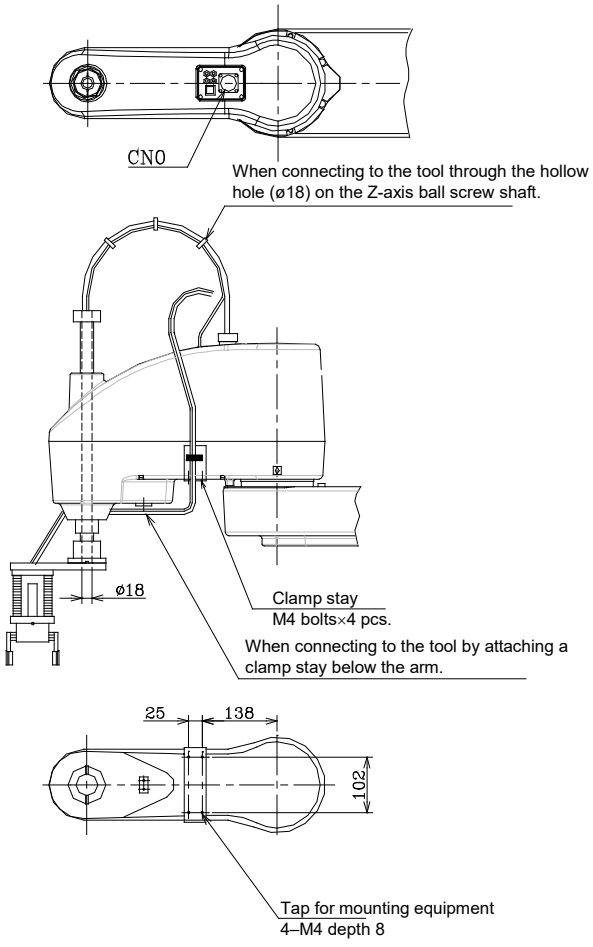


Fig. 4.8 TH650/TH850/TH1050 wiring method

4.2.4 Tool Air Piping

The robot is provided with four (4) air lines for the tool.
The outer diameter of the air pipelines is 4 mm for the TH450 and TH550, and 6 mm for the TH650 and TH850 and TH1050. The air piping drawing is shown in Fig. 4.9 and 4.10.
The air control unit (oiler, regulator with gage and filter) for the solenoid valves should be provided by the user.

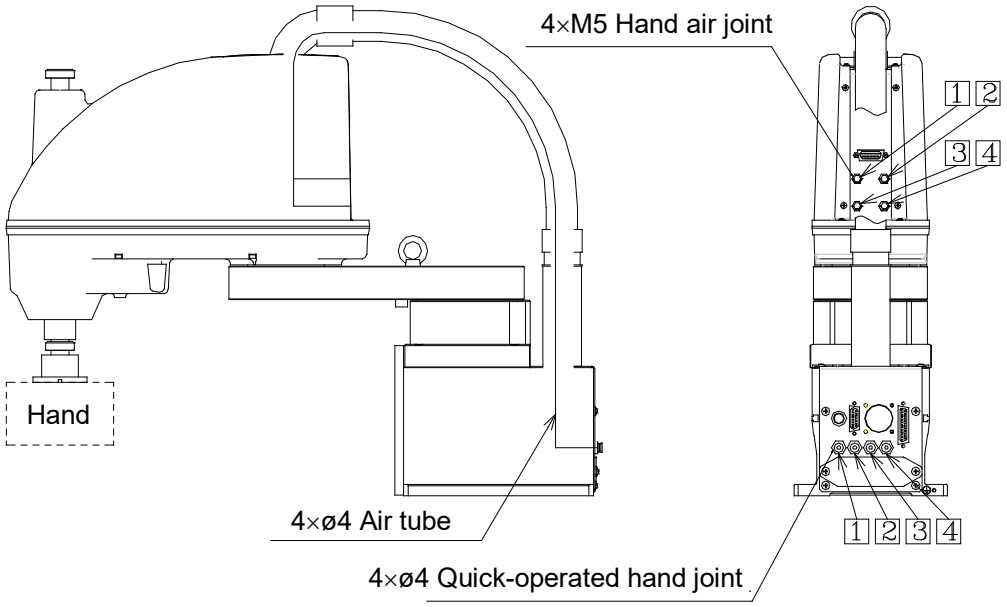


Fig. 4.9 TH450, TH550 tool air piping

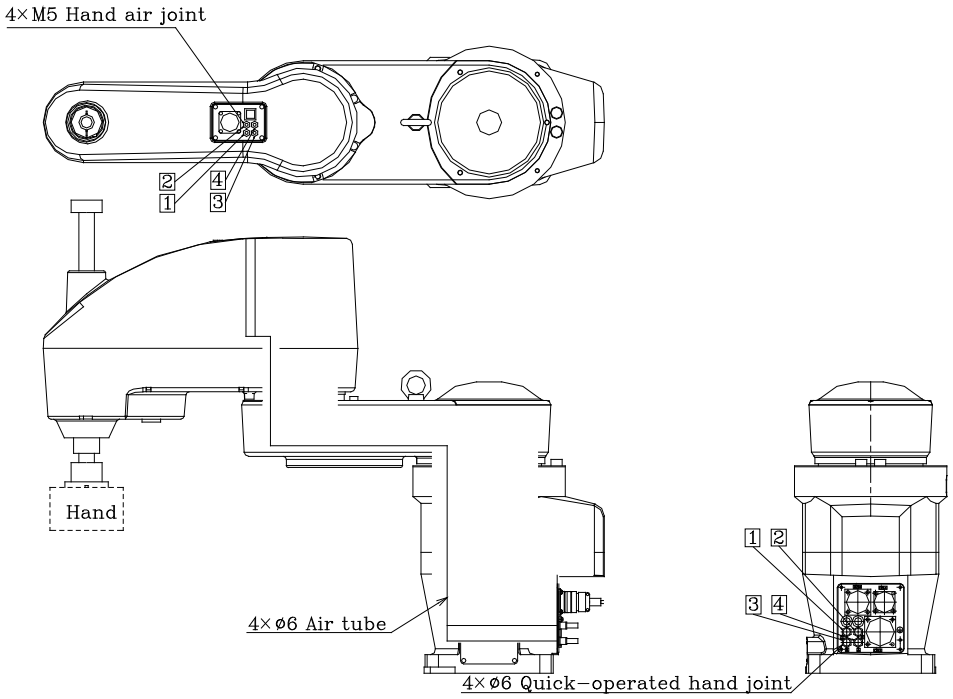


Fig. 4.10 TH650/TH850/TH1050 tool air piping

4.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and how to set up the program according to the load.

4.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and offset value of tool center of gravity from the center of the tool shaft, as shown in Fig. 4.11. The load conditions permitted for the TH series are shown in Table 4.1.

Table 4.1 Permissible load conditions

Model	Conditions	Permissible values
TH450/TH550	Mass of load	Max. 5 kg
	Moment of inertia	Max. 0.05 kg·m^2
	Center-of-gravity offset of load	Max. 100 mm
TH650	Mass of load	Max. 10 kg
	Moment of inertia	Max. 0.10 kg·m^2
	Center-of-gravity offset of load	Max. 100 mm

! CAUTION

- The robot life and safety cannot be guaranteed if the robot is operated under the load conditions exceeding the permissible values.

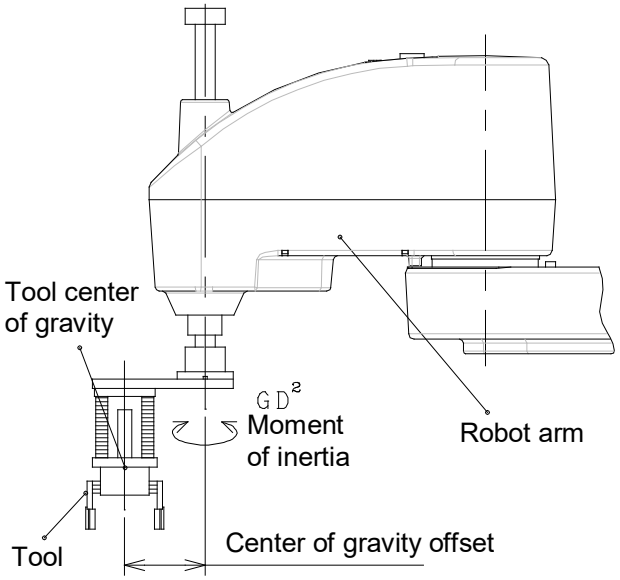


Fig. 4.11 Robot tool

4.3.2 Load Conditions and Program Setting

Even if the tool load conditions are within the permissible values listed in Table 4.1, the maximum speed, acceleration/deceleration time and servo gain that are adequate for the load conditions should be set. If this setting is inadequate, the robot may vibrate or overshoot, resulting in reduction of the robot life.
The TH series can automatically change the maximum speed, time for acceleration/deceleration and servo gain by using the PAYLOAD command in the program according to the load conditions.
Compared with the previous models, the TH series is strictly adjusted to realize high-speed operation. Under any conditions, be sure to use the PAYLOAD command.
The specific method for using this function is explained below.

a) PAYLOAD command
The PAYLOAD command format is written as shown below if the tool mass is M kg and the center of gravity is L mm.

PAYLOAD = {M, L}

M : Load mass (unit: kg)
L : Center of gravity offset (unit: mm)

The PAYLOAD command has the following functions.

- The maximum speed and acceleration/deceleration time of each robot axis are automatically changed according to the set load conditions.
- The servo gain of each robot axis is automatically changed according to the set load conditions.

b) Program examples
Basic program examples using the PAYLOAD command are shown below.
For further information, see the Robot Language Manual.

(Program example 1)
The robot is moved under the load conditions of 5 kg mass and 100 mm center of gravity offset.

```
PROGRAM SAMPLE
  SPEED= 100
  PAYLOAD={5,100}
```

```
 MOVE A
MOVE B
STOP
END
```

(Program example 2)

When the hand mass is 3 kg and the center of gravity offset is 30 mm, and the mass is 5 kg and center of gravity is 50 mm when the workpiece is grasped. Pick-and-place operation is executed under the above conditions.

```
PROGRAM SAMPLE
  PAYLOAD={3,30}
  ACCUR=COARSE
  ENABLE NOWAIT
  RESET DOUT
  MOVE  TAIKI
DOUT(1)
WAIT DIN(1)
LOOP:
  MOVE A+POINT(0,0,100)
  IF DIN(-1)THEN GOTO FIN
  MOVE A
  WAIT MOTION>=100
  DOUT(213)
  DELAY 1
  PAYLOAD={5,50}
  MOVE A+POINT(0,0,100)
  MOVE B+POINT(0,0,100)
  MOVE B
  WAIT MOTION>=100
  DOUT(-213)
  DELAY 1
  PAYLOAD={3,30}
  MOVE B+POINT(0,0,100)
  GOTO LOOP
FIN:
  MOVE  TAIKI
  DOUT(1)
  STOP
END
```

c) Setting of PAYLOAD command

In the default state, or when the PAYLOAD command is not used, the maximum speed and acceleration/deceleration time are set to 100 % and the servo gain is set to the value under the minimum load. (See Para. 4.3.3.)

! CAUTION

Be sure to use the PAYLOAD command. Otherwise, the robot will vibrate or overshoot, resulting in shortening the life of the mechanisms.

! CAUTION

The load moment of inertia should be within the tolerances given in Table 4.1. Even if there is no center-of-gravity offset of load, if the moment of inertia is large, the robot may vibrate. When this happens, figure out the virtual center-of-gravity offset (L mm) from the following equation, using the moment of inertia (J kg·m^2) and mass (M kg).

$L = \sqrt{(J \times 10^6 / M)}$

Then, designate the following.

PAYLOAD = {M, L}

! CAUTION

When guiding manually, the robot may vibrate if the load mass or center-of-gravity offset is large. This is because the servo gain is not appropriate. When this happens, perform the following operation while setting the load conditions in the test run mode.

DO PAYLOAD = {5, 100}
EXE

The servo gain is changed then to the value which meets the load conditions.

4.3.3 Setting Robot's Maximum Speed and Acceleration/Deceleration Time for Load Conditions

a) TH450

The maximum speed and acceleration/deceleration time of the robot are automatically changed according to the load conditions when the PAYLOAD command is used. The maximum speed and acceleration/deceleration time change with the mass of load, as shown in Fig. 4.12.

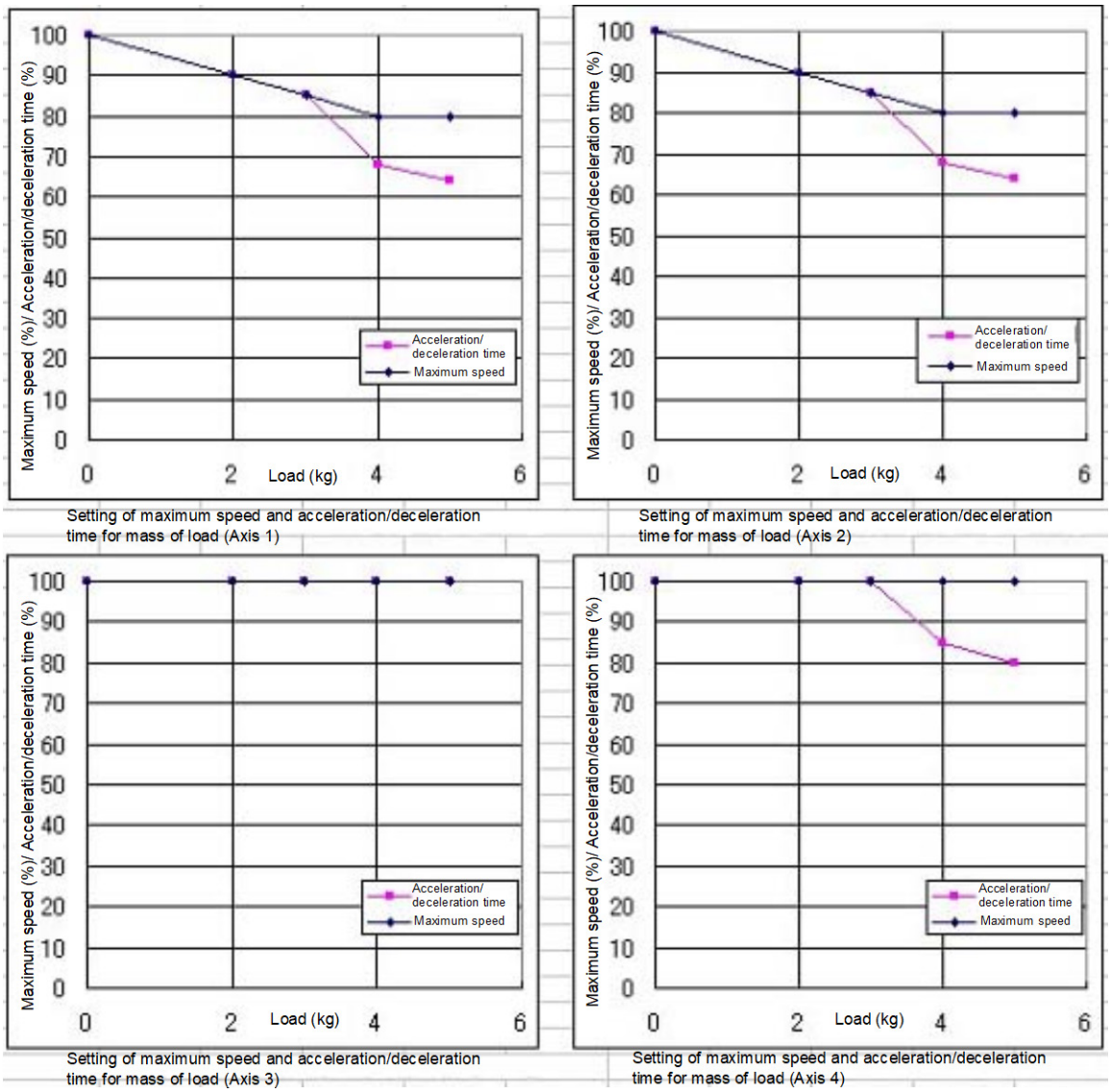


Fig. 12 Setting of maximum speed and acceleration/deceleration time for mass of load (TH450)

Additionally, if there is a center-of-gravity offset of load, the maximum speed and acceleration/deceleration time change as shown in Fig. 4.13. For the axis 3, however, no limitation is imposed on the offset.

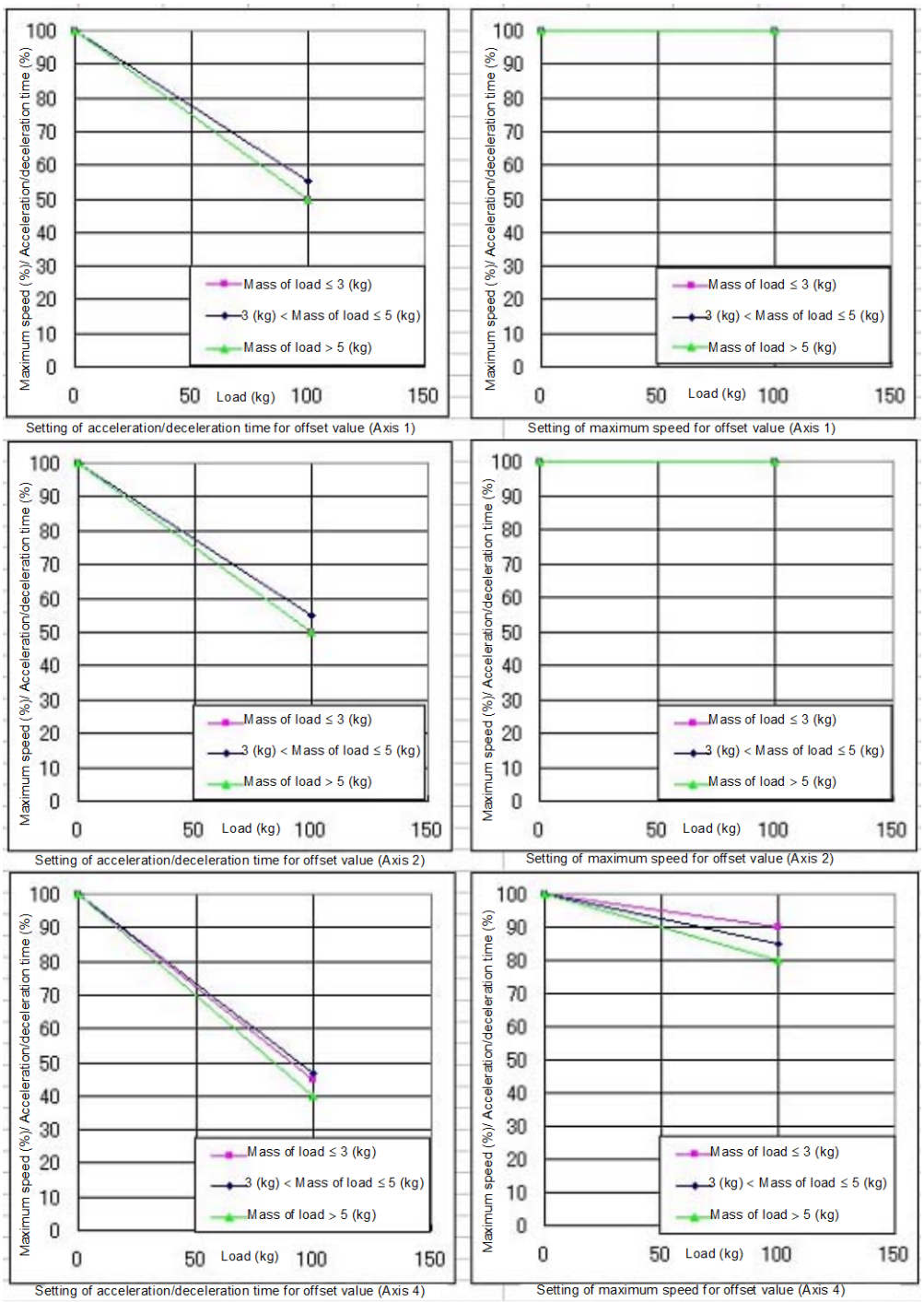


Fig. 4.13 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH450)

b) TH550

The maximum speed and acceleration/deceleration time of the robot are automatically changed according to the load conditions when the PAYLOAD command is used. The maximum speed and acceleration/deceleration time change with the mass of load, as shown in Fig. 4.14.

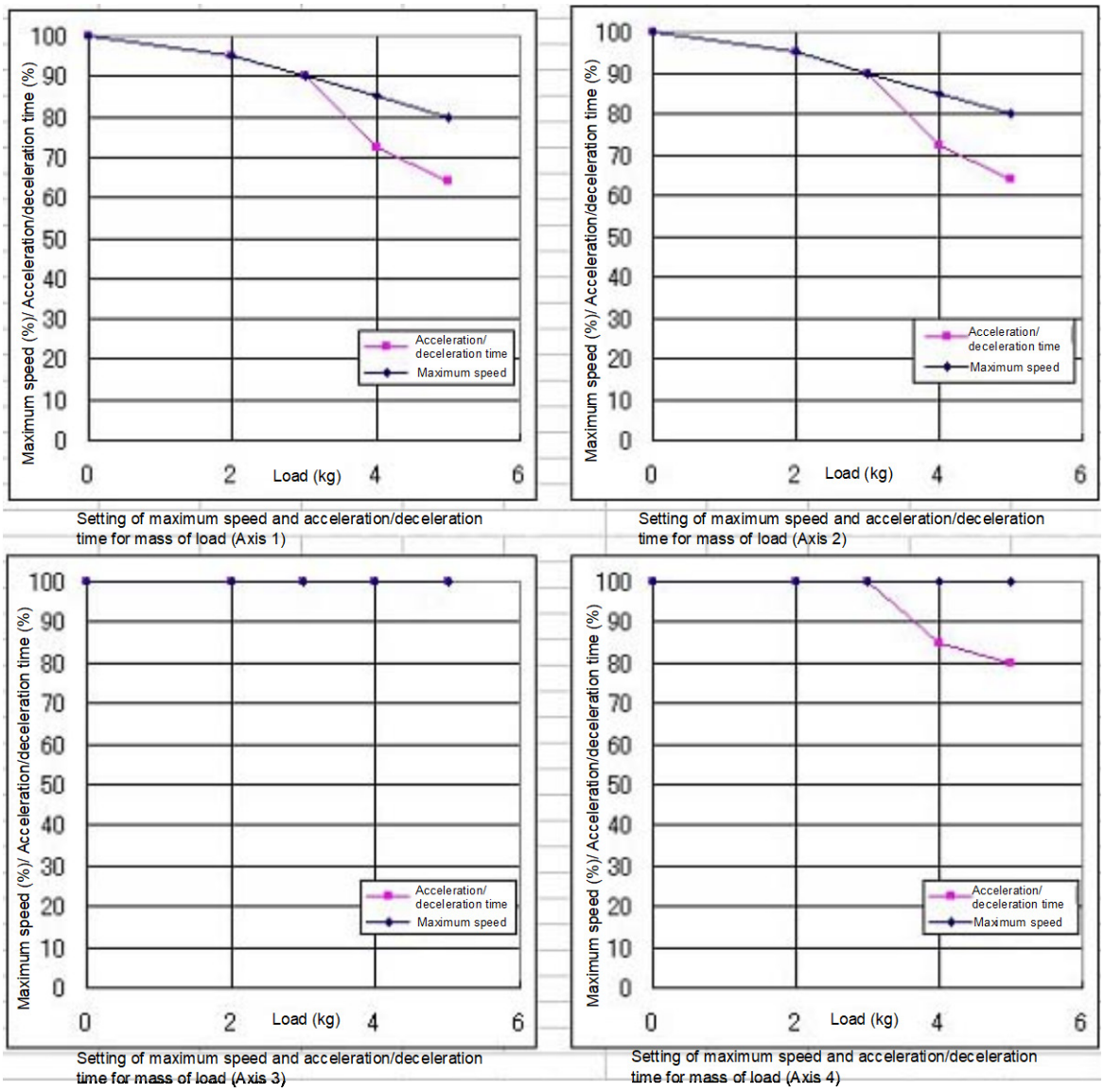


Fig. 4.14 Setting of maximum speed and acceleration/deceleration time for mass of load (TH550)

Additionally, if there is a center-of-gravity offset of load, the maximum speed and acceleration/deceleration time change as shown in Fig. 4.15. For the axis 3, however, no limitation is imposed on the offset.

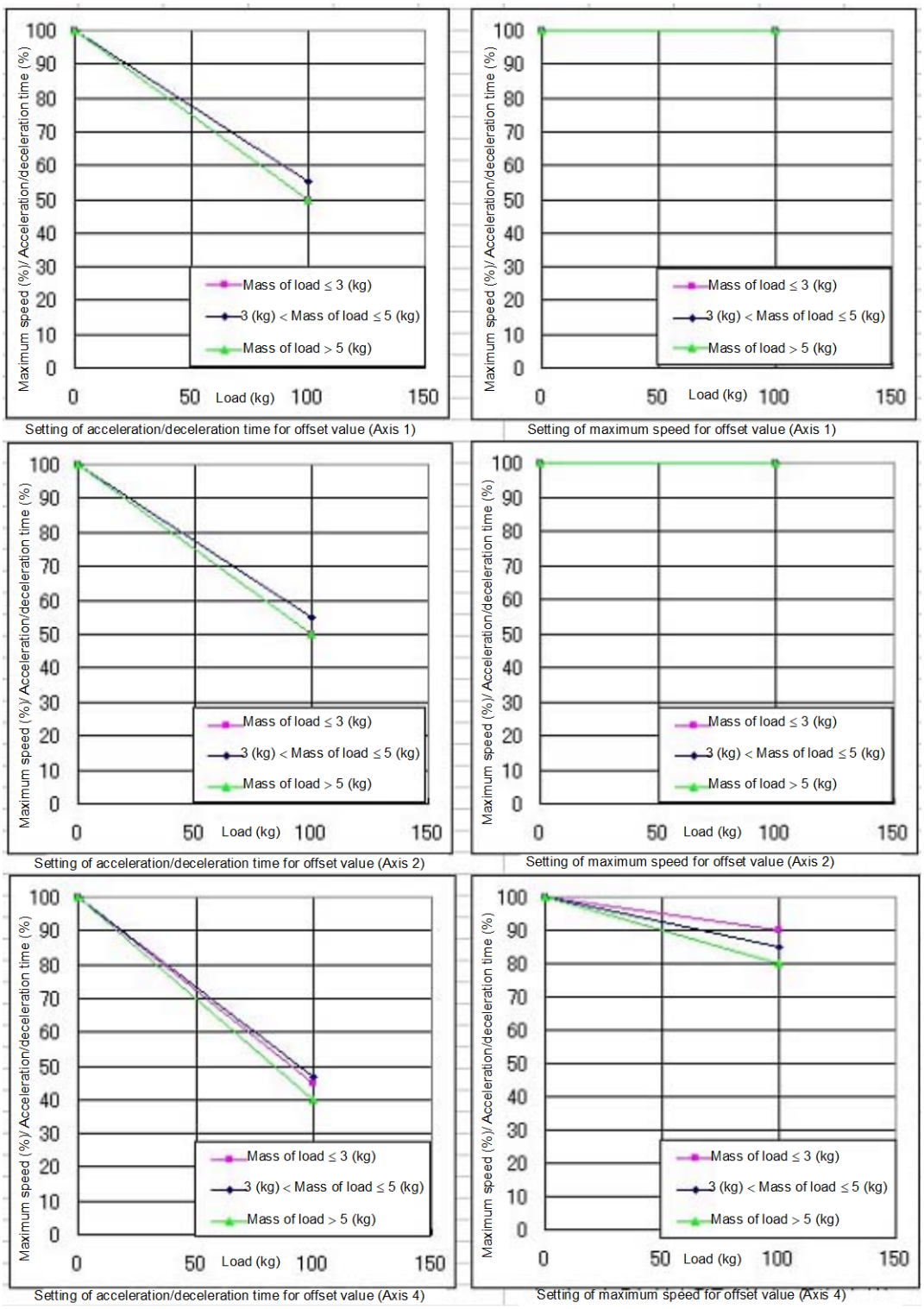


Fig. 4.15 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH550)

c) TH650

The maximum speed and acceleration/deceleration time of the robot are automatically changed according to the load conditions when the PAYLOAD command is used. The maximum speed and acceleration/deceleration time change with the mass of load, as shown in Fig. 4.16.

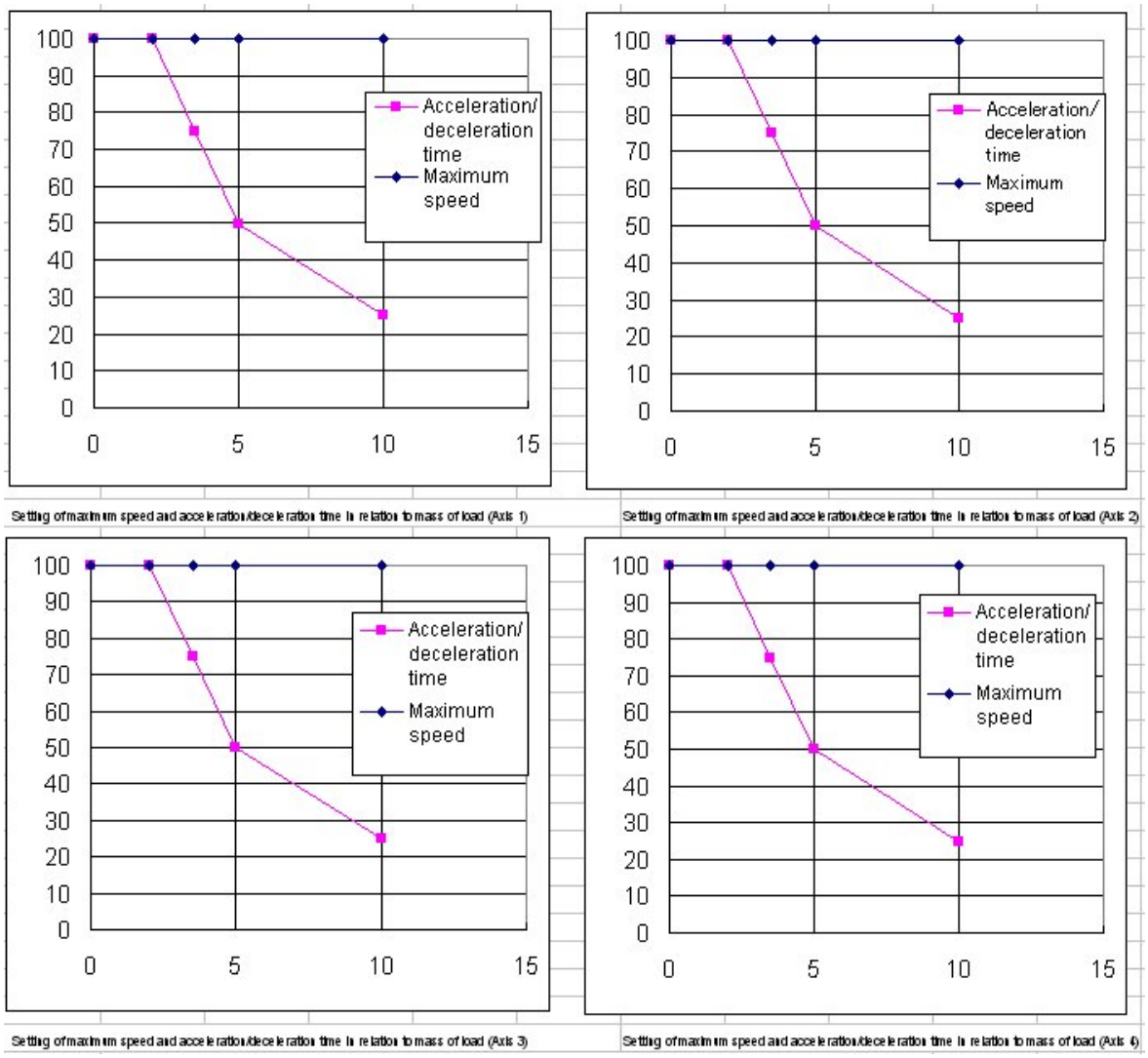


Fig. 4.16 Setting of maximum speed and acceleration/deceleration time for mass of load (TH650)

Additionally, if there is a center-of-gravity offset of load, the maximum speed and acceleration/deceleration time change as shown in Fig. 4.17 and Fig.4.18. For the axis 3, however, no limitation is imposed on the offset.

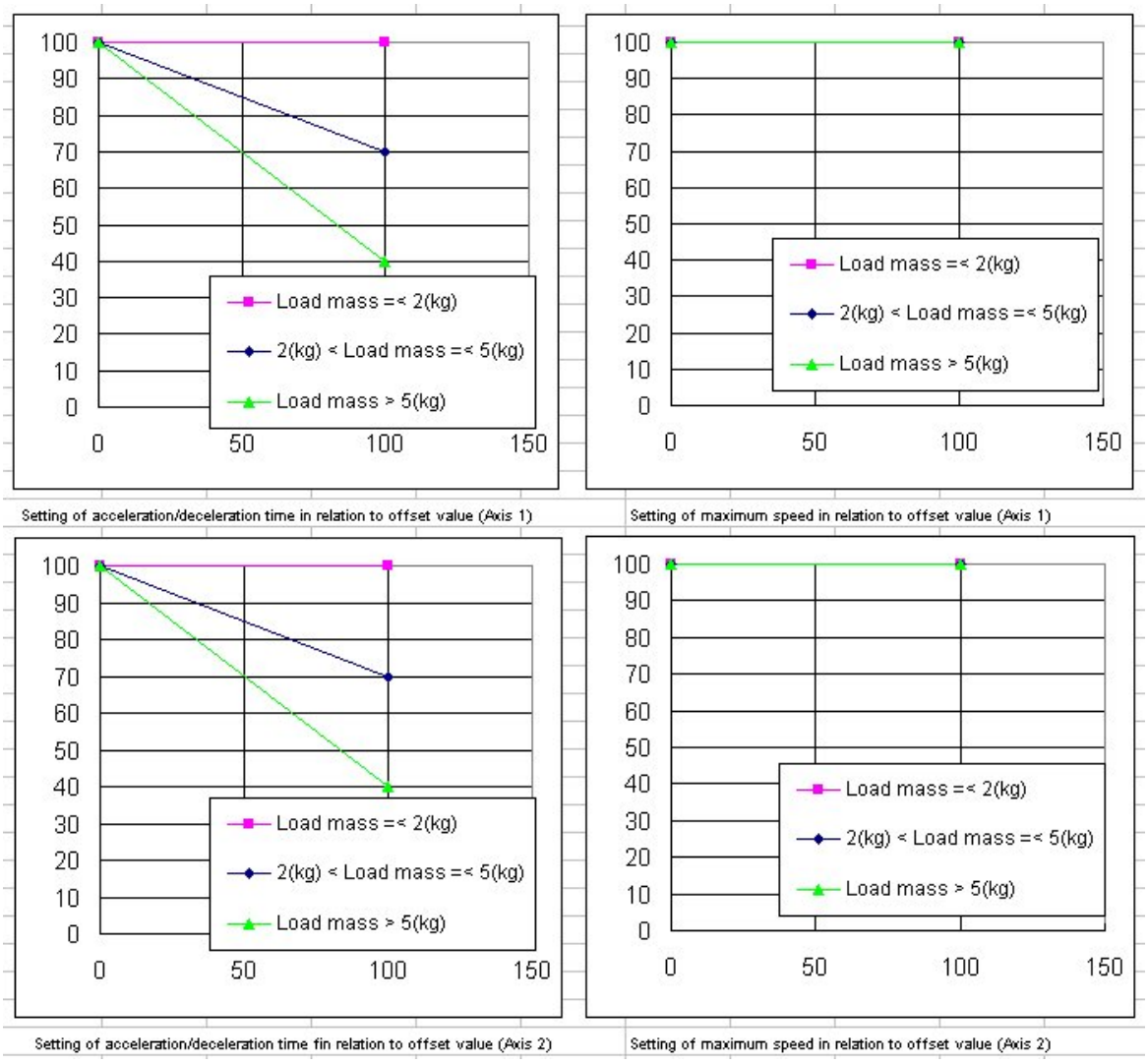


Fig. 4.17 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH650-axis1/axis2)

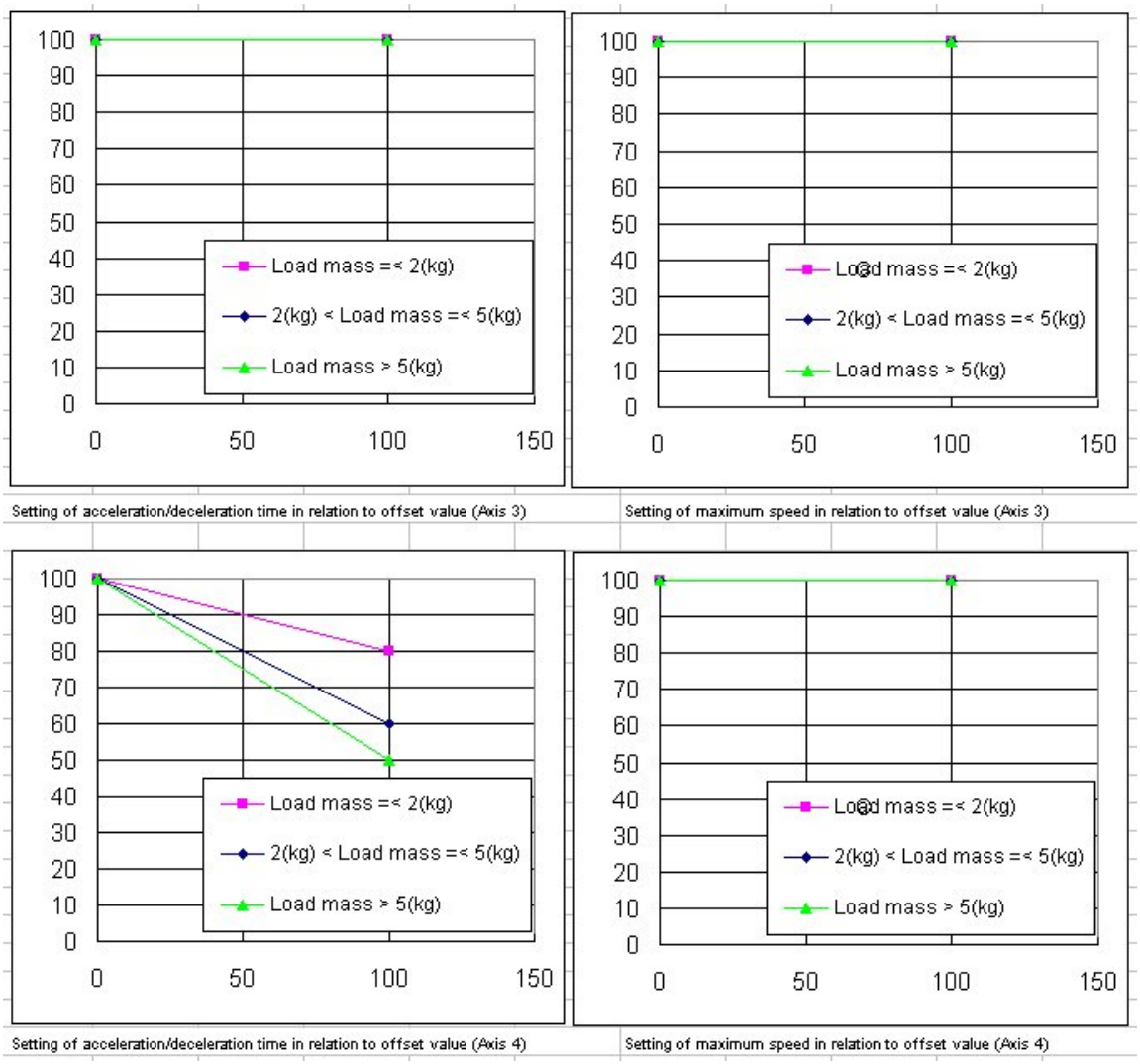


Fig. 4.18 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH650-axis3/axis4)

d) TH850

The maximum speed and acceleration/deceleration time of the robot are automatically changed according to the load conditions when the PAYLOAD command is used. The maximum speed and acceleration/deceleration time change with the mass of load, as shown in Fig. 4.19.

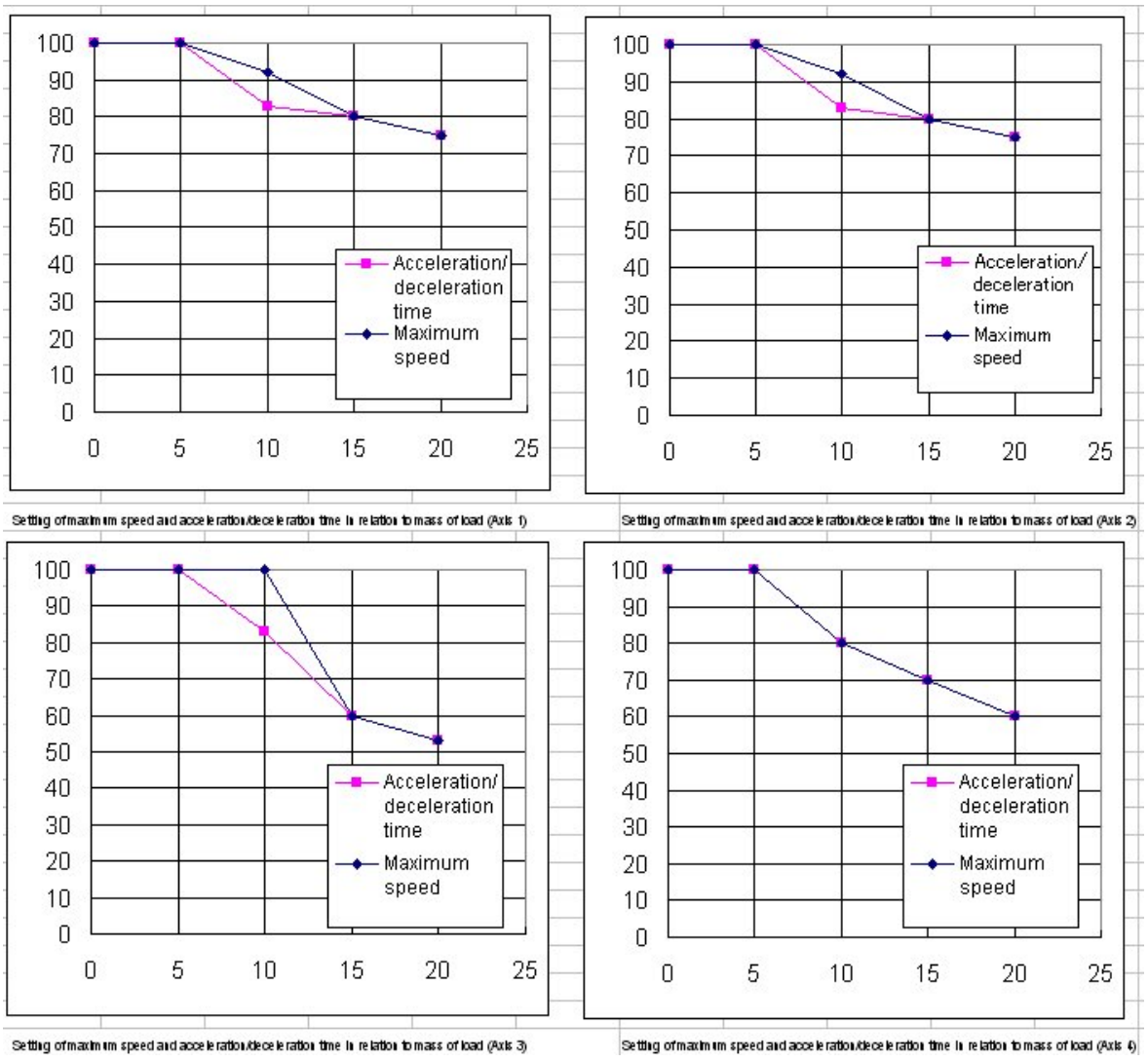


Fig. 4.19 Setting of maximum speed and acceleration/deceleration time for mass of load (TH850)

Additionally, if there is a center-of-gravity offset of load, the maximum speed and acceleration/deceleration time change as shown in Fig. 4.20 and Fig.4.21.

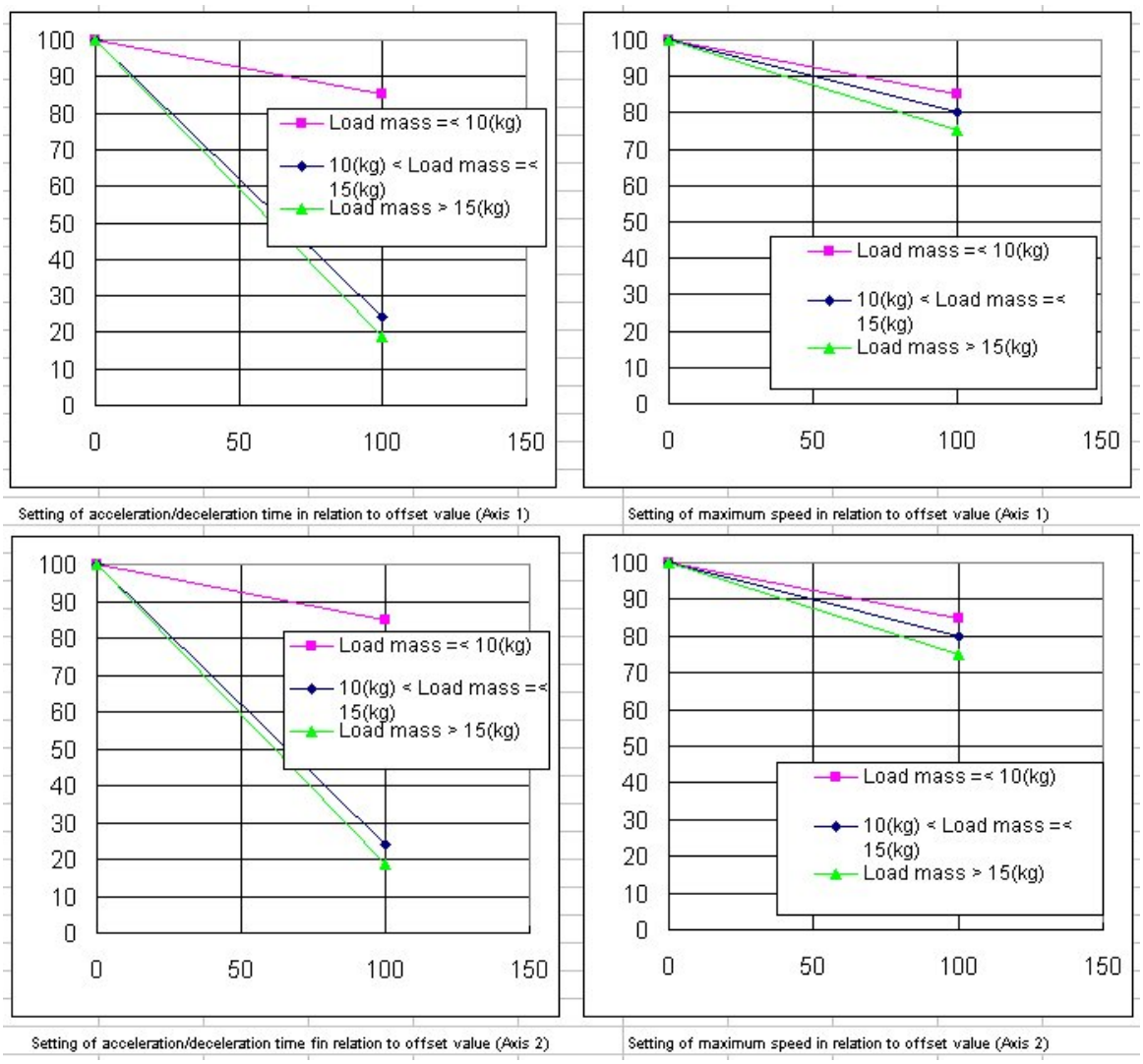


Fig. 4.20 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH850-axis1/axis2)

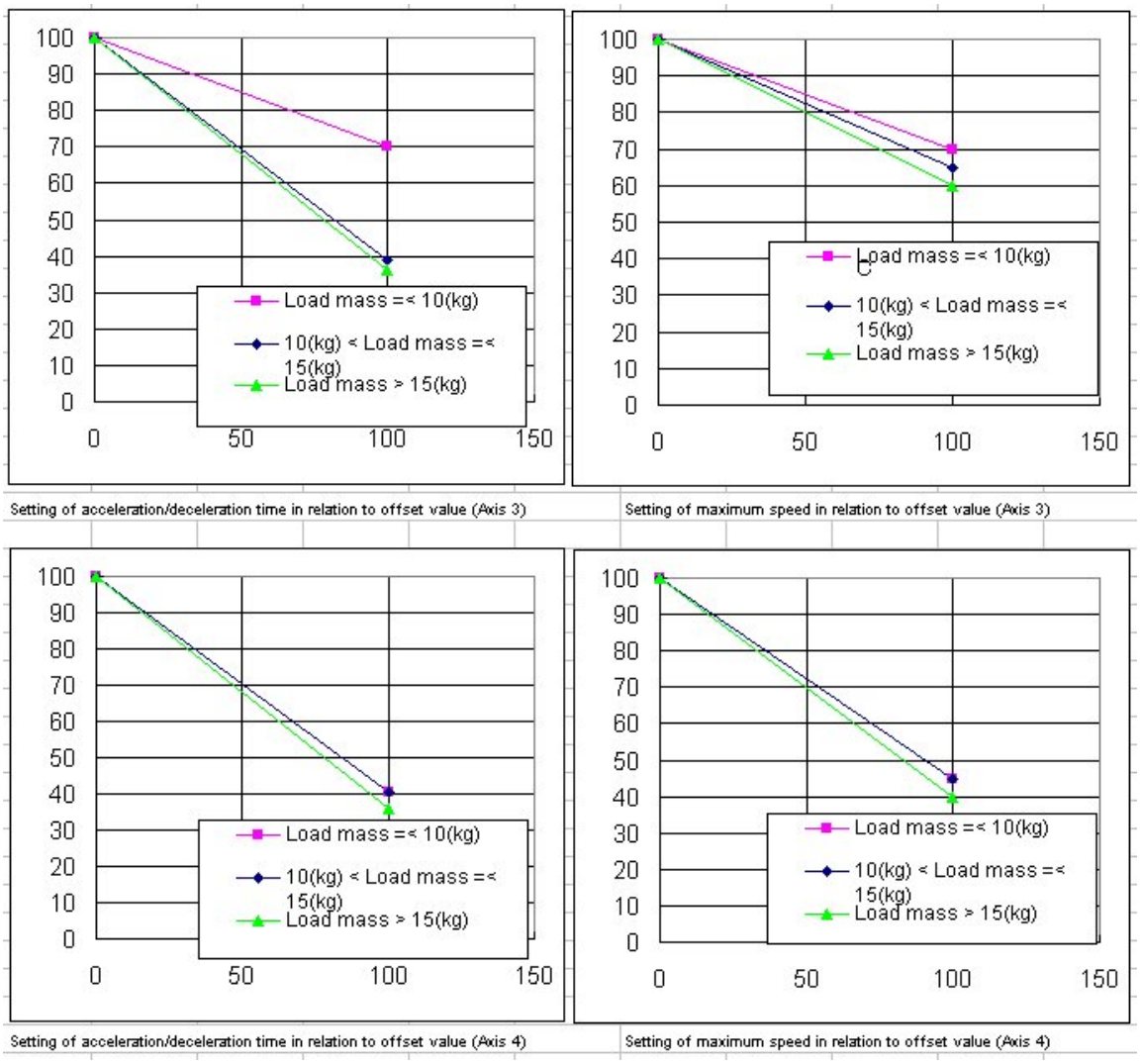


Fig. 4.21 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH850-axis3/axis4)

e) TH1050

The maximum speed and acceleration/deceleration time of the robot are automatically changed according to the load conditions when the PAYLOAD command is used. The maximum speed and acceleration/deceleration time change with the mass of load, as shown in Fig. 4.22.

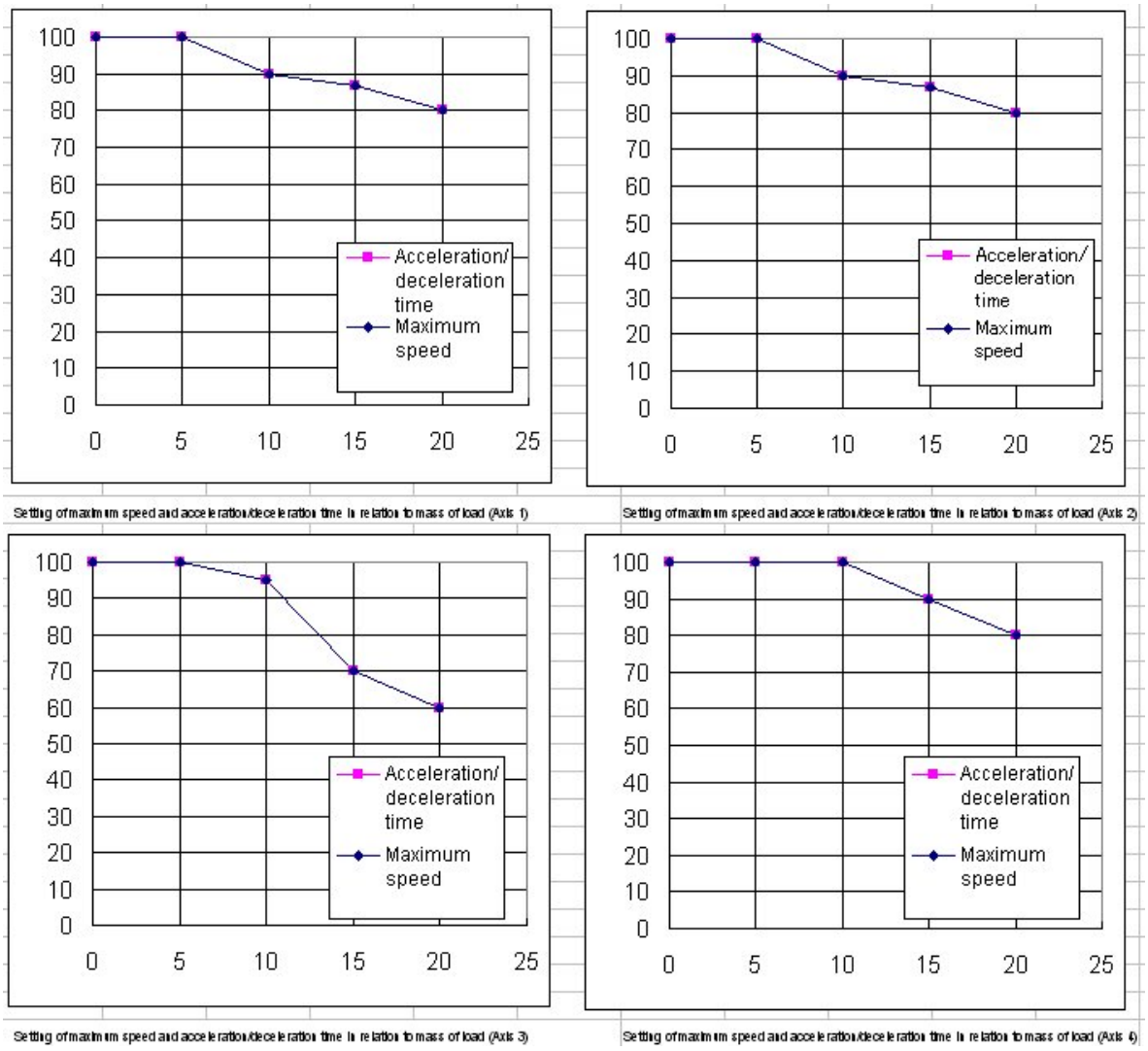


Fig. 4.22 Setting of maximum speed and acceleration/deceleration time for mass of load (TH1050)

Additionally, if there is a center-of-gravity offset of load, the maximum speed and acceleration/deceleration time change as shown in Fig. 4.23 and Fig.4.24.

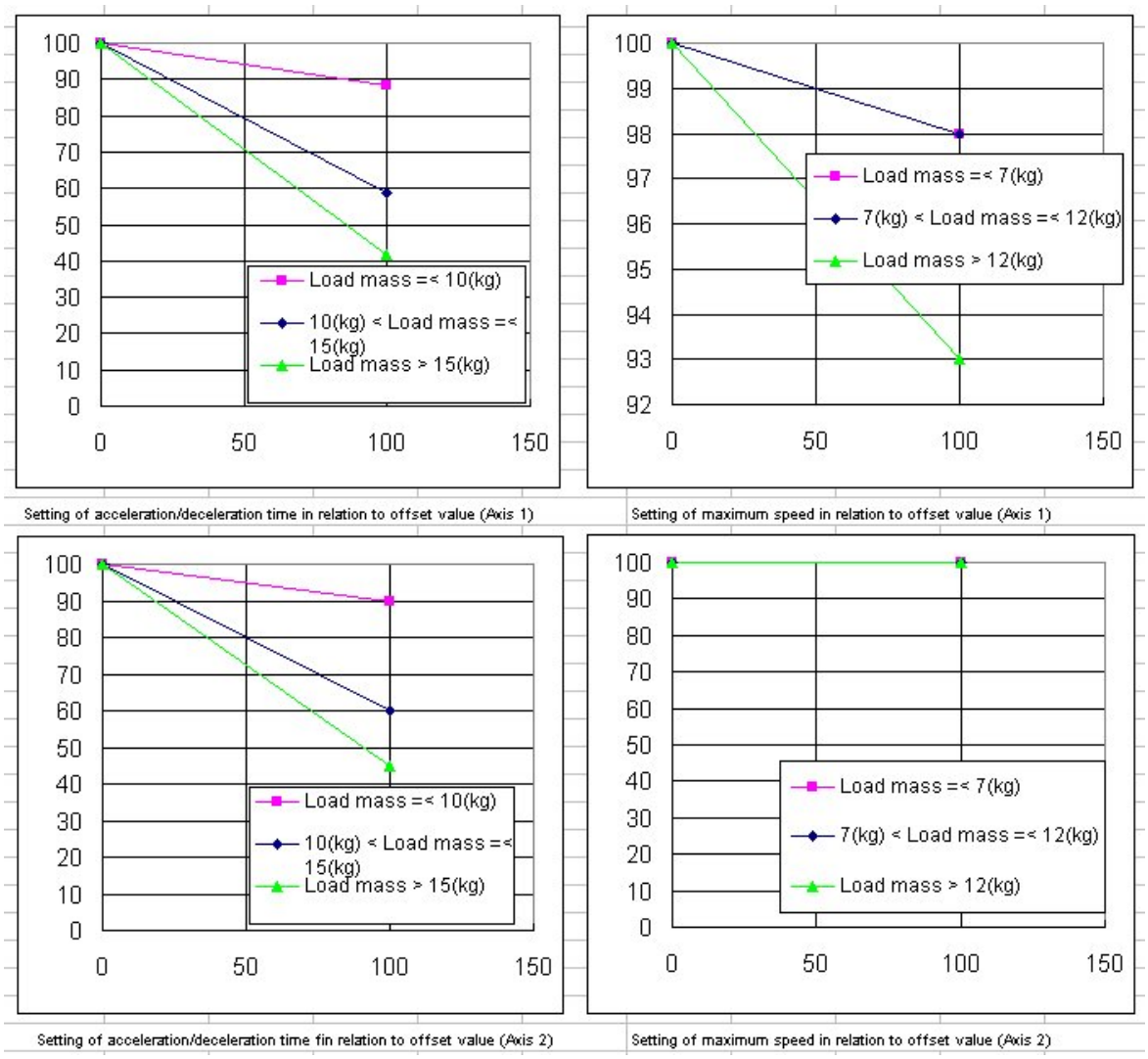


Fig. 4.23 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH1050-axis1/axis2)

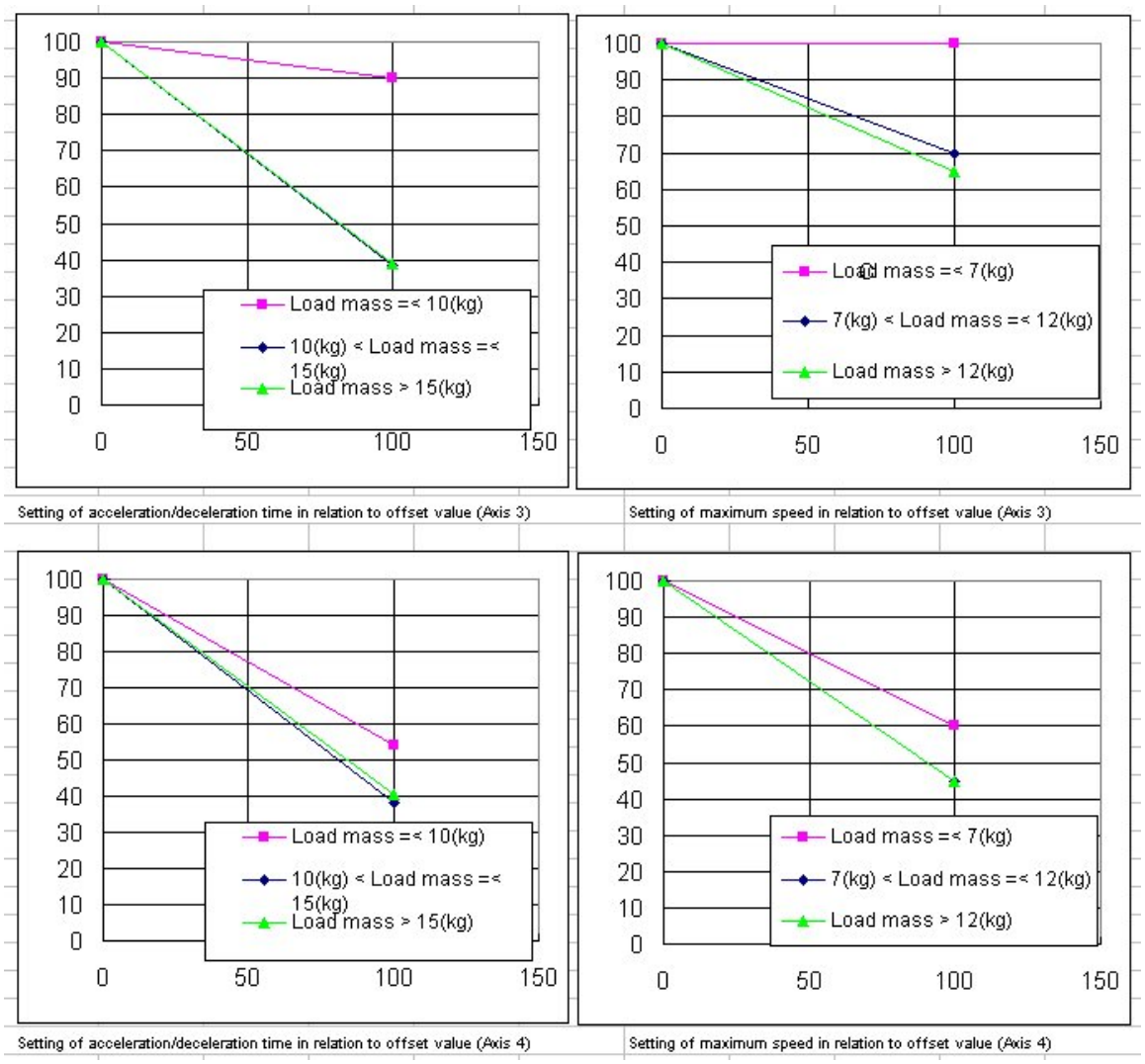


Fig. 4.24 Setting of maximum speed and acceleration/deceleration time for center-of-gravity offset (TH1050-axis3/axis4)

Section 5

Specifications

5.1 Specifications Table

Item		Specifications	
Structure		Horizontal multi-joint type SCARA robot	
Model		TH450	TH550
Applicable controller		TS2000	
Mass of actuator		27 kg	29kg
No. of controlled axes		Four (4)	
Arm length		450 mm (200 mm + 250 mm)	550 mm (300 mm + 250 mm)
Motor capacity	Axis 1	1000 (W)	
	Axis 2	400 (W)	
	Axis 3	200 (W)	
	Axis 4	200 (W)	
Operating range	Axis 1	±120 (deg)	
	Axis 2	±145 (deg)	
	Axis 3	150 (mm) [option:300(mm)]	
	Axis 4	±360 (deg)	
Maximum speed (*1)	Axis 1	600 (deg/s)	375 (deg/s)
	Axis 2	600 (deg/s)	
	Axis 3	2000 (mm/s)	
	Axis 4	2000 (deg/s)	
	Composite speed of axes 1 and 2	7.33 (m/s)	6.21 (m/s)
Rated payload mass		2 (kg)	
Maximum payload mass		5 (kg)	
Permissible load inertia (*1)		0.05 (kg·m^2)	
Repeatability	X, Y	±0.015 (mm)	
	Z	±0.01 (mm)	
	C	±0.005 (deg)	
Cycle time (*2) (When payload mass is 2 kg)		0.33 (sec)	

Drive system	By means of AC servo motors
Position detection method	Absolute

*1: When the mass of load exceeds 2 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*2: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

! CAUTION

Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).
If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

Item		Specifications
Structure		Horizontal multi-joint type SCARA robot
Model		TH650
Applicable controller		TS2100
Mass of actuator		55 kg
No. of controlled axes		Four (4)
Arm length		650 mm (300 mm + 350 mm)
Motor capacity	Axis 1	1000 (W)
	Axis 2	600 (W)
	Axis 3	400 (W)
	Axis 4	400 (W)
Operating range	Axis 1	±160 (deg)
	Axis 2	±143 (deg)
	Axis 3	200 (mm) [option:400(mm)]
	Axis 4	±360 (deg)
Maximum speed (*1)	Axis 1	337.5 (deg/s)
	Axis 2	600 (deg/s)
	Axis 3	2000 (mm/s)
	Axis 4	1700 (deg/s)
	Composite speed of axes 1 and 2	7.49 (m/s)
Rated payload mass		2 (kg)
Maximum payload mass		10 (kg)
Permissible load inertia (*1)		0.1 ($kg \cdot m^2$)
Repeatability	X, Y	±0.015 (mm)
	Z	±0.01 (mm)
	C	±0.015 (deg)
Cycle time (*2) (When payload mass is 2 kg)		0.36 (sec)
Drive system		By means of AC servo motors
Position detection method		Absolute

*1: When the mass of load exceeds 2 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*2: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

CAUTION

Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).
If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

Item		Specifications	
Structure		Horizontal multi-joint type SCARA robot	
Model		TH850	TH1050
Applicable controller		TS2100	
Mass of actuator		72 kg	75kg
No. of controlled axes		Four (4)	
Arm length		850 mm (350 mm + 500 mm)	1050 mm (550 mm + 500 mm)
Motor capacity	Axis 1	1000 (W)	
	Axis 2	1000 (W)	
	Axis 3	600 (W)	
	Axis 4	600 (W)	
Operating range	Axis 1	±160 (deg)	
	Axis 2	±145 (deg)	
	Axis 3	200 (mm) [option:400(mm)]	
	Axis 4	±360 (deg)	
Maximum speed (*1)	Axis 1	300 (deg/s)	281 (deg/s)
	Axis 2	411 (deg/s)	
	Axis 3	2000 (mm/s)	
	Axis 4	1147 (deg/s)	
	Composite speed of axes 1 and 2	8.03 (m/s)	8.73 (m/s)
Rated payload mass		5 (kg)	
Maximum payload mass		20 (kg)	
Permissible load inertia (*1)		0.2 ($kg \cdot m^2$)	
Repeatability	X, Y	±0.03 (mm)	
	Z	±0.02 (mm)	
	C	±0.015 (deg)	
Cycle time (*2) (When payload mass is 2 kg)		0.51 (sec)	0.65 (sec)
Drive system		By means of AC servo motors	
Position detection method		Absolute	

*1: When the mass of load exceeds 2 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*2: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

CAUTION

Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).
If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

Appendix A Calculating Control Power Capacity

The power capacity of the controller can be calculated in the following manner. The power connected to the controller should exceed the capacity obtained from the following equation:

(Power capacity) = (Power capacity of controller) + (Power capacity of servo driver)

a) Power capacity of control section
The power capacity of the controller excluding the servo driver is as follows:
Power capacity of controller..... 105 VA
Common to the TS2000 and TS2100.

b) Power capacity of servo driver
The power capacity of the servo driver is the total of the power capacities of the servomotors used in the robot. The following table lists the power capacity of each motor.

Motor capacity (W)	Power capacity (VA)
100	143
200	286
400	571
600	857
750	1071
1000	1429

(Ref.) The power capacity of each servo motor is figured out from the following equation.

(Power capacity of servo driver) = MW / (Cos$\theta \times \eta$)
MW : Motor rated output (W)
Cos$\theta \times \eta$: Servo amplifier efficiency (including power-factor 0.7)

c) Example of calculating power capacity

The power capacity of TH series robot is listed in the following table.

Power capacity of each robot

Power capacity [VA] = Sum of power capacities of servo amplifiers + Controller power capacity

Example: (TH450 power capacity)
= 1071 + 571 + 286 + 286 + 105
= 2319 [VA]

Type of robot	Axis 1 motor [W]	Axis 2 motor [W]	Axis 3 motor [W]	Axis 4 motor [W]	Power capacity [VA]
TH450	750	400	200	200	2319
TH550	750	400	200	200	2319
TH650	1000	600	400	400	3534

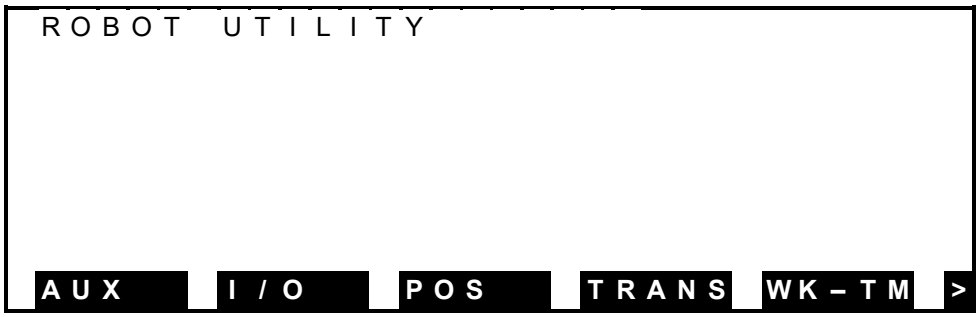
Note: With respect to the power capacity required, it is assumed that all axes are continuously operated at 100% load. If the power capacity is insufficient when the robot is operated, troubles such as a drop in power voltage, a malfunction, or resetting of the control power may take place. Thus, the robot may not be correctly operated.
Also, a current three (3) times the rated current may be required at the start of the robot. Thus, it is necessary to select a power supply in which the peak current of the power is three (3) times the rated current for the duration of 100 ms.

Appendix B Encoder Error Reset

If the battery error or position detector error occurred and it will occur repeatedly even after the cause has been removed, reset the error flag (Err-df) in the encoder according to the following procedures.

[Procedures]

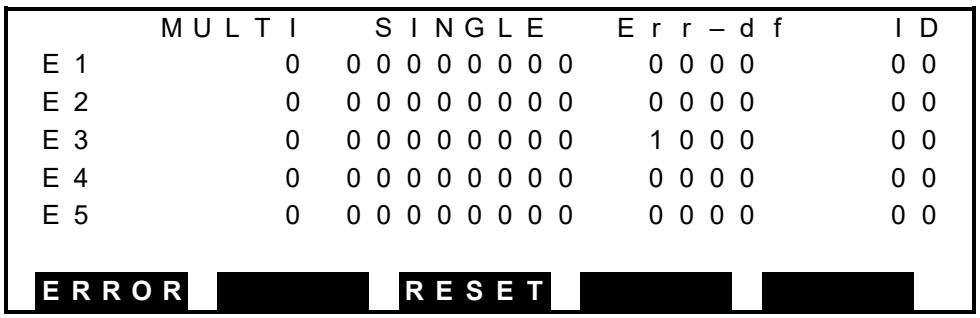
1. Press the UTILITY key, and the following screen appears.



2. Press the NEXT key.



3. Press the ENC key.



The encoder data of each axis is displayed.
Move the cursor to the position where the error flag is set to "1", using the ⇩ and ⇧ keys. Then press the RESET key, and the error flag is reset compulsively. If the error still remains even after this operation has been done, it is yet to be cancelled. When this happens, investigate into the cause again.
Even if the error has been reset, make sure of the robot home position before operating the robot.

Appendix C Restoring Home Position Data

Batteries which back up the positional information of the position detectors (i.e., encoders) are contained in the robot base unit. Unless the robot is used for a long period of time, the battery voltage drops and the home position data will be lost. Also, if the voltage of the battery which backs up the internal memory of the X8HC printed board drops, programs, parameters and absolute coordinate data set by the user will be lost. To avoid this, replace the batteries on a regular basis. The yardstick of each replacement period is shown below.

- Batteries for position detectors ⋯ Two and a half (2.5) years after delivery (when the operating time per day is eight (8) hours).
- Battery for X8HC printed board SRAM ⋯ Five (5) years after delivery

If the home position data has been destroyed due to battery voltage drop, etc., observe either of the following procedures (a ~ c) to restore it.

a) How to restore the home position, using REORG command

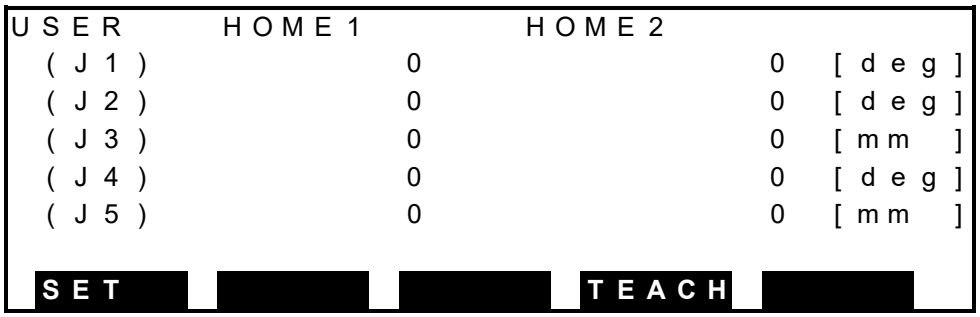
(1) Outline of REORG

This function is useful to restore the original home position data by memorizing each axis coordinate data beforehand, if the machine home position data has been destroyed.

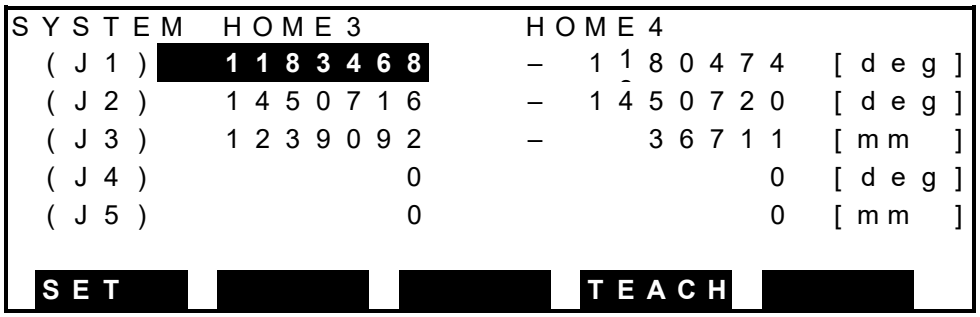
The set points can be assigned to HOME1 ~ 4. The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the 0° position according to the match-mark. In HOME1 and HOME2, the user can specify any desired positions.

We recommend to set there the positions which allow mechanical positioning of the robot end at the time of system construction.

[Page 1]



[Page 2]

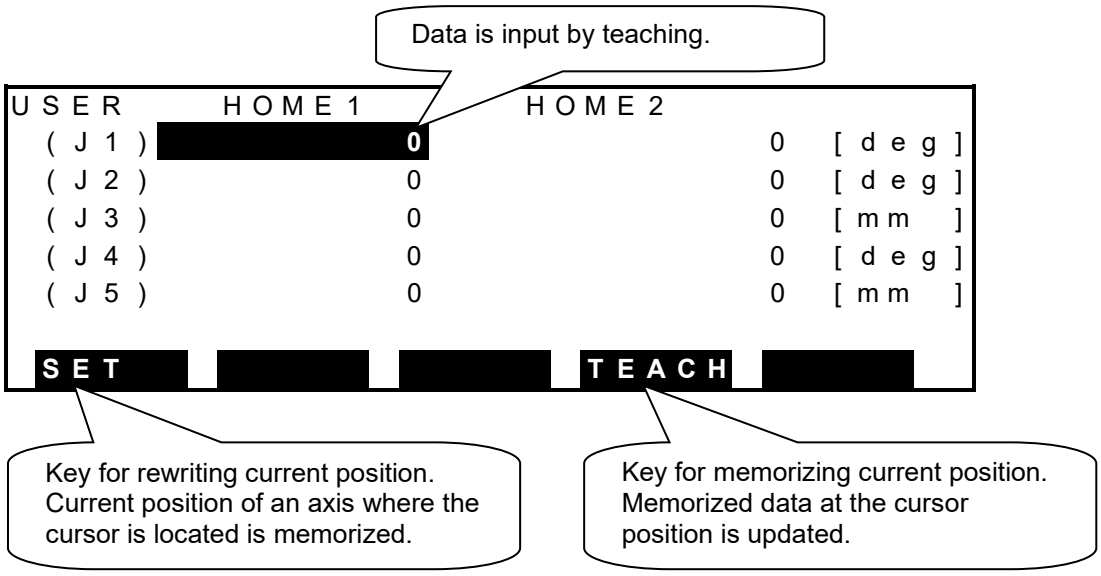


Note: The above are the screens for restoring the machine home positions. NEVER operate them at other than the restoration.

(2) Operating procedures

I. How to set the data in HOME1 and HOME2:
This paragraph describes the data setting in HOME1. (The same procedures are applied to HOME2.)

[1] Guide the robot to secure the tool shaft at a desired position.
[2] Turn off the servo power.
[3] Press the UTILITY key on the teach pendant.
[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.



[5] When the above screen appears, memorize the position where the robot was secured in the [TEACH] mode, by moving the cursor to each axis.

II. How to restore the home position data by HOME1 (or HOME2):
If the machine home position coordinate data has been destroyed, the machine home position can be restored by means of the HOME1 function.

[1] Guide the robot to the position set by HOME1 and secure the robot there.
[2] Turn off the servo power.
[3] Press the UTILITY key on the teach pendant.
[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.
[5] Move the cursor to each axis in the HOME1 mode of the REORG screen. Press the [SET] key and write the position where the robot was secured to the coordinate value set in HOME1. Thus, the machine home position can be restored.

! CAUTION

Cautions on restoring home positions of axis 3 and axis 4 (REORG)

Our SCARA robot (TH series) will cause an axis interference between the axis 3 and axis 4. Unless the home positions are restored according to the predetermined procedures, the current robot position may shift. If you restore the axis 3 (or axis 4) home position, using the REORG function, be sure to restore the axis 4 (or axis 3) home position also.

<Operating procedures>

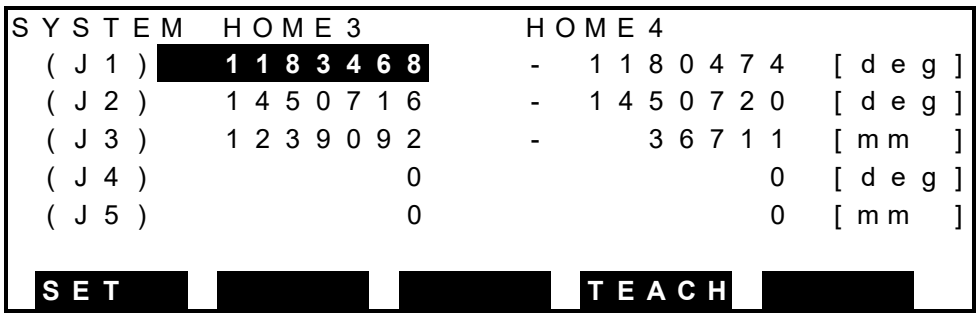
[1] Secure the axis 3 and axis 4 to the mechanical stoppers or specified positions.

[2] Set the cursor to (J3) and press the [SET] (F1) key to reproduce the axis 3 home point.

[3] Likewise, set the cursor to (J4) and press the [SET] (F1) key to reproduce the axis 4 home point.

III. How to restore the position data by HOME3 (or HOME4):
The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the match-mark position.
The machine home position can be restored, using these data.
This paragraph deals with the restoring procedures, using HOME3.

[1] Turn off the servo power.
[2] Press the UTILITY key on the teach pendant.
[3] Press the NEXT key until [REORG] appears, then press the [REORG] key.
[4] Press the NEXT key to display Page 2.



[5] Contact the axis 1 of the robot to the plus (+) side mechanical stopper by hand.
[6] Move the cursor to (J1) of HOME3, then press the [SET] key. Now, the machine home point of axis 1 is restored.
[7] Likewise, contact the axis 2 of the robot to the plus (+) side mechanical stopper by hand.
[8] Move the cursor to (J2) of HOME3, then press the [SET] key. Now, the machine home point of axis 2 is restored.
[9] Press the [ESC] key to escape from the REORG screen once.
[10] Turn on the servo power, and set the axes 3 and 4 in the servo-free state.
[11] Set the axis 4 to the match-mark.
[12] Contact the axis 3 of the robot to the plus (+) side mechanical stopper by hand.
[13] Turn off the servo power.
[14] Repeat Steps [2] to [4] inclusive to call the HOME screen again.
[15] Move the cursor to (J4) of HOME3, then press the [SET] key. Now, the machine home point of axis 4 is restored.
[16] Move the cursor to (J3) of HOME3, then press the [SET] key. Now, the machine home point of axis 3 is restored.
[17] Press the [ESC] key to escape from the HOME screen. Now, the operation has finished.

! CAUTION

Cautions on restoring home positions of axis 3 and axis 4 (REORG)

Our SCARA robot (TH series) will cause an axis interference between the axis 3 and axis 4. Unless the home positions are restored according to the predetermined procedures, the current robot position may shift. If you restore the axis 3 (or axis 4) home position, using the REORG function, be sure to restore the axis 4 (or axis 3) home position also.

<Operating procedures>

[1] Secure the axis 3 and axis 4 to the mechanical stoppers or specified positions.
[2] Set the cursor to (J3) and press the [SET] (F1) key to reproduce the axis 3 home point.
[3] Likewise, set the cursor to (J4) and press the [SET] (F1) key to reproduce the axis 4 home point.

b) How to restore the home positions by multi-turn data clear (special method)

1) A home point match-mark is put on each of the axis 1, axis 2 and axis 4. If only the multi-turn data of either of these axes has been cleared, move the robot to the home point mark of each axis.
For the axis 3, move the axis to the appropriate position for each TH series, referring to Para. c)–(2).

2) Turn the servo off, select the TEACH operation mode, and press the UTILITY key on the teach pendant.

3) When the UTILITY screen is called on the display, press the NEXT key once followed by the [ENC] (F5) function key.

4) When the encoder information screen is displayed, press the ALT and 0 keys at the same time to cancel the editing lock mode.

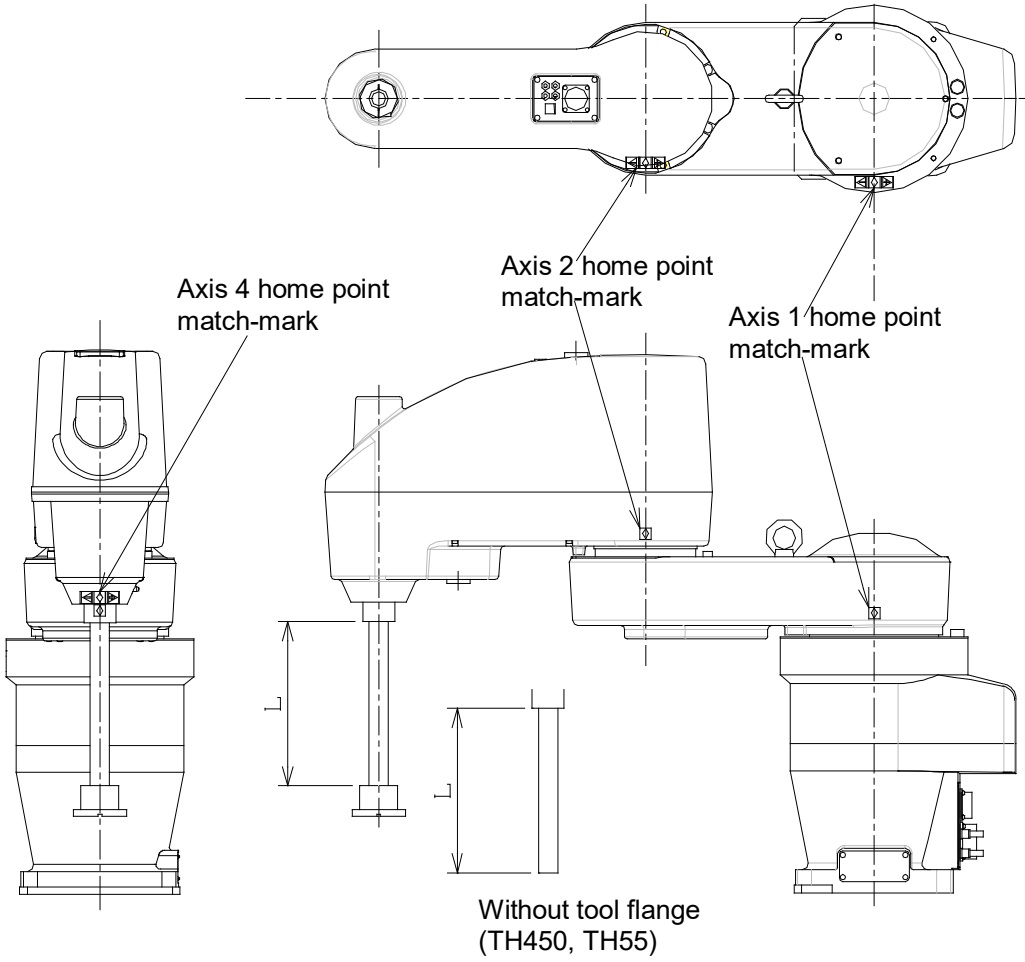
5) Set the cursor to the axis whose multi-turn data you wish to reset to zero (0), and press the [RESET] (F3) function key.

6) Move the robot to the teach points and make sure that the robot can be located at each point exactly. (This method can completely restore the robot to the previous condition.)

c) How to restore the home positions by aligning each axis with the home point match-mark (simple method)

This is the simple method for restoring the home positions. A minor error may be caused according to the adjustment.

(1) Turn on the controller power and align the home point match-marks of each axis in the manual mode. If this is not possible, mark the reference position of each axis and jot down its coordinate beforehand.



Locations of home point match-marks of respective axes

(2) Move the axis 3 to near the lower limit and set the length "L" as shown in the figure to the following value.

TH450, TH550	:	192 mm
TH650	:	215 mm

(3) Setting of new home position data (To be set through teach pendant.)

1) Select the TEACHING mode by means of the master mode switch.

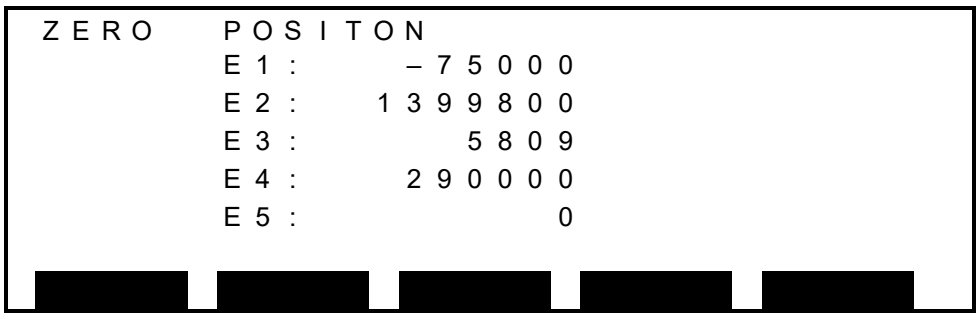
2) Press the UTILITY key on the teach pendant.

3) Press the NEXT key on the teach pendant twice.

4) Turn the servo power off.

5) Press the [F3] key on the teach pendant to select the ZEROP mode.

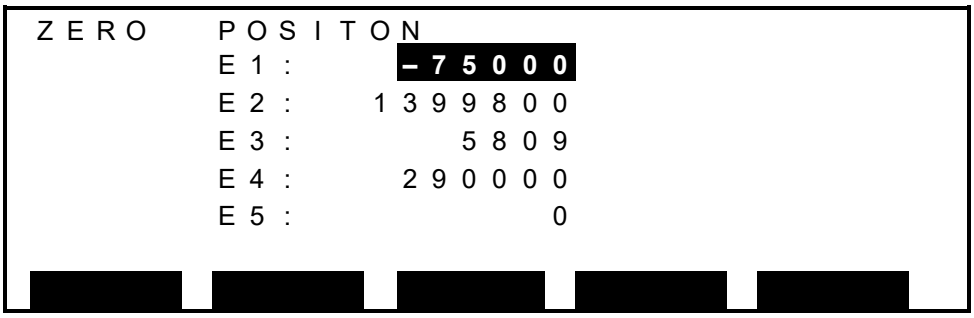
6) The following home point setting screen is called.



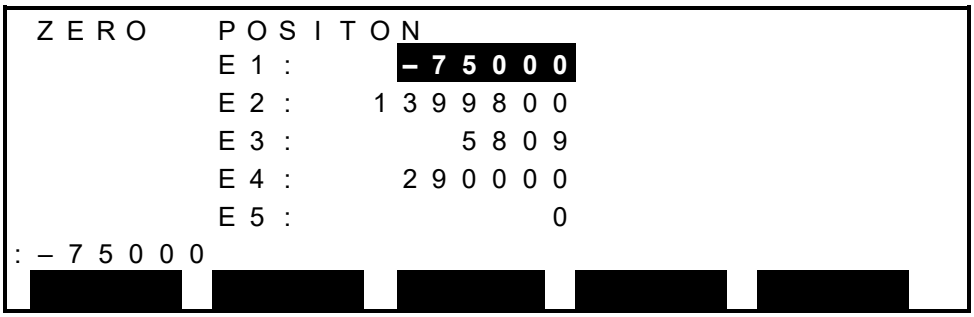
7) Press the ALT and 0 keys on the teach pendant.

8) The following home point editing screen appears.

9) The numerical value of E1 (axis 1) is inversely displayed.

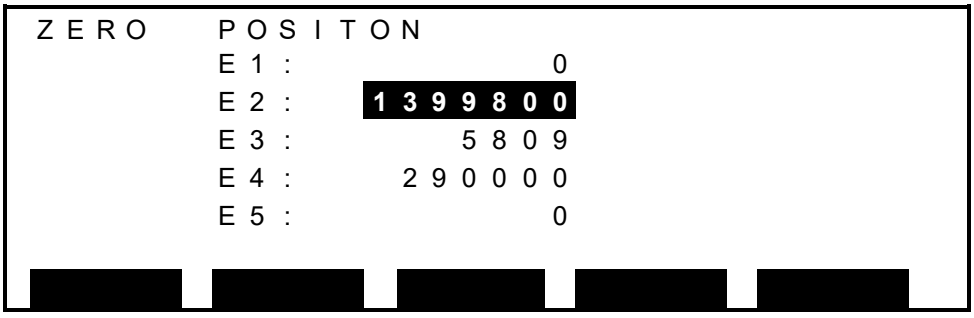


10) Press the EXE key on the teach pendant, and the value of "E1" is displayed at the lower left side. Set "0" in lieu of this value and press the EXE key. Then the E1 value becomes "0".
Unless the axis could be located at the home point mark, enter the coordinate value registered beforehand.



11) When the "↓" key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.

Caution: For the SCARA robot (TH series), be sure to set the home point in the order of E1 → E2 → E4 → E3 because E3 (axis 3) and E4 (axis 4) are interlocked with each other.



(4) Confirmation of home position

1) Make sure of the robot current position according to the POS data of the UTILITY mode. Guide the robot to the approximate home position by manual operation and confirm the position. Unless it could be located at the home point mark, make sure that the approximate position is the reference position marked in advance.

2) Execute the following program and make sure of the robot machine home position. Unless the robot could be located at the home point mark, DO NOT use the following program.
Create a program in which the robot moves to the position recorded beforehand, referring to the following program. Then make sure that the robot moves to the reference position mark according to the program thus created.

```
PROGRAM ZERO
 SPEED=10
 MOVEA 3,100
 MOVEA 1,0
 MOVEA 2,0
 MOVEA 3,0
 MOVEA 4,0
END
```

(5) Data saving
When the home points have been established, the contents of system file "MACHINE.PAR" (machine parameter) in the robot controller are updated.
Connect the personal computer with the controller, then back up the "MACHINE.PAR" file saved in the robot to the personal computer, using TSPC (option).
Additionally, copy the backed up file to the system disk attached to the controller. (Should the system file have been destroyed, you can perform the restoring operation, using the system disk.)

(6) Now, the operation completes.